

THE CHAUTAUQUAN

*A Monthly Magazine Devoted to
the Promotion of True Culture.*

*Organ of the
Chautauqua Literary and
Scientific Circle.*

Vol. V
May, 1885.
No. 8.

*Theodore L. Flood, D.D., Editor
The Chautauqua Press*

THE CHAUTAUQUAN

*A Monthly Magazine Devoted to
the Promotion of True Culture.*

*Organ of the
Chautauqua Literary and
Scientific Circle.*

Vol. V
May, 1885.
No. 8.

*Theodore L. Flood, D.D., Editor
The Chautauqua Press*

The Project Gutenberg eBook of The Chautauquan, Vol. 05, May 1885, No. 8

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Chautauquan, Vol. 05, May 1885, No. 8

Author: Chautauqua Literary and Scientific Circle
Chautauqua Institution

Editor: Theodore L. Flood

Release date: August 10, 2017 [eBook #55331]
Most recently updated: October 23, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/55331

Credits: Produced by Emmy, Juliet Sutherland and the Online
Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK THE
CHAUTAUQUAN, VOL. 05, MAY 1885, NO. 8 ***

Transcriber's Note: This cover has been created by the transcriber and is placed in the public domain.

THE CHAUTAUQUAN.

*A MONTHLY MAGAZINE DEVOTED TO THE PROMOTION OF TRUE
CULTURE. ORGAN OF THE CHAUTAUQUA LITERARY AND
SCIENTIFIC CIRCLE.*

VOL. V.

MAY, 1885.

No. 8.

Officers of the Chautauqua Literary and Scientific Circle.

President, Lewis Miller, Akron, Ohio. *Chancellor*, J. H. Vincent, D.D., New Haven, Conn. *Counselors*, The Rev. Lyman Abbott, D.D., the Rev. J. M. Gibson, D.D.; Bishop H. W. Warren, D.D.; Prof. W. C. Wilkinson, D.D.; Edward Everett Hale. *Office Secretary*, Miss Kate F. Kimball, Plainfield, N. J. *General Secretary*, Albert M. Martin, Pittsburgh, Pa.

Contents

Transcriber's Note: This table of contents of this periodical was created for the HTML version to aid the reader.

REQUIRED READING

ENGLISH AS A UNIVERSAL LANGUAGE	<u>435</u>
SUNDAY READINGS	
[May 3]	<u>438</u>
[May 10]	<u>438</u>
[May 17]	<u>438</u>
[May 24]	<u>439</u>
[May 31]	<u>439</u>
HOME STUDIES IN CHEMISTRY AND PHYSICS	
Physics of Earth	<u>441</u>
THE EYES BUSY ON THINGS ABOUT US	<u>443</u>
EASY LESSONS IN ANIMAL BIOLOGY	
Chapter II.—Sub-Kingdom VII.—Articulata	<u>445</u>
HOW TO WIN	
Chapter III.	<u>450</u>
THE LIFE OF MINERALS	<u>453</u>
THE MACHINERY OF OUR FOREIGN SERVICE	<u>455</u>
MADURA AND ITS PAGODA	<u>458</u>
GEOGRAPHY OF THE HEAVENS FOR MAY	<u>460</u>
THE HOMELIKE HOUSE	
Chapter IV.—The Bedroom	<u>461</u>
“CONSIDER THE LILIES”	<u>463</u>
A BIRD’S EYE VIEW OF FORESTRY	<u>464</u>
GOVERNMENT EMPLOYMENT FOR WOMEN	<u>467</u>
THE ART OF FISH CULTURE	

Part II.	<u>470</u>
HONESTY IN THE C. L. S. C.	<u>473</u>
OUTLINE AND PROGRAMS	<u>474</u>
LOCAL CIRCLES	<u>475</u>
THE C. L. S. C. CLASSES	<u>481</u>
EDITOR'S OUTLOOK	<u>483</u>
EDITOR'S NOTE-BOOK	<u>486</u>
C. L. S. C. NOTES ON REQUIRED READINGS FOR MAY	<u>488</u>
TALK ABOUT BOOKS	<u>491</u>
CHAUTAUQUA, 1885	<u>493</u>
THE FLORIDA CHAUTAUQUA	<u>496</u>
SPECIAL NOTES	<u>496</u>

REQUIRED READING FOR MAY.

ENGLISH AS A UNIVERSAL LANGUAGE.

BY PRESIDENT D. H. WHEELER, D.D., LL.D.

At the beginning of this study we must determine as clearly as possible the meaning attached to the word *universal* when it is applied to a language. For this purpose, the word is employed in a restricted sense. It is not meant that men are returning to the conditions of speech which prevailed before the building of Babel. To that condition men may or may not return in some far off century; so far as we can judge, that reunion of mankind is very distant. Max Müller^[1] says that 900 languages are spoken by the human race; and if dialects are added the number may be three or four times as large. Most of our race speak dialects having no literature, and it is found to be a very slow task to substitute the language of a nation for the dialects of its people. Classical Greek was probably spoken by only a few of the Greeks. We know that the Latin of Cicero was not, not even in Rome, the speech of the people. At this day more than a score of dialects are spoken in Italy, and the majority of the population of that Peninsula can neither speak nor understand Italian. Public education and other unifying influences make very little progress against the diversity of speech in the various provinces. An Italian gentleman is bred in a nurse's language (the dialect) and educated in Italian. He speaks two tongues, dialect and Italian; his servants speak but one, and that is the dialect. To break up the power of the nurse, and of the local influences which she represents, is a Herculean task which must be accomplished in every country except the United States, the Canadas, and a part of Great Britain, before mankind will speak *only* 900 tongues; the time required to reduce nine hundred to one can only be guessed at. If the problem of a universal language were how to provide a common language to be exclusively used by all men, this problem would be one for pure speculation, in the solving of which the imagination would be more active than the reason.

There is another sense in which scholars speak of a universal language. (1) There has always been in our Aryan^[2] tribe a leading literary language. Once it was Sanskrit;^[3] perhaps at a later time it was Persian; later it was Greek; later it was Latin. This Aryan tribe of ours has made the greater part of the history of the last 3,000 years; to-day it is the history-making and literature-making tribe. Great tracts of older history—Babylonian, Arabian, Egyptian—lie outside of the Aryan movements; but Persian, Greek, Roman, German, French, Spanish, English and American history lie in the Aryan line. Now, then, within this line some one Aryan language has always enjoyed a literary predominance. (2) For other than literary purposes, some one of our family of languages has at one time or another had an extended currency. For something like two centuries French, for example, has been the language of diplomacy. We are probably passing out of a period which has lasted for half a century, of the predominance of German as a language of research, especially of metaphysical and grammatical study. These examples will suffice to show what is meant by our problem. In trade and practical invention English is, in this modified sense, a universal language. What I undertake to measure is this: the probability that, at a not distant time, English will be universal in more senses (used by people of many countries for more purposes) than any other Aryan language was ever before used. Some careful observers believe that this is the present position of the language—that it is now universal to an extent quite beyond all precedent. I think that as a literary, political and commercial language, English has a fair prospect of universal use within the Aryan tribe, and a better prospect than any other tongue of coming into use for these purposes all over the globe.

1. The mere arraying of numbers carries with it a kind of presumption in favor of English. Did 5,000,000 ever at one time use either classic Greek or the Hellenic form of that speech? I seriously doubt that at any time 5,000,000 of people could have understood each other in any one form of that tongue. Did ever at any one time 15,000,000 of people speak the Latin of our classic authors? I doubt it. Turn to English, and we find not less than 110,000,000 of people speaking it at the present time. And this comparison is made much stronger when we remember that, one hundred years ago, our number did not probably exceed fifteen millions. Our vast growth is related to forces and conditions still existing, and now giving us an accelerating

progress in numbers. I shall return to these forces and conditions in a later paragraph. Looking at our competitors, we see that the Germans come next to us, with probably not more than 45,000,000. The German empire counts 45,000,000, and there are 5,000,000, perhaps, of Austrian Germans; but the empire envelops millions of non-Germans. They sometimes zealously claim the Low Dutch; but we could claim them for English with better right. Next come the French with 37,000,000, about one third only of our strength. I think it fair for these purposes to count the populations under the dominion of the several tongues. French, English and German are spoken all over the world. Among the three runs the competition for the first place in the Aryan tribe. I do not commit the folly of making race lines and speech lines the same; but it is a convenient mode of describing the field of this competition as that of the Aryan race, although there have been so many fusions of tribes that perhaps there is no Aryan race at this late day.

2. Another presumption in favor of our language arises from the history of the competition for the first place which has gone on since the rise of the modern European nations. After the Dark Ages,^[4] Italian first came to prominence as a polished literary tongue. Its poets converted a dialect into a language, and from Italian literature our Chaucer, our Shakspeare, and even our Milton drew inspiration as well as materials for English poetry. As a literary language, Italian was for two centuries a universal language. Spain had the next opportunity for the primacy. Her flag was first planted on the outposts of this continent; the conquests of her kings at one moment placed Europe and America in her hands. Her poets and philosophers made her language as honorable in literature as her sailors and soldiers made it in discovery and conquest. If Spain had retained her grasp on the Mississippi valley and on Mexico and South America, she, and not we, would now be supreme on this continent; and Spanish, not English, would be the dominant speech of North America. But Spain may be said to have retired voluntarily from the primacy of the Aryan tribe; a great language and a great literature descended to generations of Spaniards who had ceased to be great. Next, the French language took the lead. During the eighteenth century the language of France was undoubtedly the leading Aryan tongue. Since the beginning of this century German has come strongly into the competition, and English has gradually attained a preponderance in numbers over our two great competitors.

3. A third presumption in our favor arises from the diversity of great functions which English fills. It is a political, a literary, a common, and a commercial speech. In this brief space I can dwell only on the fact that it is, in a peculiar sense, a political language. The English race has led the way towards modern liberty and towards the political institutions and habits which are the safeguards of freedom. English words even are in use among the continental nations to describe political things which they have borrowed from us. The word *meeting* is a good example. But the mere diffusion of political terms means comparatively little. The only really great universal language of the past was the language of Rome, and its preëminence grew from its political character. There are in modern Europe two sets of political ideas and juridical institutions; one is Roman, and the other is English; and the two face each other in the modern world, and are really in silent but effective competition with each other. Outside of England, Europe is Roman except as it has been made English. The continental nations have parliaments which are pale copies of the vigorous legislative bodies which cover the English-speaking world. Rome is mighty yet, for the juridical systems of the continent of Europe have their foundations in the Roman civil law. The greatness of Latin did not arise from the genius of the Roman poets and orators. It laid its vigorous hand on human society, and in some sort it is still administering civil society. English has a political and a judicial system of its own, and these systems are conquering mankind. The adoption of the parliament and the trial by jury by continental nations prove the high value of the English inventions in government and law. The French and the German are neither of them political languages. French is the language of a nation which will long command the respect of mankind; but its language counts for nothing in its political structure. German is the language of a great people who make good citizens, but have until recently shown little capacity for political affairs, and have yet to make a great national record. The genius of Bismarck has created a great empire out of a great people; but it remains to be seen whether any aggressive political ideas are to be born of this German empire. The English political system is aggressive. Englishmen are nation-builders; a shipload of them cast upon a desert island would form a government as instinctively and as easily as they would build themselves houses. In this respect English is like Latin; it is the language of a distinct,

original political system, whose obvious advantages are enticing men to copy it all over the world.

4. Another presumption in favor of our speech arises from certain peculiarities of its grammar and of its enunciation. In its grammar, it has cast off more of the inflexional burdens than either of its competitors. Both of them retain the fictitious distinction of genders by terminations, and they are in other respects weighted with inflexional incumbrances. English has, unfortunately, a small inheritance of strong nouns and verbs; but it is far simpler in its grammar than either French or German. And it is not by any means a small matter that the grammarian has never obtained the control of the language. In this respect the Germans enjoy more freedom than the French; but in neither language is the grammarian so little powerful as he is among us. The reason is that English is spoken for all purposes by all our people. I do not forget the English dialects as an exception; but they are a very small exception in contrast with the fact that neither literary French nor literary German are used by the mass of either people for all purposes. There is “bad German” for daily life, and there is dialect French for the mass of Frenchmen outside of the great towns. In our language many efforts have been made to fix precise rules which would please the grammarian and render good English difficult, and therefore not common. These efforts have not prevented the progress of English toward simplicity, and the vigor which simplicity imparts; and the result is that children and foreigners readily learn and use the language which is employed in our books and periodicals. By flouting the grammarian we have dethroned the nurse, and have not one language for the home and another for the book.

The enunciation of English is characterized by a simplicity which has not been enjoyed by any other great language. The English word has but one vocal qualification, and that is the accent on a single syllable. And such is the simplicity of this accent that it may be heavy or light, and still the word is intelligible. The vowels all tend to a nearly common form in most situations; and if the accent be in the right place the word is usually understood. There are, of course, many exceptions, such as an Italian *a* at the end of many words; but though one says *feyther*, *faather*, or *fauther*, we still know what he means. The force of this consideration will best appear by contrast. We do not know how classic Greek was enunciated; but we have reason to suppose that the vocalization was as elaborate and artificial

as the music of light opera. It was, of course, easily spoken after one had learned it, but the learning of it must have been a great task for a foreigner. Latin was less elaborately artificial, but there was doubtless an intonation of the Latin sentence, and a marking of the varying lengths of vowels; and these features must have made good Latin difficult and rare. We know that Italian spreads slowly throughout Italy under a system of public instruction, and that it defies the foreigner's industry and patience, on account of the demand it makes for well defined vowels and a certain sentence rhythm. French is hardly less, if at all less difficult, because it makes similar demands upon the attention and vocal skill of the speaker. A good illustration of all this is found in the fact that the English ear is pleased, while the French and Italian ears are displeased by a "brogue," or a variation of the position of a word in a sentence. We all like to hear foreigners speak our tongue, precisely because they set it to a new music and show us the range of its grammatical simplicity. There is doubtless a *best* arrangement of the words in any English sentence. If the grammarians had had their way, this best arrangement would have become the only proper arrangement. The fact that we allow the foreigner to put our words into a different order than the best, or even the second best—and are still pleased with his work—shows, I think, that English has in its vocal and syntactical systems a simplicity and a flexibility which adapt it to universal use.

5. There is also a presumption in favor of English which arises from its history and from certain results of that history. It was originally Teutonic, and made its first alliance with Scandinavian, and so by swallowing up the speech of "the Danes" it came to represent the northern branch of the Aryan race with more breadth than High German shows. Then came the Norman Conquest, by which other "Danes" who had conquered Normandy and adopted the French language became the masters of England. The contest between French and English ended as that between Scandinavian and English had ended—English swallowed French. This event gave us our composite English, and gave the world a language in which the northern and southern streams of Aryan speech mingle their waters and collect their far-gathered wealth. Ours is a composite and compromise speech. In the Danish and Norman struggles, our tongue proved its tenacity, its "holdfast." In each struggle the conqueror adopted the dialect of the conquered. But in

these conflicts, English showed itself to be not only stubbornly vital, but also remarkably charitable and catholic. It remained English, but it gave liberal hospitality to Danish and French words. And ever since, it has been a great borrower of foreign words. It is in this respect utterly unlike French, which will hardly tolerate even a foreign name, but translates these names whenever it is possible. I have sometimes wondered whether our speech might not become a world-speech by mere swallowing of the vocabularies of other languages. It is, of course, not a subject for rational conjecture; but neither would that great feast of French words have been a rational forecast in the eleventh century. We have borrowed from all the great tongues of our tribe, and from many other great and small dialects. We have always been borrowing; we borrow every year. It would be an entertaining and not unprofitable task to collect these harvests of the last fifty years. The readiness with which we naturalize a foreign word has made many a grammarian sick of his profession; but borrowing words is an English propensity which nothing can “reform” out of us. And this catholicity looks in several ways toward universality. Neither of our competitors has any corresponding merit. French is a grand speech, but it will not tolerate foreign words. German is a grand speech; but it is German, and nothing else, except as it has been infiltrated with literary and judicial Latin. I must pass hurriedly over some other important presumptions in our favor.

6. The primacy in commerce is freely conceded to our tongue. For such purposes it is spoken in all the great seaports of the world. In addition to this trade use, we may note that a great variety of practical inventions carry their English names and terms around the world.

7. We also have the great reading public of the world. Germans read; but their 45,000,000 do not consume more than one eighth as much literary food as England and America, the Canadas and Australia. The English literature market is not only the greatest in the world—exceeding in one year all the literary produce of antiquity which has come down to us—but it is a market which grows rapidly. If a German or a Frenchman wishes to reach the largest number of minds, he must write in the English language. We have made more than half the fame of German authors by translating their works into our tongue. Some of the German authors have been wise enough to learn and to use our speech in their books.

8. French and German each had a special advantage over English fifty years ago. Frenchmen have long had a rare power of popularizing knowledge—a power of which they have made much less use than would have been expected. French is very rich in the power of accurate and plain statement of scientific truth. This power Englishmen have been for some time borrowing, and the fruits of such copying appears in the good fame and liberal incomes of our Huxleys and Proctors. There is no reason to doubt that we shall overtake the French in this path of progress; for we have an inexhaustible demand for popularizations of science. Thirty years ago Germans had a primacy in research. They made the world come to their universities to study and master the German *method* in investigation. I believe that, though this German primacy still exists, it has nearly reached its end. English students have not gone in droves to Germany to come back empty handed. Many of them have conquered the German method and transferred it to their own home and tongue.

9. France is already out of the race. She has but thirty-seven millions, and grows only at a snail's pace. A Frenchman has recently described the stationary condition of his people in the *Révue des deux Mondes* in stronger terms than I could use. German is our only serious rival for the primacy. Is it a serious rivalry? The Germans are a prolific race. They lose millions by emigration, and still increase to an extent which makes all Frenchmen sad. But how shall forty-five millions shut up in old Europe overtake the one hundred and ten millions who have the great open fields of the world? If North America were as thickly populated as Germany we should count our hosts as three or four hundred millions. They have room at home; but we have vastly more room. In a century our North American English population will number 250,000,000. The rest of the English speaking world can hardly fail to grow enough to make our grand total 400,000,000. The next doubling of our tribe—not more than one hundred and fifty years from now—would put us so far in front of all competition that no language would contest the primacy with us. Besides, we have great possibilities of gains outside of our own tribes. South America is more and more under the influence of English and American ideas; the East is being anglicised by the English dominion in India, and by American and English missionary schools. Africa is an open question; but a vast English speaking population

on the Dark Continent is a far more probable addition to our numbers than this American-English population was in 1492, or even in 1700.

Such seems to be the outlook. In a recent letter forecasting the English-speaking primacy of the world, Mr. Gladstone said: “Mr. Barham Zincke, no incompetent calculator, reckons that the English-speaking peoples of the world one hundred years hence will probably count a thousand millions.... A century back I suppose they were not much, if at all, beyond fifteen millions.” This primacy, he adds, “would demand no propaganda, no superlative ingenuity or effort; it ought to be an orderly and natural growth.... To gain it will need no preterhuman strength; to miss it will require some portentous degeneracy.” I have made much more modest estimates than those quoted by Mr. Gladstone. I attach most importance to the political value of English and the nation-building instinct of our tribe. The great rush forward from fifteen to one hundred and ten millions *in a century* is a result of our political facility.

SUNDAY READINGS.

SELECTED BY CHANCELLOR J. H. VINCENT, D.D.

[*May 3.*]

The heart here, the Father yonder, and the universe of man and matter as the meeting place between them, is the whole scope and the whole poetry of the Sermon on the Mount. The preacher shears off all the superfluities and externals of worship and of action, that he may show, in its naked simplicity, the communion which takes place between the heart as worshiper and God as hearer. The righteousness he inculcates must exceed that “of the Scribes and the Pharisees.” The man who hates his brother, or calls him “Raca,” is a murderer in deed.... Oaths are but big sounds; the inner feelings are better represented by “yea, yea, nay, nay.” That love which resides within will walk through the world as men walk through a gallery of pictures, loving and admiring, and expecting no return. The giving of alms must be secret. The sweetest prayer will be solitary and short. One must fast, too, as if he fasted not. The enduring treasures must be laid up within. Righteousness must be sought before, and as inclusive of all things; life is more precious than all the means of it. The examination and correction of faults must begin at home. Prayer, if issuing from the heart, is all powerful. The essence of the law and the prophets lies in doing to others as we would have others do to us. Having neglected the inner life, the majority have gone to ruin, even while following fully and devotedly external forms of faith and worship. The heart must, at the same time, be known by its fruits. It is only the good worker that shall enter the heavenly kingdom. These truths, in fine, acted upon, these precepts from the Mount, heard and kept—become a rock of absolute safety, while all beside is sand now, and sea hereafter.

Such is, in substance, this sermon. It includes unconsciously all theology and all morals, and is invested, besides, with the beauty of imagery—

theology, for what do we know, or can we ever know, of God, but that he is “our Father in heaven,” that he accepts our heart worship, forgives our debts, and hears our earnest prayers—morals, for all sin lies in selfishness, all virtue lies in losing our petty identity in the great river of the species, which flows into the ocean of God; and as to imagery, how many natural objects—the salt of the sea, the lilies of the valley, the thorns of the wilderness, the trees of the field, the hairs of the head, the rocks of the mountain, and the sand of the seashore—combine to explain and beautify the deep lessons conveyed! Here is, verily, the model—long sought elsewhere in vain—of a “perfect sermon,” which ought to speak of God and of man in words and figures borrowed from that beautiful creation, which lies between, which adumbrates the former to the latter, and enables the latter to glorify at once the works and the author.—*Gilfillan*.[\[1\]](#)

[May 10.]

The Hebrew poet was nothing if not sacred. To him the poetical and the religious were almost the same. Song was the form instinctively assumed by all the higher moods of his worship. He was not surprised into religious emotion and poetry by the influence of circumstances, nor stung into it by the pressure of remorse.... Religion was with him a habitual feeling, and from the joy or the agony of that feeling poetry broke out irrepressibly. To him, the question, “Are you in a religious mood to-day?” had been as absurd as “Are you alive to-day?” for all his moods— ... whether wretched as the penitence of David, or triumphant as the rapture of Isaiah—were tinged with the religious element. From God he sank, or up to him he soared. The grand theocracy around ruled all the soul and all the song of the bard. Wherever he stood, under the silent starry canopy, or in the congregation of the faithful—musing in solitary spots, or smiting, with high, hot, rebounding hand, the loud cymbal—his feeling was, “How dreadful is this place! this is none other but the house of God, and this is the gate of heaven.” In him, surrounded by sacred influences, haunted by sacred recollections, moving through a holy land, and overhung by a heavenly presence, religion became a passion, a patriotism, and a poetry. Hence the sacred song of the Hebrews stands alone, and hence we may draw the deduction that its equal we shall never see again, till again religion

outshine the earth with an atmosphere as it then enshrined Palestine—till poets, not only as the organs of their personal belief, but of the general sentiment around them, have become the high priests in a vast sanctuary, where all shall be worshipers, because all is felt to be divine. How this high and solemn reference to the Supreme Intelligence and Great Whole comes forth in all the varied forms of Hebrew poetry! Is it the pastoral? The Lord is the shepherd. Is it elegy? It bewails his absence. Is it ode? It cries aloud for his return, or shouts his praise. Is it the historical ballad? It recounts his deeds. Is it the penitential psalm? Its climax is, “Against *thee* only have I sinned.” Is it the didactic poem? Running down through the world, like a scythed chariot, and hewing down before it all things as vanity, it clears the way to the final conclusion, “Fear God, and keep his commandments, for this is the whole duty of man.” Is it a burden, “tossed as from a midnight mountain, by the hand of lonely seer, toward the lands of Egypt and Babylon?” It is the burden of the Lord; his the handful of devouring fire flung by the fierce prophet. Is it apologue, or emblem? God’s meaning lies in the hollow of the parable; God’s eye glares in the “terrible crystal” over the rushing wheels. Even the love-canticle seems to rise above itself, and behold! a greater than Solomon, and a fairer than his Egyptian spouse, are here. Thus, from their poetry, as from a thousand mirrors, flashes back the one awful face of their God.—*Gilfillan*.

[*May 17.*]

They say it is an ill mason that refuseth any stone; and there is no knowledge but in a skillful hand serves, either positively as it is, or else to illustrate some other knowledge, ... because people by what they understand, are best led to what they understand not.

But the chief and top of his knowledge consists in the Book of books, the storehouse and magazine of life and comfort, THE HOLY SCRIPTURES.... In the Scriptures he finds four things: precepts for life, doctrines for knowledge, examples for illustration, and promises for comfort. These he hath digested severally.

But for the understanding of these, the means he useth are: First, A HOLY LIFE; remembering what his Master saith, that *if any do God’s will he shall*

know of the doctrine (John viii), and assuring himself that wicked men, however learned, do not know the Scriptures, because they feel them not, and because they are not understood but with the same spirit that writ them. The second means is PRAYER; which, if it be necessary even in temporal things, how much more in things of another world, where *the well is deep, and we have nothing of ourselves to draw with!* Wherefore he ever begins the reading of the Scripture with some short ejaculation, as *Lord, pen mine eyes, that I may see the wondrous things of thy law.* The third means is A DILIGENT COLLATION of Scripture with Scripture. For, all truth being consonant to itself, and all being penned by one and the self-same Spirit, it can not be, but that an industrious and judicious comparing of place with place must be a singular help for the right understanding of the Scriptures. To this may be added the consideration of any text with the coherence thereof, touching what goes before and what follows after; as also the scope of the Holy Ghost. When the apostles would have called down fire from heaven, they were reprov'd as ignorant of what spirit they were. For the law required one thing and the gospel another; yet as diverse, not as repugnant; therefore the spirit of both is to be considered and weighed. The fourth means are COMMENTERS AND FATHERS, who have handled the places controverted; which the parson by no means refuseth. As he doth not so study others as to neglect the grace of God in himself, and what the Holy Spirit teacheth him; so doth he assure himself, that God in all ages hath had his servants, to whom he hath revealed his truth, as well as to him; and that as one country doth not bear all things, that there may be a commerce, so neither hath God opened, or will open, all to one, that there may be a traffic in knowledge between the servants of God, for the planting both of love and humility. Wherefore he hath one comment, at least, upon every book of Scripture; and, plowing with this and his own meditations, he enters into the secrets of God treasured in the holy Scripture.—*Herbert.*[\[2\]](#)

[May 24.]

It is exceedingly important, therefore, that all the Christian gifts and graces should be possessed in purity of spirit, uncontaminated by any unholy mixtures of an earthly nature. The mere suggestion that they have merit of themselves and separate from the God who gives them, if it be

received with the least complacency, necessarily inflicts a deep wound. They are accordingly held in purity of spirit, and with the divine approbation, only when their tendency is to separate the soul from everything inward and outward, considered as objects of complacency and of spiritual rest, and to unite it more closely to God.... We do not find the parent, who has that degree of affection for his child which may be called entire or perfect love, making his love a distinct object of his thoughts, and rejoicing in it as such a distinct object; that would not be the genuine operation of perfect love. If his love is perfect, he has no time and no disposition to think of anything but the beloved object toward which his affections are directed. His love is so deep, so pure, so fixed and centered upon one point, that the sight of self, and of his own personal exercises, is lost. It ought to be thus in the feelings which we exercise toward God; and undoubtedly such will be the result, when the religious feeling has reached a certain degree of intensity; that is to say, when the feeling is perfect, the mind is not occupied with the feeling itself, but with the object of the feeling. The heart, if we may so express it, seems to recede from us; it certainly does so as an object of distinct contemplation; and the object of its affections comes in and takes its place. O, the blessedness of the heart, that, free from self and its secret and pernicious influences, sees nothing but God; that recognizes, even in its highest gifts and graces, nothing but God; that would rather be infinitely miserable with God, if it were possible, than infinitely happy without him!

In connection with these remarks we are enabled to understand and appreciate the state of mind which is described in some primitive writers as a state of cessation from “reflex acts.” By REFLEX ACTS, as we employ the phrase here, and as it appears to be employed by the writers referred to, we mean those acts of the mind in which the soul turns inward upon itself, and, ceasing for a time to regard the mere will of God as the only good, takes a self-conscious satisfaction in its own exercises. Such acts, when they are indulged in, stand directly in the way of the highest results of the religious life. On the other hand, he who has entirely ceased to put forth acts of this kind, and loves God to the entire forgetfulness of self, losing sight even of his own exercises, in consequence of being fully occupied with an infinitely higher object, has reached the broad and calm position of spiritual rest, the region of inward and abiding peace—a region where there is no noisy

clamor, no outcries and contests of the passions; no contrivances of prejudice, interest, and ambition; no rebellious sighing and tears of the natural spirit; but all is hushed and lost in the one deep conviction that there is nothing good, nothing permanently true, nothing desirable—no, not heaven itself—but pure and everlasting union with the will of God.—*Prof. Upham.*[\[3\]](#)

[*May 31.*]

All science is simply a perception of the laws of God—a discovery of what he designed when he spread out the heavens and gemmed the infinity of space with its myriad of worlds. The laws of light are simply the power with which the Creator invested it. All we can do is to find what he has written on its wings. The law of magnetism is the subtle power and the mode of action with which God has touched the loadstone. The laws of astronomy, what are they but the thoughts of God, as he projected worlds into space, and gave to them their orbits and their periods?... Of nature in all its expanse, of all created powers, visible and invisible, hath not God said, “All are yours?” Are we not “heirs of God and joint heirs of Jesus Christ?”

I can accord the scientist nearly all he can claim, without in the slightest degree affecting the foundations of my faith.... There are many things which are claimed in evolution, to which I must give the verdict of the Scotch jury, “Not proven.” Yet were I to admit them all, they would not affect my faith in the wisdom and skill and power of the great Father. I admire the skill of the watch-maker who produces an accurate timepiece; but how much more would I admire his skill if he so made one watch that it was capable not only of keeping time, but also of evolving a series of watches, each keeping better time than that which produced it, so that from the plainest, simplest form of a watch there should be eventually evolved a magnificent chronometer, with jeweled holes, whose time would not vary from the true time a second in a million of years! If the great Creator created but a germ, but in that germ were all possibilities of form, and motion, and magnitude, of atoms and of worlds, with their laws of motion so impressed on each that it should take its place in due time, my

admiration for his wonderful skill would be only enhanced.... These men who talk of evolution claim an infinity of time. I ask, how long since this protoplasm developed into a turtle, the turtle into a monkey, the monkey into a man? They admit there is no positive record anywhere. Since human history began there is no instance of any animal ascending in the scale of man. If at all, it must have been far back in the distant ages. Then, I ask, why not give Christianity similar time? She is changing the face of creation; she is transforming sinners into saints, savages into enlightened men. She took them naked, rude and uncultured, and has clothed, taught, and refined them. She has taken man that bowed down to stocks and stones, and has elevated him until he uses the world as a workshop, and all elements as his instruments, until he feels that he is a son of God, and his vicegerent upon earth. Why shall Christianity be called a failure, because it has not yet reached all the sons of men, or transformed them into sons of science? Give her at least as much time to change millions of savages into enlightened humanity, millions of sinners into saints, as, according to their own asking, it takes to change one species into another. We promise that the whole world shall be brought to the foot of the cross before the evolutionist shall find even a single monkey transformed into a man.—*Bishop Simpson*.

HOME STUDIES IN CHEMISTRY AND PHYSICS.

BY PROF. J. T. EDWARDS, D.D.
Director of the Chautauqua School of Experimental Science.

PHYSICS OF EARTH.

Our earth, as a whole, may be compared to a ship sailing on a smooth sea. Like the ship, it has its own motions with reference to other objects, and is affected by forces exterior to itself. The ship feels the influence of winds, currents and tides.

The earth also yields to forces outside of itself; to such an extent, indeed, that philosophers have more than once been led to look for an invisible power, which strangely affected it. For example, Adams^[1] and Leverrier^[2] were led to prophesy the existence of the planet Neptune, because the earth seemed to yield to the touch of some unknown body, which was afterward discovered at the enormous distance of 2,746,271,000 miles.

The ship, in addition to its motion in regard to distant points and susceptibility of being moved by outside forces, has its own peculiar construction, a complex adjustment of planks, timbers, bolts, spars, sails and ropes. It also has its inhabitants, living beings which move to and fro, quite independent, for the time being, of all other parts of the universe, save that on which they reside.

As it glides past some island we look from our cabin window and discover that the *island* seems to be moving by us in an opposite direction. Even so, as our earth sweeps round its axis with a speed of a thousand miles an hour, it seems motionless, while the starry spheres above appear gliding westward.

In spite of the impassioned and reiterated exclamation of Galileo, “It does move!” men would not believe it, until his experiment at the leaning tower of Pisa; and the beautiful demonstration of Foucault^[3] from the dome of the Pantheon in Paris, proved beyond question the earth’s rotation. Let us briefly outline these two experiments. If the world has no motion, a heavy weight dropped from the top of a tall shaft would strike exactly at its base. But if the earth were rotating, the top of the tower must move faster than the base. The weight at the top would have the same motion as the top of the shaft, and would keep it, in falling, in accordance with a well known law, and consequently would strike *beyond* the base. The ball dropped by Galileo did thus strike. Foucault argued: If the earth does *not* rotate, a pendulum suspended so that a needle fastened to its lowest point would trace a line in sand sprinkled on the surface beneath, would forever move along the same line. But if the earth is rotating, the needle will trace different lines on the sand. If the pendulum was suspended at one of the poles, it would, in twenty-four hours, trace a series of lines like the spokes of a carriage wheel about the pole. Foucault showed that the needle *did* trace varying lines in the sand, therefore the earth moves. This experiment is repeated annually at Paris, and I have performed it in the Amphitheater at Chautauqua.

Sir Isaac Newton, as all the world knows, discovered the relation and mutual dependence of all matter in the universe. The law of gravitation has been called “Newton’s Darling Child.” It states, in brief, that every body attracts every other directly as the mass, and inversely as the square of the distance.

The sun has three hundred thousand times the amount of matter contained in our earth; its power of attraction is therefore proportional. A man on its surface would be crushed by his own weight, but even Brobdignag,^[4] or any other giant, could live comfortably on an asteroid. Indeed, he would weigh so little *there* that he could leap like the mountain goat.

As an illustration of the second part of the law: one body three times as far away as another from the attracting power, will be held by a force but one ninth as great. A body near the surface of the earth, or 4,000 miles from its center, will fall sixteen feet in a second. The moon is sixty times further

from the center of the earth than such a body. Newton found that the moon fell toward the earth, or varied from a straight line, 1.36 of sixteen feet in a second. Now, the square of sixty is 3,600, therefore the moon proves that the force of gravity decreases as the square of the distance increases.

Nothing in nature is more beautiful than the adjustment of forces by which our earth is kept forever revolving in its appropriate orbit. In perihelion,^[5] or its nearest approach to the sun, it is 3,000,000 miles nearer than in aphelion, when farthest away from it. Of course it will follow from the law that the attraction of the sun would then be greater. If there were no counteracting influence, there could be but one result—the earth would fly into the sun and be consumed. The very proximity of the earth to the sun, however, increases its speed, and therefore its tendency to fly off on a tangent. Kepler^[6] has expressed this truth in one of his laws: “The radius vector (a line drawn from the sun to the earth) passes over equal areas in equal times.” The average speed of the earth in its orbit is 1,100 miles a minute—3,300 times that of the fastest steamship—but it varies throughout its course, and how wonderful that system of breaks by which its motion is regulated! Divine wisdom alone could invent such a plan.

TERRESTRIAL GRAVITY.

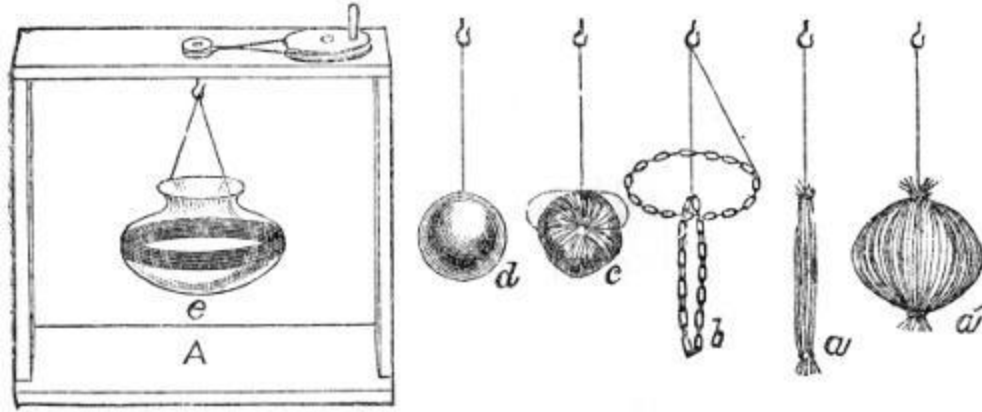
To the ordinary observer, our earth seems, in general, a flat surface, here and there varied by hills and valleys. In reality it is a sphere, with a curvature of eight inches to the mile. The equatorial diameter is twenty-six and five-elevenths miles greater than the polar. The irregularities of the earth’s surface are relatively far less than they seem. Very thin letter paper, spread over a globe sixteen inches in diameter, would by its thickness adequately represent the highest mountain ranges. The greatest ocean depth is about equal to the height of the highest mountain; we see by this that the earth is essentially a smooth, round body.

Its shape is proven in four ways: First, two different navigators may start from the same point, one sailing east and the other west, and reach the same destination. Second, navigators have sailed around the world, Magellan having first performed the task. Third, in moving toward elevated objects, their upper portion first strikes the eye. Fourth, the shadow of the earth, when it falls upon the moon, is round.

The enlargement of the equatorial diameter is supposed to be due to the fact that the earth was once in a plastic state, and the centrifugal force, which is directly proportioned to the rotating speed of a body, caused the matter in the equatorial region to bulge. This action can easily be shown by revolving rapidly a flexible steel hoop, or other mobile substance. All bodies tend to revolve around their shortest axis. A great variety of interesting experiments showing this can easily be performed, some of which are indicated in an accompanying picture.

There is no magical power in the center of our earth, as some have supposed from the fact that all bodies seek that point. Indeed, that is the one spot where there is no attraction, and where all substances would weigh nothing. The path described by a plummet, or any falling body, is simply a resultant motion produced by *opposite* particles of the earth making it pass half way between their lines of attraction. This will ordinarily be toward the center of the earth. As attraction of gravity is in proportion to mass, a body suspended near a mountain will be deflected toward it.

This has been shown by an experiment performed by Dr. Maskelyne,[\[7\]](#) near Mt. Schehallien, in Wales. Upon suspending a light body on opposite sides of this mountain, he observed that it swerved from the perpendicular toward the mountain. The amount of this variation measured the attraction of the mountain, as compared with the attraction of the earth. As the geological structure of this eminence was known, it was not difficult to compute its mass, and a comparison was made between it and the earth. From this calculation the entire weight of the earth was obtained, proving its specific gravity to be five times that of an equal bulk of water. Dr. Cavendish[\[8\]](#) afterward arrived at precisely the same result by experimenting with a pendulum.



A is a rotating machine; *a* is a skein of thread; *á* is the skein rotated; *b* is a chain; *c* is an onion; *d* is an apple; *e* is a glass fish aquarium, one tenth full of water, and rotated. A stick, hoop, shingle, or any such body suspended by a cord, when rapidly rotated will rise and revolve around its shortest axis.

Terrestrial gravity is constantly affecting the motion of bodies. Motion is the act of changing place, and always indicates the presence of some force; force or energy being that which tends to produce motion or rest. Motion in curved lines is produced by two or more forces acting upon a body, one of which must be constant. Example: A cannon ball is acted upon by the sudden explosion of the powder, the resistance of air, and the constant downward attraction of gravity. Nature seems to delight in curved motion; the waves, the flight of birds, the running brooks, the clouds, even the waving trees and grasses, all furnish illustrations of this. A little reflection upon any such instances will show that they are *usually* produced by the united action of an instantaneous and constant force.

The center of gravity is that point around which the opposite particles of a body balance each other. This point does not necessarily coincide with center of figure or center of motion, the former of which is a point equally distant from opposite parts of a regular body, while the latter is a point in a substance around which it revolves.



Ex.—A lead pencil poised on the finger. This experiment can be varied in many ways, showing the nature of stable and unstable equilibrium.

If the sun and all the planets could be strung on a rod passing through their centers, with the planets to the east, the center of gravity of the solar system would be somewhere in the sun, east of its center. As the planets assume various positions with reference to the sun, it must follow that the center of gravity in our system must vary accordingly.

The same is true of objects on the earth. The center of gravity may be elevated or depressed, moved to the right or left. We instinctively adjust our bodies so that a perpendicular let fall from the center of gravity will constantly fall within the base. The most surprising exhibition of this power of automatic adjustment was seen in Blondin, in his performances on the tight rope.

Stability in structures is usually secured by lowering the center of gravity in one of two ways: either by broadening the base or by making it of heavy materials.

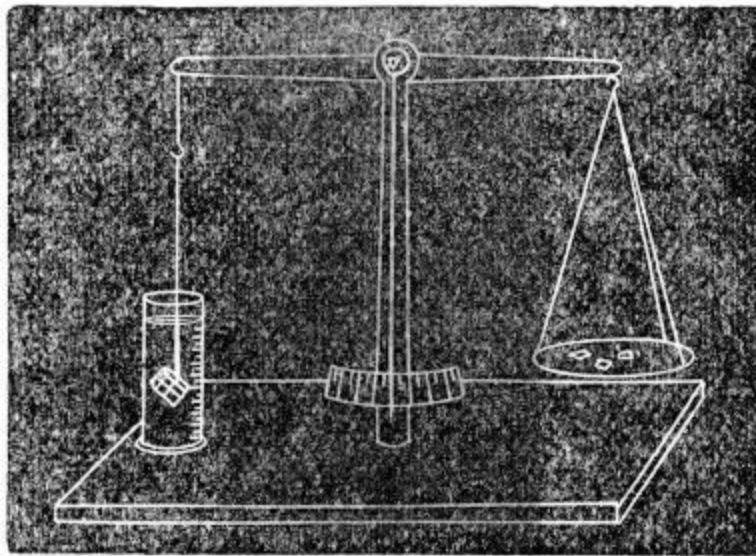
Specific gravity is the weight of a substance as compared with an equal bulk of something taken as a standard; *water* having been selected as the standard for solids and liquids, and *air* for gases.

Gravity furnishes more *units of measure* of various kinds—weight, work, heat, tenacity—than any other force of nature.

It will be remembered that *Physics* is that branch of science that considers the general properties of matter, and the character of those forces which affect matter without destroying its molecule. It includes many

subdivisions. In addition to those already mentioned, we find Molecular Attraction, or the operation of forces that act at insensible distances; Hydrostatics, which treats of liquids at rest; Hydraulics, of liquids in motion; Pneumatics, of gases; Machines, of means for applying force; Acoustics, of the laws of sound; Heat; Light; and Electricity.

As many physical properties have been mentioned in the articles on Air, Water, and Fire, they will not now be considered. Our discussion here applies more especially to those substances which, at ordinary temperatures, are solid.



Ex.—A body buoyed up in water displaces its own weight of the liquid. The glass is nicely graded, and as the water rises in the vessel, the registration at once indicates the amount of water displaced. This proves the truth of the “Law of Archimedes”^[9], ascertained while he was investigating the problem of the golden crown.

The most characteristic properties of solid bodies are the following: Hardness, tenacity, malleability, ductility, and crystalline form. Hardness is the resistance which a body offers to being scratched. Tenacity is the resistance offered by a body to a separation of its parts. Malleability is that property of a body which makes it capable of being rolled into sheets. Ductility is capacity for being drawn into wire, and crystalline form is the property which causes it to assume regular shapes.

As will be observed, these peculiarities are closely dependent upon cohesion and adhesion. By the former we understand the force which holds together the similar molecules of a substance; and by the latter, the force which unites the surfaces of different materials. Familiar as we are with these two agencies, their nature is not yet understood. We can easily discover that they are very dependent upon heat, by the application of which most solids pass from the stable form, to one in which, instead of cohesive force between the molecules, there is repulsion; as in the conversion of ice into water, and then into steam.

This movement of molecules is also dependent upon pressure. The most interesting illustration of this is seen in the action of glaciers. It has been ascertained that the melting temperature of ice lowers one two hundred and fiftieth of a degree for every fifteen pounds of pressure to the square inch.

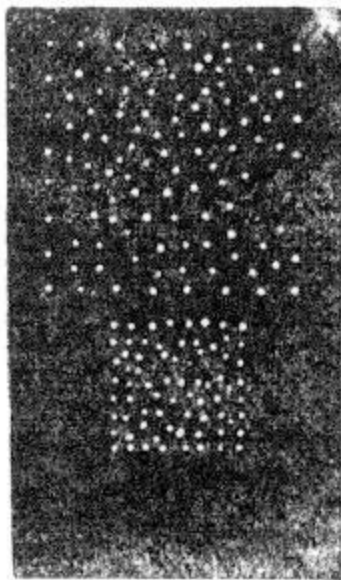
The immense superincumbent mass of ice must, in many places, set free so much latent heat that a portion of the ice melts, so that here and there cells and liquid veins would be opened in the interior of the glacier. But the particles which separate these thin layers of water would almost immediately close up. This is the brilliant demonstration of Prof. Tyndall, who has given the operation the name of “regelation.” It has been thus described: “This phenomenon takes place at every point in the thickness of the glacier. Particles of ice approach one another, and unite across little veins of water, which permeate it in every direction; fresh liquid films are formed under the pressure from above; fresh unions take place between the divided morsels of ice; and, by this continual process of change, the air contained in the mass of that which once was snow, is gradually expelled. Thus it happens that the whole mass ultimately assumes an almost perfect transparency and a beautiful azure color.”

CRYSTALLIZATION.

One of the most beautiful illustrations of cohesive attraction is seen in crystallization. In every instance in which substances pass into the form of a solid, they tend to assume regular shapes called crystals. Each material has its own characteristic form, so that a crystal is a type of a species in the mineral world, even as a plant or an animal is in the organic kingdom. A crystal is a substance bounded by plain surfaces and symmetrically

arranged about imaginary lines called axes. The final form depends upon certain smaller forms in its interior structure. They possess lines of division, often in three directions, called “cleavage.”

While there are millions of crystals, they have all been classified under six systems, as follows: 1. Monometric, where the three axes are equal. 2. Dimetric, having one axis unequal to the other two, which are equal to each other. 3. Trimetric, having no two axes equal. 4. Monoclinic, having one axis inclined. 5. Triclinic, in which all the three intersections are oblique and the axes unequal. 6. Hexagonal, which has the form of a regular hexagonal prism.



Ex.—Showing change of volume. The upper part of the figure represents a substance expanded. There are no more molecules here than below, but they are pushed further apart. This is supposed to be the way in which all bodies are enlarged by heat.

While contemplating the thousand beautiful forms in which molecules are arranged into crystals, whereby many economic purposes are served, as well as taste manifested, one can not resist the conviction that such displays of wisdom, benevolence and love of beauty can alone emanate from the eternal Mind.

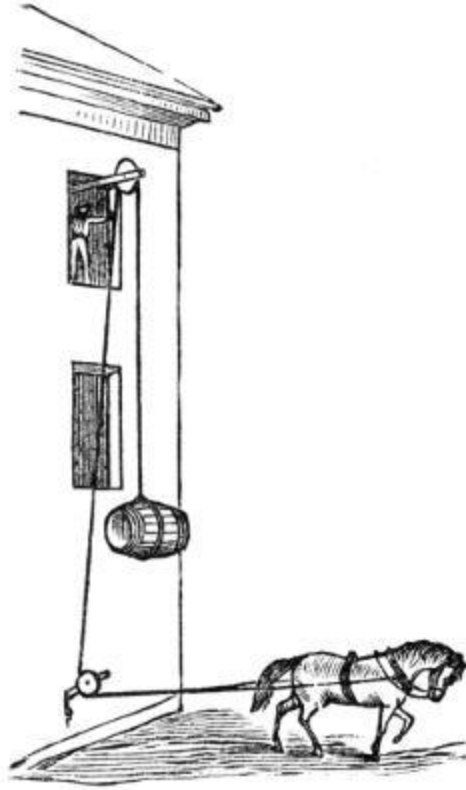
Another wide-spread effect of cohesion is seen in

PETRIFACTIONS.

Everywhere in fossiliferous rock may be found organic remains in which the material of which they were originally composed has been replaced by some mineral substance. Some have supposed that these plants and animals have actually been converted into stone by a change of their elements. This is of course absurd. Carbon can never be anything but carbon, nor indeed, can any element ever become anything other than itself. This dream of the alchemist was long since dissipated. No, strange as it may seem, the molecules of these fossilized organisms must actually pass out, and silica, lime, clay, or some such matter pass in and take their places. Beautiful specimens of petrified wood, found especially on the Pacific coast, are often hard as glass. One very handsome variety, called “opalized” wood, clearly indicates that petrification was either accompanied or followed by crystallization.

Myriads of shells, bones and plants scattered through the earth’s strata have been transformed in the manner indicated. Although petrification is usually a long process, there is reason to believe that it sometimes takes place rapidly. This operation must not be confounded with incrustation, which is often mistaken for it, and takes place where substances, like bending twigs, have deposited upon them layer after layer of lime, salt, sulphur or ice.

The molecules of *solids*, even, are in intense and ceaseless motion. As has been said, “A continuous and restless, nay, a very complicated activity is the order of Nature throughout all her individuals, whether these be living beings or inanimate particles of matter. Existence is, in truth, one continued fight, and a great battle is always and everywhere raging, although the field in which it is fought is often completely shrouded from our view.”



Ex.—A simple illustration of the convenience of machinery in applying force and changing direction.

The motto of the brave Huguenots in the time of Louis XIV. was “Ever burning, but never consumed.”

Nature’s motto, both for matter and energy is, “Ever changing, but never destroyed.” Let us next notice some instances of the

CONSERVATION OF ENERGY.

Energy is the power to do work or overcome resistance. It is of two kinds—potential and kinetic. The former is the energy or force due to position, but it is latent or inactive. The latter is the energy of a body which is in motion. A stone resting on a mountain top, the water in a quiet mill pond, a coiled spring, are all examples of potential energy.

The stone, crushing through the cottage of a peasant, the water turning a factory wheel, the spring turning the wheels of a clock, are examples of actual or kinetic energy.



Ex.—Lay a magnet down on iron filings. They will gather in greatest abundance about the poles, and diminish toward the center, where there are none; thus showing the nature of polarity.

Energy often disappears to reappear under a different name. If we lift our hand to strike the palm of another, our vital energy becomes motion, and that in turn is changed into heat.

In the Bell telephone the sound-waves in the mouthpiece are converted into electric vibrations in the wire, and these, in turn, induce sound-waves in the receiving instrument at the other end of the line.

In dynamo-electric machines we have a chain of transmutations of force—chemical affinity in the fire-box, expansion in the boiler, becoming in turn, motion, magnetism, electric currents, until it appears as resplendent light and intense heat between the carbon points.

Potential energy slumbers in the raindrop, and, anon, as kinetic energy, flashes in the lightning.

In short, the sum of all the energies of nature is a constant quantity, although it manifests itself in a thousand different ways. The foregoing reflections indicate that the researches of modern science all point to a grand unity in God's universe. Let us conclude by briefly referring to some instances of plan or design in the

GROUPING OF LAWS.

The most characteristic feature of all science is that it arranges facts in an orderly manner, under principles or laws.

Nature seems to delight, likewise, in doing a variety of things under one general principle. Note a curious *trinity* in her method: We have three great departments of nature—animal, vegetable and mineral; three parts to our being—physical, mental and moral; three divisions of the mind—intellect, sensibilities and will; three parts to all plants—root, stem and foliage; there

is earth, sea and sky; three great classes in all mechanism—lever, cord, and inclined plane—and many others that might be mentioned.

Observe another group of laws in physics: Variation, in accordance with an exact proportion.

Gravity varies inversely as the square of the distance; heat varies inversely as the square of the distance; light varies inversely as the square of the distance, and sound varies also in exactly the same ratio.

Who can contemplate this exact mathematical arrangement, extending through many departments of matter, without concluding that “Nature is but the name for an effect whose cause is God?”

THE EYES BUSY ON THINGS ABOUT US.

BY JOSEPHINE POLLARD.

A distinguished writer has said: "The eyes are of no use without the observing power," and surely no faculty we possess is capable of so much cultivation as the sight. The facility with which the eye can express the emotions of the soul has been the theme of poets of all ages, who have not hesitated to confess which style of eyes pleased them the most. Says one:

"I everywhere am thinking
Of thy blue eye's sweet smile;
A sea of thoughts is spreading
Over my heart the while."

And others:

"His eyes are songs without words."

"A suppressed resolve will betray itself in the eyes."

"An eye can threaten like a loaded and leveled gun, or can insult like hissing or kicking; or, in its altered mood, by beams of kindness, it can make the heart dance with joy."

"Eyes are bold as lions, roving, running, leaping, here and there, far and near. They speak all languages; wait for no introduction; ask no leave of age or rank; respect neither poverty nor riches, neither learning nor power, nor virtue nor sex, but intrude, and come again, and go through and through you in a moment of time. What inundation of life and thought is discharged from one soul into another through them!"

There are

“True eyes
Too pure and too honest in aught to disguise
The sweet soul shining through them;”

and “eyes that have murder in them, whose flash is the forerunner of thunder.” One has “an eye like Mars, to threaten and command,” and other eyes are “the homes of silent prayer.”

But the variety in color and expression of the eye is as nothing compared to difference in the power of observation. Those ancient companions, “Eyes and No-Eyes,” the story of whose wanderings conveyed a valuable lesson to young and old, were but prototypes of people who go through the world to-day, some of whom see everything, while others see nothing at all. Poets, who could write so beautifully of the eyes, must first have trained their own vision to perceive the beauty or baseness they described, and it is the exercise of this far-seeing, penetrating, analytical power that is the prerogative of genius.

The specialist devotes himself to the closest examination of details. The naturalist does not let the smallest insect escape him, and his trained eye perceives the least peculiarity that denotes the varieties of species.

A person with ordinary eyesight takes up a rose, a lily, or a daisy, and only admires color, shape, or perfume; while the botanist examines the flower in every part, and tells who was its grandfather or grandmother, and feels as tender an interest in it as if it were a human being.

The artist has to train his eye to look for beauty where apparently none appears. He must have an eye for color, for form, for expression, for whatever line he proposes to follow, and he will never rise to eminence if he is satisfied with a hasty, careless, superficial glance.

Turner^[1] was one day painting a landscape with the richness of color that was his specialty, when an English girl who was painting near him left her easel and came to look over his shoulder. “Why, Mr. Turner,” said she, “I don’t see any of those colors in the grass or the trees.”

“No?” said Turner. “Don’t you wish you could?”

It is astonishing that with so much of beauty as there is around us, so many people are found who travel through the world without having used their eyes to any profit whatever. The training needs to be begun in early life; children should be taught how to observe; and as some are duller than others they need to have things pointed out to them, until the habit of examining closely becomes fixed, and like second nature.

What a wonderful field for study there is in the sky above us! Look at the clouds; here, in great, heavy masses; there assuming strange shapes, and taking on an infinite variety of coloring. See the setting sun; never twice alike; a marvel of beauty and grandeur; a feast for even young eyes.

Let us go down by the seashore and watch the great waves come in. The sea is broad, and grand, and deep; but is that all? Note how it reflects the color of the sky; mark the waves that rise afar, and show their white manes like wild horses of the sea, and dash on the shore like a charge of cavalry. How they come galloping, galloping on! Watch for the ninth wave, and look out for yourself! Observe the height that each succeeding wave obtains when the tide is on the rise, and how the character of the beach is changed after a severe storm of wind or rain. There is a volume of interesting study in a handful of sand, a tuft of moss, a small patch of grass, or a bunch of seaweed.

Ruskin,^[2] that exceedingly close observer of art and nature, and eminently sharp critic of men and things, gives us some excellent instruction in the art of looking below the surface. “There is no bush,” he says, “on the face of the globe exactly like another bush; there are no two trees in the forest whose boughs bend into the same network, nor two leaves on the same tree which could not be told one from the other, nor two waves in the sea exactly alike. And out of this mass of various yet agreeing beauty, it is by long attention only that the conception of the constant character—the ideal form—hinted at by all, yet assumed by none, is fixed upon the imagination for its standard of truth. Ask the connoisseur, who has scampered over all Europe, the shape of the leaf of an elm, and the chances are ninety to one that he can not tell you, and yet he will be voluble of criticism on every painted landscape from Dresden to Madrid, and pretend to tell you whether they are like nature or not. A man may recognize the

portrait of his friend, though he can not, if you ask him apart, tell you the shape of his nose or the height of his forehead.

“The color of plants is constantly changing with the season, and that of everything with the quality of light falling upon it; but the nature and essence of the thing are independent of these changes. An oak is an oak, whether green with spring or red with winter; a dahlia is a dahlia, whether it be red or crimson; but let one curve of the petals, one groove of the stamens be wanting, and the flower ceases to be the same. Two trees of the same kind, at the same season, and of the same age, are of absolutely the same color; but they are not of the same form, nor anything like it.”

How few of us observe these things! and how much we miss daily and hourly through lack of this special training of the eye!

A geologist was with a party of friends in the Yosemite valley and called their attention to the play of the light from a campfire on the underside of the leaves of the trees above them. It was a beautiful revelation, and all wondered that they had never noticed it before.

If you are living in the country you should educate the eye to study nature in all its phases, and every day add something to your store of knowledge. Observe the habits of birds, and their haunts; watch the ants and other insects; familiarize yourself with plant life so that you can tell a weed from a flower, and a medicinal herb from a poisonous plant.

If a dweller in the town, observe varieties of architecture, the materials used in the manufacture of houses; compare modern with ancient styles; and lose no opportunity of obtaining information in regard to all that is new and strange. Wherever you are, be less intent on reading novels than in observing wherein you can improve your surroundings. The slattern, with her nose in a book, is blind to the cobwebs that hang from the ceiling, and the rags and dirt visible to every one else. She is cultivating the eyes of her imagination, and reveling in scenes of fairy-like splendor, and has no eyes for the common things of every day life. Her powers of observation are exceedingly limited, and her home is no better for her being in it. She is content to lead an idle life, and does not see in how many ways she might amuse and improve herself.

The trained housekeeper has made good use of her eyes, and by noticing trifles has brought her department to a high state of perfection. It is not enough that she has a natural taste for it; she must be continually looking after things with the searching gaze of an inspector-general. Her practised eyes see when the table-cloth is awry, or the dishes not in their places; when the furniture needs renovating, or the dust has accumulated, and she feels that her reputation is at stake if the defects are not speedily remedied.

An expert in precious stones can tell almost at a glance the value and weight of each gem, and is not easily deceived by counterfeits.

The physician can so train his eye that he has merely to look closely at the patient to determine the nature of his disease; while the microscopist, the geologist, and the astronomer acquire such accuracy from their close and long continued investigations that they can detect the least change in the appearance of the heavens above or the earth beneath.

But the astronomer may have his eyes so fixed on the stars that he can not observe what is going on below; the geologist may be able to analyze a stone and tell to which stratum it belongs, and yet take no interest in anything that is above ground; and the devoted student of the microscope may be so entranced by the wonders continually opening before him, that he is utterly oblivious to all else surrounding him. Without this habit of observation, the world would have had no Galileo, no Humboldt, no Newton, no Agassiz, no Hugh Miller, no Edison,[\[3\]](#) and no progress. But all are not gifted in the same way; and often the sphere we move in or the place in which we are born, determines and decides our calling, and controls our habits to a very great extent. It is natural that one accustomed to an open country should have his eyes attracted toward the heavens, which are constantly revealing new wonders; and that one brought up among the rocks should take to hammering them to bits, boy-like, to see of what they are made, or how they look inside.

The differences between men consist in a great measure in the intelligence of their observation. The Russian proverb says: "He goes through the forest and sees no firewood." "The wise man's eyes are in his head," says Solomon, "but the fool walketh in darkness." It is the mind that sees as well as the eye. Where unthinking gazers observe nothing, men of

intelligent vision penetrate into the very fiber of the phenomena presented to them, attentively noting differences, making comparisons and recognizing their underlying idea. Many before Galileo had seen a suspended weight swing before their eyes with a measured beat; but he was the first to detect the value of the fact.

One of the vergers^[4] in the cathedral at Pisa,^[5] after replenishing with oil a lamp which hung from the roof, left it swinging to and fro; and Galileo, then a youth of only eighteen, noting it attentively, conceived the idea of applying to it the measurement of time. Fifty years of study and labor elapsed before he completed the invention of his pendulum—the importance of which, in the measurement of time and in astronomical calculations, can scarcely be overrated. In like manner, Galileo having heard that a Dutch spectacle-maker had presented to Count Maurice, of Nassau,^[6] an instrument by means of which distant objects appeared nearer to the beholder, began to inquire into the cause of such a phenomena, and this led to the invention of the telescope, and proved the beginning of the modern science of astronomy.

While Captain (afterward Sir Samuel) Brown^[7] was occupied in studying the construction of bridges, with the view of contriving one of a cheap description to be thrown across the Tweed, near which he lived, he was walking in his garden one morning when he saw a tiny spider's web suspended across his path. The idea immediately occurred to him that a bridge of iron ropes or chains might be constructed in like manner, and the result was the invention of his suspension bridge.

So James Watt,^[8] when consulted about the mode of carrying water by pipes under the Clyde, along the unequal bed of the river, turned his attention one day to the shell of a lobster presented at table, and from that model he invented an iron tube, which, when laid down, was found effectually to answer the purpose.

Sir Isambard Brunel^[9] took his first lessons in forming the Thames tunnel from the tiny ship-worm; he saw how the little creature perforated the wood with its well-armed head, first in one direction and then in another, till the archway was complete, and then daubed over the roof and sides with a kind of varnish, and by copying this work on a large scale,

Brunel was at length enabled to construct his shield and accomplish his great engineering work.

It is the intelligent eye of the careful observer which gives these apparently trivial phenomena their value. So trifling a matter as the sight of seaweed floating past his ship enabled Columbus to quell the mutiny which arose amongst his sailors at not discovering land, and to assure them that the eagerly sought New World was not far off.

It is the close observation of little things which is the secret of success in business, in art, in science, and in every pursuit in life. When Franklin made his discovery of the identity of lightning and electricity, it was sneered at, and people asked, "Of what use is it?" To which his reply was, "What is the use of a child? It may become a man!" The great Cuvier^[10] was a singularly accurate, careful, and industrious observer. When a boy he was attracted to the subject of natural history by the sight of a volume of Buffon,^[11] which accidentally fell in his way. He at once proceeded to copy the drawings, and to color them after the descriptions given in the text. At eighteen he was offered the situation of tutor in a family residing near Fécamp, in Normandy. Living close to the seashore, he was brought face to face with the wonders of marine life. Strolling along the sands one day he observed a stranded cuttle-fish.^[12] He was attracted by the curious object, took it home to dissect, and thus began the study of the molluscæ, in the pursuit of which he achieved so distinguished a reputation. He had no books to refer to excepting only the great book of nature which lay open before him. The study of the novel and interesting objects which it daily presented to his eyes made a much deeper impression on his mind than any written or engraved descriptions could possibly have done. Three years thus passed, during which he compared the living specimens of marine animals with the fossil remains found in the neighborhood, dissected the specimens of marine life that came under his notice, and, by careful observation, prepared the way for a complete reform in the classification of the animal kingdom.

The life of Hugh Miller furnishes another illustration of the advantage of making a good use of the eyes. While Hugh was but a child, his father, who was a sailor, was drowned at sea, and he was brought up by his widowed mother. He had a school training after a sort, but his best teachers were the boys with whom he played, the men among whom he worked, the friends

and relatives with whom he lived. With a big hammer which had belonged to his great-grandfather, an old buccaneer, the boy went about chipping the stones and accumulating specimens of mica, porphyry, garnet, and other stones. Sometimes he had a day in the woods, and there, too, his attention was excited by the peculiar geological curiosities which came in his way. While searching among the rocks on the beach, he was sometimes asked, in irony, by the farm-servants who came to load their carts with seaweed, whether he was getting “siller in the stanes,” but was so unlucky as never to be able to answer in the affirmative. When of a suitable age he was apprenticed to the trade of his choice—that of a working stone cutter—and he began his laboring career in a quarry looking out upon the Cromarty Firth.^[13] This quarry proved one of his best schools. The remarkable geological formations which it displayed awakened his curiosity. The bar of deep-red stone beneath, and the bar of pale-red clay above, were noted by the young quarryman, who even in such unpromising subjects found matter for observation and reflection. Where other men saw nothing, he detected analogies, differences, and peculiarities which set him thinking. He simply kept his eyes and his mind open; was sober, diligent and persevering, and this was the secret of his intellectual growth.

His curiosity was excited and kept alive by the curious organic remains, principally of old and extinct species of fishes, ferns, and ammonites,^[14] which were revealed along the coast by the washings of the waves, or were exposed by the stroke of his mason’s hammer. He never lost sight of the subject, but went on accumulating observations and comparing formations, until at length, many years afterward, when no longer a working mason, he gave to the world his highly interesting work on the “Old Red Sandstone,” which at once established his reputation as a scientific geologist. But this work was the fruit of long years of patient observation and research.

We learn from these interesting records that, no matter how or where one is situated, he will always find opportunities for observation if he will only keep his eyes open and his mind open at the same time. It is the brain behind the eyes that makes seeing of any value. Every gift may be perfected by self-culture, and by keeping our eyes busy on things about us, by observing and comparing, we color our future lives, increase our intelligence, and are never at a loss for new worlds to conquer.

What the world needs to-day is less *outlook* and more *insight*; more careful observance of what is needed in our homes by those we love and those who love us. We need eyes to see our own duty in every department of life, to note our own faults, and to observe the beauty rather than the blemishes of others; to see wherein we can be of service, and in what way we may enlarge our opportunities, and in order to acquire any skill or proficiency we need continually to pray, "Lord, open thou the eyes of our understanding."

EASY LESSONS IN ANIMAL BIOLOGY.

CHAPTER II.

SUB-KINGDOM VII.—ARTICULATA.

This subdivision of the animal kingdom, containing articulated or jointed animals and insects, exceeds every other in the number and diversity of the species. The articulation may belong to their bodies, limbs, or outer covering. The tough shells of some, formed by a secretion of a hard, horn-like substance, have numerous segments, or rings, either closely joined and firmly cemented, as those about the head and thorax, or loosely cemented, as those which encompass the abdomen. The skeleton of some is external, and consists of these articulated segments, which serve the double purpose of framework and covering. The muscles, or elastic cartilages holding them together, are striated, or furnished with small grooves in the sheath or shell. If the animal has limbs, they also are jointed, and hollow.

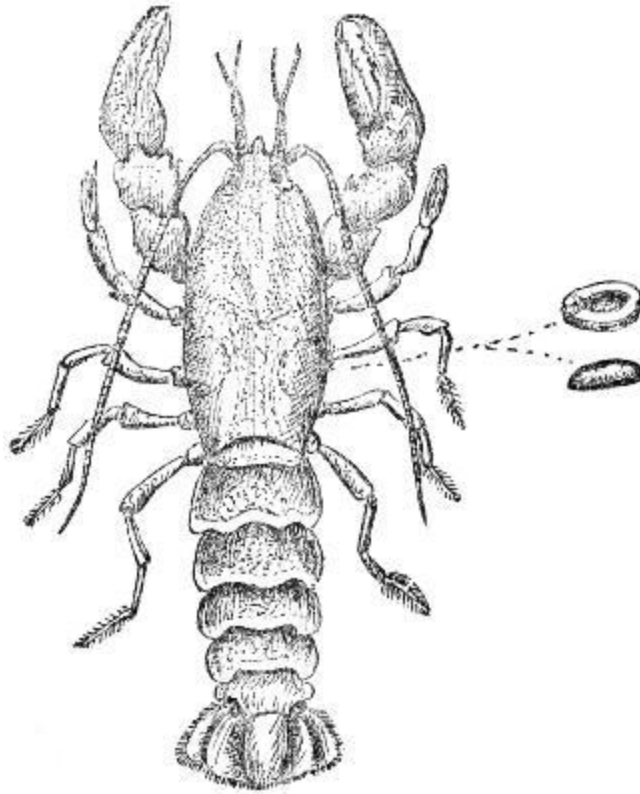
CLASS I.—*Crustacea*, so called from the crust in which their soft bodies are encased. They are a very large family, mostly of air breathing animals, with enough in common to indicate their relationship, yet distinguished by a great diversity in their forms and modes of life. Some are very small, and are as numberless as the sands on the shore. Others, when their members are all extended, can stretch themselves over a circle several feet in diameter.

The chief orders of the Crustacea are the Barnacles,[\[1\]](#) the Water-flea, the Fourteen-footed Crustacea,[\[2\]](#) and Ten-footed Crustacea.[\[3\]](#)

The Crayfish may be taken as a type of the structure of the Crustacea. The body has two principal sections. The anterior, called the cephalo-thorax,[\[4\]](#) extends to the first distinctly marked ring, and the shield, thus far, is comparatively smooth, the segments fitting so closely as to be practically one. In front and between the two pairs of antennæ, or feelers, is a small

pointed process in the place of the nasal organ, but serving some other purpose. At the base of each of the smaller antennæ, on the under side, is a minute sac, the mouth of which is protected with delicate hairs. These are the organs of hearing, and near them, on the outer side, are the organs of smell. The sense of touch is in the fine cilia that fringe the mouth and the antennæ.

There are numerous appendages. Of the five pairs of legs, the first two are provided with claws, or nippers. The fore-legs, or arms, have, in the place of hands, strong pincers, similar, but not entirely alike; the one with sharp edge and smaller teeth is used for cutting, the other for mashing, or grinding the food. The other legs terminate in feathery points, and are used, in part, for locomotion, and by the female for carrying her eggs. The posterior pair, called swimmerets, together with the expansion of the last segment of the abdomen into a kind of caudal fin, are the main dependence for swimming. The segments are so loosely jointed that the “tail” can be moved freely, and by flapping it the animal moves easily. As there is no neck, in order to see objects in different directions, the eyes are not sunk in the head, but placed at the extremities of little muscular processes, or “eye stalks,” which are movable, making even hind-sight practicable when backward motion is desired.



THE CRAYFISH.

The crayfish breathes through branchiæ, or gills, situated at the sides of the thorax, protected by the carapace,^[5] or horny covering, under the edges of which the water and air reach the gills. Here a very curious appendage is attached, called the “gill bailer,” which moves back and forth, creating a current of water through the gills that finds its way out through an opening near the mouth.

Under the welded sheath or cover of the head are the mandibles, or jaws, between which the mouth opens; a short passage, leading to the capacious, gizzard-like stomach, is provided with grinders, to still further masticate the food before it passes into the intestine. The eggs are small, and attached by glutinous threads to the appendages until they are hatched; the young are also attached, until sufficiently developed to live apart from the parent.

This class of animals undergoes periodic changes which are attended with some degree of violence. The crustaceous covering is a kind of epidermis,^[6] having beneath it the true skin. It is formed by some process of exudation from the growing body. This sheath, while soft, expands slowly,

but when hardened, the growth is retarded, and in time it is found too small for convenience, so it is cast off, and a new and larger one supplied to take its place. In this process of moulting the animal attempts to put off its outer covering, not in fragments or parts, but in one piece, though many delicate attachments have to be sundered, membranes rent, and sometimes even a limb torn off in the resolute effort to undress. This can not be done at all times, or at any time, without special preparation. A period of apparent sickness precedes, and the muscular parts of the limbs become shrunken, so that they are more easily extricated. The loss of a leg is not so serious a matter, since the damage is repaired by a new one with the same form and articulations. As the work of repairing the limb begins at the joint nearest the body, if the member is torn between that and the extremity, the partially mutilated animal has the strange power of throwing off all that remains beyond that joint.

Of other crustaceans, the common lobster is in most respects so similar to that shown in the first diagram as to need no further description than to say the cephalo-thorax is comparatively smaller, while the forearms and claws are larger.

There are also marine crayfish that are very numerous about the coral reefs off the Florida coasts, and have substantially the same characteristics, only their claws are considerably less, and their ciliated antennæ larger.

Crabs are closely allied to lobsters, and belong to the highest orders of the crustaceans. The lengthened, loose-jointed abdomen of the typical crayfish is wanting, and there is a general concentration of the parts; all the most important viscera being included in the thorax, and covered by a single, closely compacted shield. There are many species of crabs, differing in other respects as well as in the form of the shell or back, which in some is nearly orbicular, in others it is oblong, longer than it is broad, or broader than it is long. They differ in the smoothness of their shells, and in the length of their legs, which they stretch out from under their horny covering. Their first pair of limbs is not fitted for locomotion, but shows a vigorous development of the strong claws and pincers of other decapod crustaceans. Though found in almost all seas, they are poor swimmers, their legs being formed for walking or creeping, rather than as oars to propel them through the water. They are found in pools, among seaweeds, and particularly in

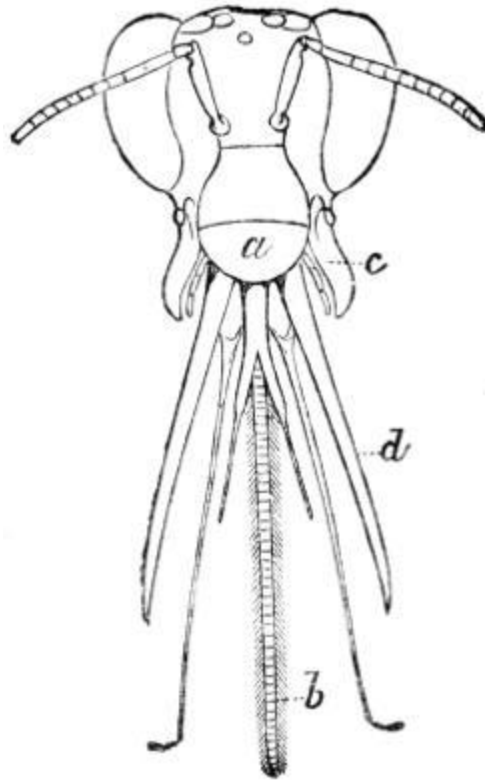
marshy places left by the receding tides. Most species live in water, some in moist places on land. Many kinds of crabs are used for food. Its black claws and broad carapace readily distinguish it from other species. From activity in seizing, tearing, and devouring their food, and from their pugnacity, crabs are interesting inmates of the aquarium. They also moult, or cast off their shells; not at regular seasons, but when the demand for more room requires it.

CLASS II.—*Arachnida* are closely related to the crustaceans, having, like them, the body divided into two sections—cephalo-thorax and abdomen. To the former are attached four pairs of legs, but the abdomen has no appendages for locomotion. There are about 5,000 species, produced from eggs, and undergoing no metamorphoses in their development.

The lowest forms, under the common name of *Acarina*,^[7] have the anterior part in a mass with the abdomen, and short legs near the head, terminated in little claws suitable for taking hold of hairs and feathers. They are mostly parasitic, and all birds and animals, even parasites themselves, are liable to suffer from acarina peculiar to their own species. *Pedipalpi*^[8] (scorpions), and *Araneina*^[9] (spiders), though much larger, belong to this class. The body of the scorpion is divided into segments, though the anterior of the abdominal part seems but a continuance of the thorax, and is as large. It, however, soon tapers off into a long, jointed, tail-like process, in the terminus of which is its hooked sting, perforated and connected with the poison sac. In striking, the tail is raised over the back and struck down. Its other weapons are the crab-like claws on the strong forearms. The *Araneina*, at least some classes of them, are well known. The soft, unjointed body is separated from the thorax by a narrow constriction or tie, and at the posterior end there are little appendages called spinnerets, through which the silken lines issue that form the web. The hinder feet are skillfully employed in arranging the gossamer threads after patterns that are instinctively followed.

CLASS III.—*Myriapoda*, Centipedes,^[10] have the thorax merged with the elongated abdomen, while the head is free. They resemble worms in form, but the skin is stiffened with chitine,^[11] and the many legs are articulated. There are two orders: the *Chilognatha*,^[12] which move slowly, and are harmless, the “thousand legged worm” is a representative, and the

Chilopoda,^[13] more active, and having a flattened body of about twenty segments, each carrying one pair of legs. Their mouths are armed with formidable fangs connected with poison glands. They are carnivorous, and may be distinguished by their general appearance, quicker movements, and by having longer antennæ than the innocent vegetarians.



THE HEAD OF AN INSECT.

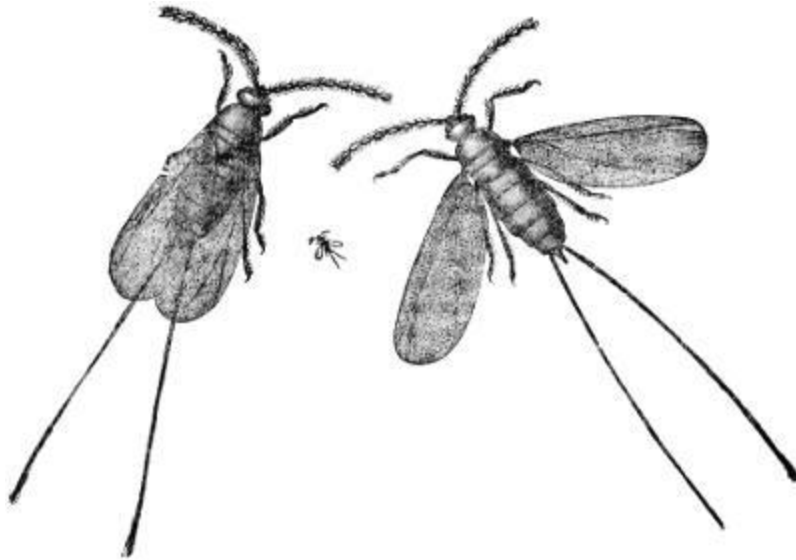
Ex.—*A*, gula, or throat; *b*, ligula, or tongue; *c* mandibles; *d*, maxillæ, or inner jaws.

CLASS IV.—INSECTA. The distinguishing characteristics of this class are that the head, thorax, and abdomen are distinct; that they possess three pairs of jointed legs, one pair of antennæ, and, generally, two pairs of wings. The skin is hardened, and to it the muscles are attached. The eyes are usually composed of a number of facets, from fifty in the ant to many thousands in the winged insects. As the eyes are not movable, these facets enable them to see in many directions.

The several parts of the head and its appendages are shown in our illustration. The sensitive palpi, or feelers, with the delicate hair-brush tips at the ends, may also be noticed. The mouth differs in different species, and is fitted for biting and masticating, or puncturing and sucking. The adaptation seems perfect. Of all animals belonging to the articulate type, the Insecta possess the highest instincts. To this class belong the following orders: I. *Neuroptera*,[\[14\]](#) or lace-winged insects, of which the Dragon-fly, or Devil's Darning Needle, is a good representative. II. The *Orthoptera*[\[15\]](#) (straight-winged). They have four wings, the front pair thick and narrow, overlapping along the back; the hind pair broad, net-veined and folded upon the abdomen. The representative forms are Crickets, Grasshoppers, Locusts, and Cockroaches. III. *Hemiptera*[\[16\]](#) (half winged). To this order belong the wingless Bed-bug, the Squash-bug, the Seventeen-year Locust, and the Cochineal.

Coccus-cacti[\[17\]](#). The Mexican cochineal insect is of great value as a dye, and from it the most beautiful scarlet and crimson colors are obtained. The female is wingless, and, as an uncomely parasite, lives and feeds on cactus plants, especially those of the nopal[\[18\]](#) species, native in Mexico and Peru. The male only is represented in the diagram, and magnified somewhat larger than life. They are comparatively few in number, and of no commercial value. The plants are cultivated for the purpose, and the care of the insects, which increase very rapidly, is an industry giving employment to thousands of laborers, and in some parts of Mexico the product of the cochineal farms is among the most valuable of their exports. Cortes[\[19\]](#) in the sixteenth century received instructions from the Spanish court to obtain cochineal in as large quantities as possible. The export became very large, both Spaniards and others becoming skilled in the use of the beautiful dye stuff, nearly a hundred years before its real nature was known. The dried insect being very small, and crushed in preparing it for market, it was supposed to be the seed of some plant; and it was not until in 1703 that its true nature was discovered by microscopic observations. The industry still flourishes in Mexico, but both plant and the insect have been taken to other countries, and do well. The annual export of cochineal from the Canary islands has, in the present century, amounted to over 6,000 tons, valued at more than \$4,000,000. The manner of collecting the insects is very simple. When of sufficient age, some already dead, and others yet alive, they are

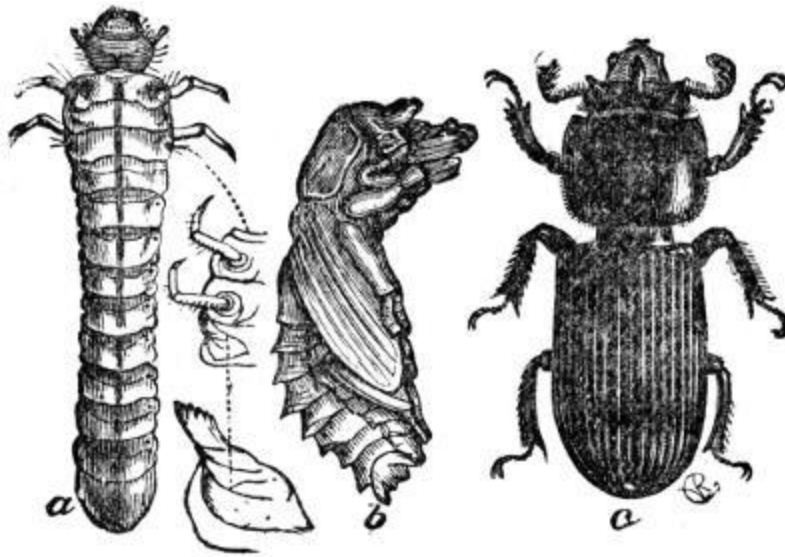
brushed off into bags, and the living killed by holding them either in boiling water or heated ovens, and then exposed in the sun till quite dry. The dried insects have the form and appearance of irregular fluted and concave grains, of which it is estimated there are 70,000 in a pound.



THE COCHINEAL.

There are several species of these insects, alike in form and habits, but not alike useful. Some, as the Scale insect, are a great annoyance to gardeners, and destructive to our house plants. Others, as the Wax insect, live on certain tropical trees, and soon entomb themselves in a mass of glutinous matter that oozes from the small twigs of the tree, and which furnishes them both food and shelter. As they are marvelously prolific, a single female, according to the estimates of entomologists, being succeeded by many millions of descendants in less than a year, when a colony has possession of a tree, every tender branch is soon punctured, and the abundant resinous juices that flow out envelop it in a coating often half an inch thick. These branches, and also what falls to the ground, are collected, and the wax, melted off, is prepared for the market. From this source the shellac of commerce, so extensively used, is obtained. This curious and useful insect, like its congener, the cochineal, secretes a coloring substance, but of different tints, and less valuable. The Lac insect is a native of Siam, Assam, Burmah, Bengal, and Malabar. For some years the average annual imports into Britain have been a little over 600 tons of the lac dye, and

more than a thousand tons of lac, including the several varieties. This industry also gives employment to many thousands of people.



METAMORPHOSIS OF A BEETLE.

Ex.—A, larva; b, pupa; c, beetle.

IV.—*Coleoptera*^[20] (sheath-winged). Beetles are innumerable, about ninety thousand species being recognized. Their anterior or upper wings, useless for flight, are composed of a hard, horn-like substance known as chitine, and meet in a straight line on the top of the back. The posterior wings are thin, membranous, and, when folded, out of sight. They have usually two pairs of laterally moving mandibles, or jaws, and in their development undergo several metamorphoses. We see the egg, the larva, or grub, in different stages of its growth, the chrysalis, and the imago,^[21] or complete beetle. Entomologists have spent much time and labor in making collections, and classifying them according to their peculiarities of form or habits. If many are repulsive, and most plain in form and color, some are beautiful, and worthy of our admiration. Beetles, especially in their larva stage of development are very voracious, and as most of their species live on fruits, leaves, and stems of plants, they are often destructive of crops, and even of forests. Millions of vigorous, valuable trees have been assailed, and stripped of their leaves fast as they appeared, or literally bored to death. All know the ravages of the potato beetle on our American tuber, that has to be assiduously defended to prevent the entire destruction of the crop. Some known as Goliath and Hercules beetles are large, often measuring six inches

in length, exclusive of their long antennæ. The “Diamond beetles” of Brazil are adorned with the most brilliant colors, showing a beautiful metallic luster, and the elytra,[\[22\]](#) or chitine sheaths, of this species are now largely used in the manufacture of personal ornaments.[\[23\]](#)

V. *Diptera* (two-winged), or Flies, number about 24,000 varieties. Among these are the Mosquito, Hessian fly, Daddy-long-legs, Flea, and common House fly. They usually have one pair of fully developed wings, the second pair being rudimentary, although a few, as the fleas, are wingless. They pass through a complete metamorphosis, the larvæ being usually footless maggots, with the breathing holes in the posterior part of the body; the pupa are either encased in the dry skin of the larvæ or are naked.

VI. *Lepidoptera*[\[24\]](#) (scaly-winged), or Butterflies and Moths, are distinguished by four wings, covered on both sides by minute scales. The butterflies fly by day and have knobbed antennæ, while the moths fly by night and have feathery antennæ. Among the moths one of the most interesting is that of the silk worm.[\[25\]](#) The physiology of the insect and its metamorphoses reveal nothing very peculiar, and its habits need not be mentioned farther than to say, the larva eats voraciously, with short intervals of abstinence, until full grown, which stage is reached in about a month. During the last ten days the silk germ is elaborated, the eggs laid, and then the spinning and winding soon begin. To complete the cocoon requires at most only about three days. The larva then becomes a chrysalis, and in due time the moth emerges from its cell.

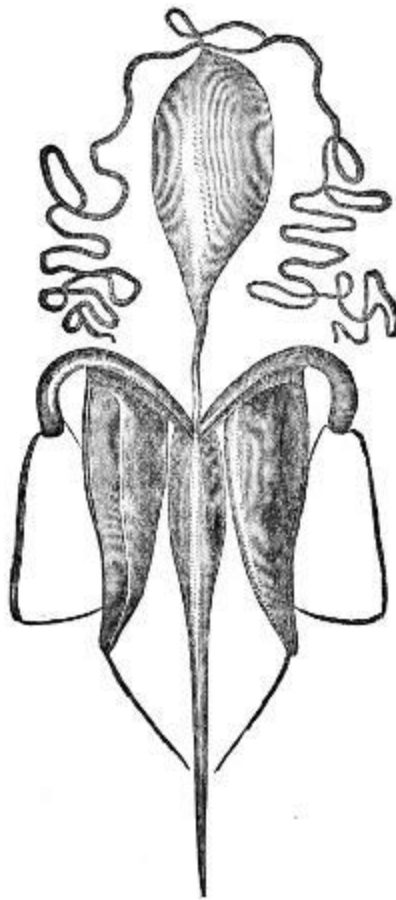
VII. *Hymenoptera* (membrane-winged) comprises the Ichneumon and Gall fly, Ants, Wasps, Bees, in all about 25,000 species. This order includes the most social of the insects. They have four wings, which in flying they fasten together by means of small hooks on the edges. The females are usually provided with a sting or borer. The Gall fly produces the gall nuts or oak balls so common on oak trees. The Ichneumon fly introduces its eggs underneath the skin of the caterpillar.[\[26\]](#)

Ants[\[27\]](#) live in communities. They are divided into fertile females, males and infertile females. Among the ants, the mining ants, which make long galleries in the earth, and carpentering ants, which perforate solid timber,

are the best known classes. Some species, like the white ants of the tropics, the termites, are famous for their ravages.

The bees are divided into queens or females, drones or males, and workers. Each community or swarm has one queen, which lays the eggs. The bee is provided with a formidable sting.

This curious weapon of attack and defense is here magnified with the adjacent parts. It consists of an extensile sheath with two needle shaped darts that are exceedingly sharp. This spear is furnished with barbs near the point, and when it pierces the skin, if thrust with violence, it sometimes remains, not only making the wound more painful, but, having been wrenched from the bee, frequently causes its death. The sting is connected with a little sac containing a poisonous liquid which is thrown into the wound and increases the pain.



STING OF A BEE.

SUB-KINGDOM VIII.—TUNICATA.

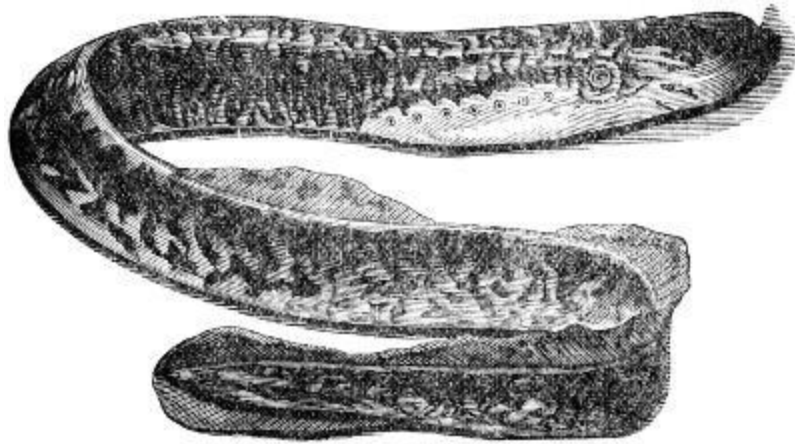
This very small class of animals is distinguished by the leathery sac-like covering, from which they take their name of Tunicata (having a tunic). The Ascidian^[28] is the best known representative. It is found fastened to rocks, shells, crabs, and other bodies. These animals are both simple and compound; the latter are often phosphorescent. They have neither feet, head nor shell, but a shapeless body with apertures at both ends.

SUB-KINGDOM IX.—VERTEBRATA.

This division includes the most perfect animals. Their chief distinguishing characteristics are an internal skeleton; a backbone; a dorsal nervous cord, separated from the body cavity; a complete circulation, and limbs not exceeding four. There are about 25,000 living species, beside the numberless host now extinct.

Lowest of the vertebrates, and closely related to the true fishes, are the Lancelet and Lamprey.^[29] The former is a lance-shaped animal having no skeleton, but boasting the rudiment of a backbone in a string-like cartilaginous cord. The organs are very simple, the heart being a long sac in which colorless blood circulates. It breathes by taking in water through the mouth and letting it out through the gill-slits. The Lamprey belongs to the pouch-gilled vertebrates or *Marsipobranchii*.^[30] It is an eel-shaped animal of about three feet in length.

The round, soft mouth is suctorial, the tongue acting as a piston. By suction it can anchor itself to a rock, and allow the long body to float freely, without being carried by the current from the place. There are seven gill openings on each side, and the whole breathing apparatus so arranged that the animal can live some days out of the water. In some parts of England it is in demand for the table, and “lamprey pie”^[31] is esteemed a great luxury. The American lamprey is similar, and the flesh good, but in less demand in our markets. During the breeding season those about the estuaries go up stream as do the shad, and by rolling stones together construct large conical nests^[32] for the protection of their young. With apparently little adaptation for such architecture they accomplish wonders in that line.



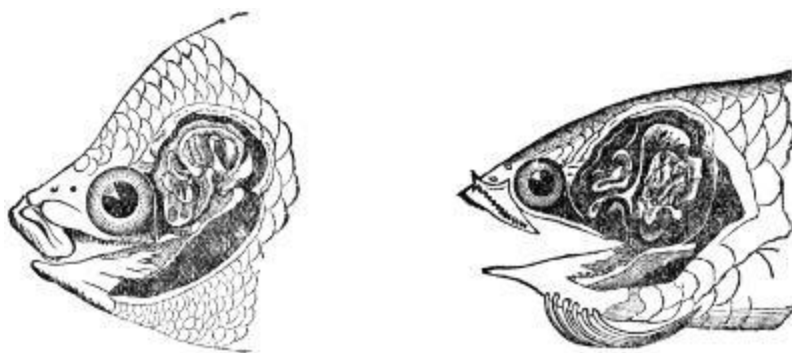
THE LAMPREY.

CLASS I.—*Pisces*. The first clearly defined division of vertebrates is that of the fishes. They are regarded as, in some respects, the lowest of vertebrate animals. They are credited with having the least intelligence and sensibility. Their eyes, though often large, are nearly motionless in their sockets, are protected by no eye-lids, and are without the expression usual even in the animal eye. If they have ears the external parts are wanting. Sounds may reach the auditory nerve and be heard through the cranium. The other senses, as taste, touch and smell, are but slightly developed. But they are admirably adapted to the element in which they live, and the mode of life for which they were created. In no other organisms is the evidence of design in the adaptation of means to the ends contemplated, more apparent. In the number of species and variety of forms they exceed all other vertebrate animals.

The skeleton of a fish is usually divided into four parts: the *head*, *respiratory organs*, *vertebral column* and *limbs*.

The head is very suggestive, and of itself presents a profitable study. It is not hung or poised on a neck, but attached immediately to the body. In most species it is large, making a large mouth possible, but pointed, to lessen the resistance met in passing through the water. In some the eyes are quite near the nasal organ, in others farther back. In some they face laterally, in others upward. As there is no nictating[\[33\]](#) membrane there is neither winking nor the shedding of tears. Both jaws are, to some extent, movable and provided with osseous teeth that are usually sharp and of a spike-like form.

From the heads here presented the operculum,[\[34\]](#) or gill cover, is removed to show those delicate respiratory organs. The branchiæ, or gills, are situated at the sides of the head just back of the eyes, and consist of numerous and very vascular[\[35\]](#) plates, arranged in double fringe-like rows, fixed or attached at the base only, and so constructed, in all respects, as to expose as much surface as possible. These gills are covered with innumerable small blood vessels, to which blood is pumped from the heart, there to receive the needed supply of oxygen. The oxygen is obtained from what air circulates through the water, and not by a decomposition of the water, as some have supposed. For some species the modicum of oxygen, thus obtained, seems insufficient, and they come to the surface for more.



HEADS OF FISHES WITH GILL-COVERING REMOVED.

Notice another peculiarity. There being no neck or long gullet, the principal digestive organs are packed in the cavity near the capacious mouth, and this leaves the whole of the posterior tapering part of the body for strong muscles, that can vigorously move the caudal extremity from side to side as a propelling oar. The spine is so jointed as to allow a free horizontal and but little vertical motion.

The bones of fishes are less compact than those of the higher orders of animals, but quite elastic; and in some species small bones are distributed through the flesh, giving additional firmness to portions that lack muscular strength. As the peculiar breathing apparatus of the fish is adapted to its element and mode of life, so is every part and appendage of the whole structure. Adaptation reigns through the whole. The elongated tapering body, its scaly covering exquisitely adjusted, the material and position of the fins, all attest the intelligent and beneficent purpose of the Creator.

The *Elasmobranchii*^[36] (strap-gilled) have a cartilaginous skeleton, rough skin, and uncovered gills. The Shark, the Saw-fish and the Ray are representatives of this order.

The *Ganoidei* (enameled scales) formed one of the largest orders in ancient geological history, but they have now but few representatives, such as the Sturgeon, Gar-pike and Mud-fish. Their characteristics are a skeleton not completely ossified, ventral fins placed far back, and the tail heterocercal, that is, having the upper lobe larger than the lower.

The *Teleosti*^[37] (perfect bone) form the largest order, including nearly all our common fishes. The characteristics of the order are an osseous or bony skeleton; gills protected by a gill cover or operculum; and an equally divided tail. Lowest among the Teleosts is the common eel. It has an elongated, cylindrical, thick-skinned body, and is destitute of ventral fins. There are many species, and they are widely distributed, living in both salt and fresh water. One species is electrical.^[38] The cat fishes of this order have long threads hanging from their jaws, and are noted for their peculiar methods of protecting their young; one species is electric, having cells arranged in layers over the body.

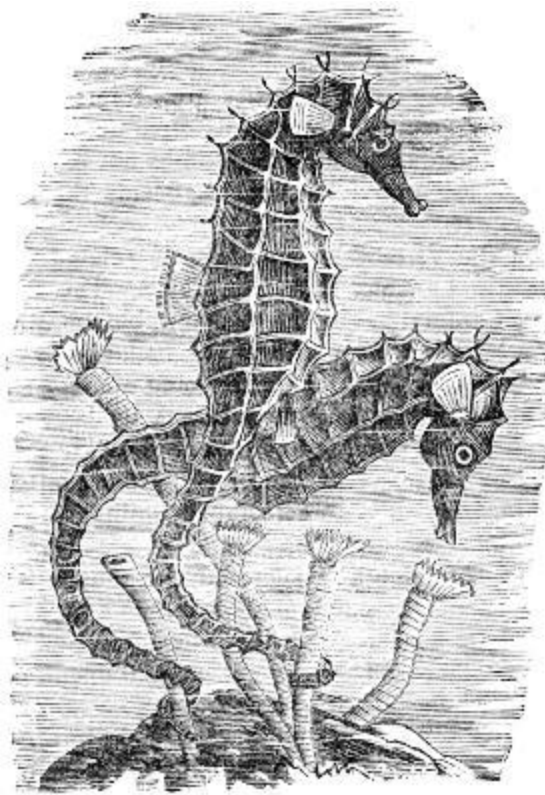
Other peculiar types are the Blind or Cave fishes,^[39] living in the waters of caves; the Lamp fishes which take their name from the luminous spots arranged along the sides, and which are supposed to light the recesses where they live, and the Flying fishes.

There are at least two genera, and more than thirty species of “flying fish,” a name given to all those which have the pectoral fins so large that they are sustained in short, seeming flights through the air. They do not really fly, as they have not muscular power in their fins to beat the air as birds do. But when extended the fins help bear them up, and the impulse received at the start, sufficient to give them the elevation they reach, may be supplemented by the use of the caudal fin as in swimming. Some naturalists claim to have noticed a movement of the other fins, but the preponderance of testimony is that these are of service only as parachutes or the wings of a paper kite. The Herring, Shad, Salmon, Pike, Perches, Bass, Mackerel and Cod are valuable food fishes, belonging to the Teleosts. The Sword fishes, in which the upper jaw is developed into a long, sword-like projection, used

as a weapon of defense and offense, is an interesting member of the order. The Climbing fishes, noted for being able to live out of water, and the Nest Builders, which make homes from the weeds of the Sargossa Sea,[\[40\]](#) are peculiar types.

Another curious Teleost is the Sea Horse. This peculiar animal has its name (*Hippocampus*) from the shape of the head, that has some resemblance to that of a horse. There is no other resemblance. The short body, without legs, is covered with angular spinous plates. Its fishy part is a long prehensile tail, but it has neither dorsal nor caudal fins.

The last order of fishes is the *Dipnoi*, or Lung fishes. They are characterized by the possession of two lungs, as well as gills. The gills are used in respiration when the fishes are under water, but when out of it, and burrowing in the mud as they often do, the lungs are put into service. They are also known as mud fishes, from their habits of encasing themselves in the mud.



THE SEA HORSE.

End of Required Reading for May.

HOW TO WIN.

BY FRANCES E. WILLARD,
President National W. C. T. U.

CHAPTER III.

But—as I was saying when the stern old gentleman was pleased to interrupt me—I am to give you reasons why you are to cultivate your specialty. And I claim, first (as has been implied already), that you should do this because you have a specialty to cultivate. (This, on the principle of the old cook book, which begins its “Recipe for Broiling Hares,” with the straightforward exhortation: “First catch your hare.”) The second reason is, because you will then work more easily and naturally, with the least friction, with the greatest pleasure to yourself and the most advantage to those around you. “Paddle your own canoe,” but paddle it right out into the swift, sure current of your strongest, noblest inclination. Thirdly, by this means you will get into your cranium, *in place of aimless reverie, a resolute aim*. This is where your brother has had his chief intellectual advantage over you. Quicker of wit than he, far less unwieldy in your mental processes, swifter in judgment, and every whit as accurate, you still have felt, when measuring intellectual swords with him, that yours was in your left hand, that his was in his right; and you have felt this chiefly, as I believe, because from the dawn of thought in his sturdy young brain, he has been taught that he must have a definite aim in life if he ever meant to swell the ranks of the somebodies upon this planet, while you have been just as sedulously taught that the handsome prince might whirl past your door “’most any day,” lift you to a seat beside him in his golden chariot and carry you off to his castle in Spain.

And of course you dream about all this; why shouldn’t you? Who wouldn’t? But, my dear friends, dreaming is the poorest of all grindstones on which to sharpen one’s wits. And to my thinking, the rust of woman’s

intellect, the canker of her heart, the “worm i’ the bud” of her noblest possibilities has been this aimless reverie; this rambling of the thoughts; this vagueness, which when it is finished, is vacuity. Let us turn our gaze inward, those of us who are not thorough-going workers with brain or hand. What do we find? A wild chaos; a glimmering nebula of fancies; an insipid brain-soup where a few lumps of thought swim in a watery gravy of dreams, and, as nothing can come of nothing, what wonder if no brilliancy of achievement promises to flood our future with its light? Few women, growing up under the present order of things, can claim complete exemption from this grave intellectual infirmity.

Somehow one falls so readily into a sort of mental indolence; one’s thoughts flow onward in a pleasant, gurgling stream, a sort of intellectual lullaby, coming no-whence, going no-whither. Only one thing can help you if you are in this extremity, and that is what your brothers have—the snag of a fixed purpose in this stream of thought. Around this will soon cluster the dormant ideas, hopes, and possibilities that have thus far floated at random. The first one in the idle stream of my life was the purpose, lodged there by my life’s best friend, my mother, to have an education. Then, later on, Charlotte Brontë’s “Shirley” was a tremendous snag in the stream to me. Around that brave and steadfast character clustered a thousand new resolves. I was never quite so steeped in reveries again, though my temptations were unusual; my “Forest Home,” by a Wisconsin river, offering few reminders to my girlish thought, of the wide, wide world and its sore need of workers. The next jog that I got was from the intellectual attrition of a gifted and scholarly woman who asked me often to her home and sent me away laden with volumes of Wordsworth, Niebuhr and the British essayists. Margaret Fuller Ossoli was another fixed point—shall I not rather say a fixed star?—in the sky of my thought, while Arnold of Rugby, to one who meant to make teaching a profession, was chief of all. Well, is it possible that any word I have here written may set some of you thinking—that’s it, *set* you, a fixed purpose rather than a floating one—about a definite object in life toward which, henceforth, you may bend a steady, earnest gaze? I am not speaking of a thorough intellectual training only. It is rather to the life-work, which only a lifetime can fully compass, that I would direct your thoughts. Rather than that you should fail to have a fixed purpose concerning it, I would that your mental attitude might be like

the one confided to me by a charming Philadelphia girl, whose letter of this morning has the following *naïve* statement:

“I feel such an aching in me to do or be something uncommon, and yet a kind of awful assurance that I never shall.”

Nor do I here refer to that general knowledge of household arts which forms the sole acquirement set forth in the regulation “Women’s Department” of the bygone age newspaper, which in many localities remains in this like the boulder of a past epoch.

It was once thought to be a high virtue for women, no matter how lofty in station or how ample of fortune, to do their own work with the needle. Homer represents Penelope spinning, surrounded by her maids, and classic art abounds with illustrations of like character. But the virtues of one age often become the mistakes of the next. When loom, needle and broom were woman’s only weapons, she did well to handle them deftly, no matter what her rank, for they were her bread-winning implements, and fortune has been proverbially fickle in all ages. But men, by their “witty inventions,” have perpetually encroached on “woman’s sphere.”

Eli Whitney, with his cotton gin, Elias Howe, with his sewing machine, and a hundred other intricate-brained mechanics who have set steel fingers to do in an hour what women’s fingers could not accomplish in a year; all these have combined to revolutionize the daily cares of the gentler sex. With former occupations gone, and the world’s welcome ready when they succeed in special vocations new to them, it becomes not only the privilege but the sacred duty of every woman to cultivate and utilize her *highest* gift. There is no more practical form of philanthropy than this, for every one who makes a place for herself “higher up” leaves one lower down for some other woman who, but for the vacancy thus afforded her in the world’s close crowded ranks, might be tempted into paths of sin. There is an army of poor girls wholly dependent, for a livelihood, upon the doing of house work. They have no other earthly resource between them and the poor house or haunt of infamy. There is another class to whom an honorable support can come only by sewing or millinery work. Whoever then fits herself for some employment involving better pay and higher social recognition, graduates

out of these lower grades and leaves them to those who can not so advance, has helped the world along in a substantial way, because she has added to the sum of humanity's well being.

To young women in wealthy homes, these considerations should come with even greater convincing force. As David Swing has wisely said to his own rich congregation:

“The rhetoric thrown at women of property for not doing ‘their own work’ could only be useful in an age of fashionable idleness, but in a busy age it is a part of nature’s law that what are called the ‘better classes’ shall leave for the poorer classes some labor to be done, just as the Mosaic law left some sheaves in the field for the gleaner. The world’s work is to be apportioned according to the need and capability of its workers, and the higher order of power must not encroach upon the task which nature seems to have set apart for the employment and support of the less capable.”

Let it not be concluded that I have meant to speak lightly of the intricate, skilled labor involved in making healthful and attractive that bright, consummate flower of a Christian civilization—the home. I have felt that this theme has been so often treated that it needed no amplification at my hands, but I will add that, having been entertained in scores of homes belonging to “exceptional women,” “women with a career,” etc., my testimony is that for wholesomeness, heartsomeness, and every quality that superadds home-making to housekeeping, I have never seen their superiors, and seldom, take them all in all, their peers. But as a rule, these women have earned the “wherewithal” to make a home, by the exercise of some good gift of brain or hand, and thus having been enabled to put a proxy in the kitchen, they direct, but do not attend to the minutiae of their daily household cares.

Cultivate, then, your specialty, because the independence thus involved will lift you above the world’s pity to the level of its respect, perchance its honor. Understand this first, last, and always: *The world wants the best thing*. It wants your best. It needs you as a significant figure to give its ciphers value; to designate as an example; to serve up in a eulogy, perchance to shine in the galaxy by whose light alone its centuries maintain their places in the firmament of history. I know this may strike you as

contradiction, for the paradox of paradoxes is this crotchety but kind, narrow-minded but just old world in which you and I are cast away, like Æneas in the domain of Dido. The effrontery of “Madame Grundy” passes all comprehension, and would be laughable if it were not so sad. She tells us women distinctly that we positively shall not do for society the thing we can do best; she declares that if we attempt it we shall be frowned down, and practically ostracised, if not utterly made away with, and then, if we go right on and succeed, she trumpets our names from sea to shore, showers us with greenbacks, and nods her conventional old head with a knowing “I told you so.” And *per contra*, while on one hand this same unreasonable old lady cripples our attempts to succeed, on the other she snubs us for not doing so. In fact, she is so poor a mathematician that she has never yet so much as tried to learn the value of the “unknown quantity.” The mute Milton is, to her, indeed “inglorious.” Her code of ethics recognizes just one crime (not mentioned in the Decalogue), and it is Failure. Her law is written on a single table—it is a table of stone—and it reads thus: “Succeed and live; make shipwreck of success, and die.”

And so, young friends, fold away your talents in a napkin if you choose; the world will not openly reprove you. She will never urge you to bring out your hidden treasure, but she knows right well when you defraud her, and the relentless old tyrant will punish you, with tireless lash, because you did not bring all your tithes into the storehouse of the common good, because you lived “beneath your privilege;” because, for yourself (which means for *her*), you did not “covet earnestly the best gifts.” She will cut you on the public street when she would have shown you all her teeth in smiles. She will send poverty on your track, when you might have sat down at her banquet an honored guest. Yes, the world wants the best thing; *your best*, and she will smite you stealthily if you do not hand over your gift. Now last, but not least (under the head of reasons for seeking to know your true vocation as a human being), let me bring forward the *rationale* of the bread-and-butter argument. In sooth, no writer or speaker may omit it with impunity, if he would retire in good order from an American audience. Briefly, then, your specialty, well trained, is your best bread-winning implement, and she who earliest grasps this, and who firmest holds it, comes off best in the race. “Be not simply good, be good for something,” said Henry D. Thoreau. A bright eyed girl of eighteen used to come to me

on Friday evenings to give me German lessons. To be sure, I have lived in Germany, and she has never been out of Illinois, but then that language is not my specialty, while it is hers. “How is it that though so young, you have made yourself independent?” I inquired of her one day. Listen to the reply: “My mother was always quoting this saying of Carlyle: ‘The man who has a sixpence commands the world—to the extent of that sixpence.’ I early laid this sentiment to heart. Besides, when I was fifteen years old, I heard a sermon on the text: ‘This one thing I do.’ Being of a practical turn of mind, I made an application of which the preacher, perhaps, had no intention. I thought, why not in everyday affairs as well as in religion do one thing well, rather than many things indifferently, and in that way secure the magic sixpence of Carlyle! My father was a rich man then, but I resolved to prepare myself to teach the German language, of which I was very fond, by way of a profession. When the Chicago fire came we lost our property, but I discovered that I could not only support myself, but help my father to many a convenient sixpence, because, in prosperous days I had forearmed myself with a cultivated specialty.”

As she told me this, I thought how, from widely different premises and conditions in life young people may reach similar conclusions. For instance, on the top of the great St. Bernard, I said to the “Hospitable Father,” a noble young monk, “How is it that you, so gifted and well taught, are spending your life away up here among eternal snows?” And I shall never forget his look of exaltation as he simply answered, “’Tis my vocation.”

After all, this is the vital question: With what sort of a weapon will you ward off the attacks of the blood-hound Poverty, which Dame Fortune is pretty sure to let on everybody’s track sooner or later, that she may try his mettle, and learn what manner of spirit he is of? In times like these, when men’s hearts are failing them for fear, when riches are saved the trouble of “taking to themselves wings” by the faithless cashiers and bookkeepers who are adepts at furnishing these flying implements, and, above all, when labor is coming to be king, the question “*What will you do?*” has fresh significance. Remember, going forth from the uncertain Eden of your dreams, into the satisfying pleasures of honest, hard work, “the world is all before you, where to choose.” Will you share some other woman’s home, and help her make it beautiful? No task more noble or more needed awaits the thoughtful worker of to-day. The world exists but for the sake of its

homes. Will you bestow your hand upon some fine æsthetic industry, as drawing, designing, engraving, telegraphing, phonographing, photographing? Will you be an architect? a printer? an editor? Will you enter one of the three learned professions? Braver women than you or I have won a foot-hold for us in each of them; as to the brain-hold, that is our affair. I will not now pursue the question further. Only the “Cyclopædia of Woman’s Occupations” (a book I recommend to your attention) can exhaust it, and with it exhaust you and the world’s work, too, for that matter!

After all, it doesn’t so much signify what you may do as that you do it well, whatever it may be. For the value of skilled labor is estimated on a democratic basis, nowadays. President Eliot, of Harvard University, the cook in the Parker House restaurant, and Mary L. Booth, who edits *Harper’s Bazar*, each receive \$4,000 per year.

Think a moment. Will you be led to say: “The good old ways are good enough for me,” and so drop into the swollen ranks of teacherdom, or rattle awhile on a martyred piano, and then set up for a musician, though you have not a particle of music in throat or finger-tips? Or will you stay at home and let papa support you until you grow tired of doing nothing and expecting nothing, and proceed to marry some man whom you endure rather than love, just to get decently out of your dilemma?

Nay, I do you injustice. Few girls who breathe the free air of our western prairies will be so cowardly. I may not construct your horoscope, but this much I will venture—that when you marry, no matter what you *find*, you will *seek* not a name, behind which to cover up the insignificance of your own; not a “good provider,” to feed and clothe one who has learned how to feed and clothe herself; not a “natural protector,” to shield you in his plaidie, the gallant, gallant laddie, from the cauld, cauld blast; but you will seek (and may heaven grant that you shall find) that rarest, choicest, most elusive prize of man’s existence, as of woman’s; one which—mournfully I say it—the modern marriage is by no means certain to involve, namely, a *mate*. At this juncture, shrewd *mater familias* whispers to *pater*: “That’s the first orthodox word she’s said.” Some youth throws down the magazine and mutters to himself: “There, I knew it would come to this! Look at the absurdity of these women! Why, they preach up all sorts of trades and professions, and then they come back, at last, to the ‘good old way’ they

have forsaken, and advise every young lady to get a situation in a school of one scholar, and her board thrown in.”

Meanwhile, heroic Hypatia sits near by, and “musing in maiden meditation, fancy free,” on a “career,” murmurs within herself, “To this complexion must it come at last!”

Peace, peace, good friends! This seeming inconsistency is readily explained. In this century, when the wage of battle has cost our land an army of her sons, when widows mourn, and unwedded thousands are forced to meet the hard-faced world (from which rose-water theorists would shield them), America is coming to the rescue of her daughters! For the nearer perfect—that is, the more Christian—a civilization has become, the more carefully are the *exceptional* classes of society provided for. All our philanthropic institutions under state or private patronage illustrate this. In less enlightened days, your ideal woman composed the single, grand class for which public prejudice set itself to provide. She was to be the wife and mother, and she was carefully enshrined at home. But, happily, this is the world’s way no longer. The exceptions are so many, made by war, by the thousand misunderstandings and cross-purposes of social intercourse, by the peculiar features of the transition period in which we live, by the absurdly extravagant customs of our day, and the false notions of both men and women—that not to provide for them would be a monstrous meanness, if not a crime. And the provision made in this instance is the most rational, indeed, the only rational one which it is in the power of society or government to make for any save the utterly incapable, namely: *a fair chance for self-help*. Nor (to pursue the line of our argument still further) can we forget that skeleton hand which, in utter disregard of “the proprieties” in destiny’s drama, thrusts itself so often into the charmed domestic circle, and snatches the beloved “provider” away forever, while it sets gaunt famine by the fireside in his stead? Can we forget that, in ten thousand families, wives are this moment waiting in suspense and agony the return of wretched husbands to homes made hideous by the drunkard’s sin—wives whose work of brain or hand alone keeps their children from want, now that their “strong staff is broken, and their beautiful rod?” There are delicate white fingers turning the page on which I print these words, that will never wear the marriage ring; there are slight forms bending over my friendly lines, which, not far down the years, will be clothed in widow’s

weeds. Alas, there are as surely others, who, when they have been wooed and won, shall find that they are worse than widowed. And what of these three classes of women, sweet and helpless? Clearly, to all of them I am declaring a true and blessed gospel, in this good news concerning honest independence and brave self-help! Clearly, also, no one is wise enough to go through the assembly of my readers, and tell me who, in future years, shall need a bread-winning weapon with which to defend herself and perchance, also, the helpless ones between whom and the world there may be no arm but hers. But it is a principle in public as well as private economy, that *the wisest foresight provides for the remotest contingency*, and thus, in its full force, all that I have been saying applies to every woman who may read this article on “How to Win.” Suppose that many of you, dear girls, are destined to a downy nest, instead of a strong-winged flight. What then? Will the years spent in making the most of the best powers with which God has endowed you be worse employed than if you had given them to fashion and frivolity? Those “*ad interim*” years which separate the graduate’s diploma from the bride’s marriage certificate, can they possibly be invested better than in the acquisition of some useful trade or dignified profession? And then, aside from this, I would help the youngest of you to remember (even in the bewildered years of her second decade) what noble Margaret Fuller said: “No woman can give her hand with dignity, or her heart with loyalty, until she has learned *how to stand alone*.” It is not so much *what comes to you* as *what you come to*, that determines whether you are a winner in the great race of life. Never forget that the only indestructible material in destiny’s fierce crucible is *character*. Say this, not to another—say it to yourself; utter it early, and repeat it often: *Fail me not, thou*.

THE LIFE OF MINERALS.

BY M. J. THOULET.

Translated for THE CHAUTAUQUAN from the *Révue Scientifique*.

The definition given to-day to mineralogy places it among the exact sciences. Long continued study has shown that it possesses all the inflexibility of chemistry, of physics, of mathematics. The work of one making a specialty of this subject is similar to that of a millwright who collects the different pieces, forged and cast and prepared in various ways by other workmen, and arranges them all in their proper relations to one another, joins them, and forms the mill with all its complication of machinery in good order, ready to run without friction, without jar. The mineralogist gathers up the facts and theories wrought out by workers in other fields of science, studies their variations, their agreements and connections, demonstrates their presence and their union in inorganic bodies, and sums up and announces all the results of his labor in the form of laws which shall be exact rules for events past, present, and future; for a science incapable of foreseeing and foretelling is not a true science. Mineralogy is not chemistry, nor physics nor mathematics, any more than the millwright is the smith or the smelter. It is a distinct science pursuing a particular aim, and which, although borrowing from other sciences certain of their results, nevertheless possesses its own individuality. It might be said to be a direct application of these three sciences, together with geology, to the study of the life of minerals.

I have just used a very significant expression: *The life of minerals*. Others have used it before me. "Not only do stones live, but they suffer from sickness, from old age, and death," wrote Cardan in the sixteenth century. And he was right. Eternal matter performs an eternal cycle; the incessant variations which it experiences; the movement which is never arrested, which from modification to modification, from transformation to

transformation draws it along without a single moment of rest; the continual births and deaths and resurrections *are life*. Every man, every animal, every plant, and every stone obey without any power to resist, and they are all borne along without relaxation or repose toward a vortex whose beginning and ending are concealed within the shadows of eternity. There is no difference between the mineral and vegetable, or animal. Inorganic life is identical with organic life, varying only in degree.

From the moment which we call birth, that is to say at the commencement of one of these periods of transformation, our eyes see, hour by hour, moment by moment, the living being develop. The atoms entering into its construction seek like atoms to which they ally themselves, and molecules combine with other molecules. What matters it about the form of being? Simple or complicated, the law is the same, and it is obeyed. The individual appears with its own chemical constitution, its own form and look, its own variations, all decided under its predetermined conditions. Among these conditions a single one is variable, but the equilibrium is constantly preserved; the individual changes from time to time in its own appointed way, but it never ceases to exist.

In the same manner as organic life bears the impress of its surroundings, so do minerals submit themselves to external influences. The one perhaps is more frail, more delicate, less able to resist, more susceptible to impressions; the natural forces of the other, more powerful because they are simpler, yield less readily to circumstances. Both alike are forced to take their part in the great concert of forces in which they fare only infinitely feeble notes; both alike are influenced by the majestic assembly of powers which act upon them, and upon which they, in turn, also act, conformably to one of the first laws of matter, that of equality between action and reaction.

Let us take any mineral whatever and subject it to a constantly increasing temperature. We notice first that it undergoes a change of form. Cease the application of heat and it will gradually resume its former shape. Let us heat it again, and more intensely. All the properties of the matter which constitutes it become changed, some quickly, some slowly, and it is incapable now of taking back its first appearance. Its crystalline form is different, and its mechanical elasticity, its hardness, and its electric properties; even its color is changed. We will still increase the heat. The

molecules disperse, following certain directions, and, following others, gather themselves together. Suddenly a limit, varying according to the chemical composition, the crystalline type, or the pressure, is broken; the solid, beginning to melt, becomes a liquid. Heat it still higher and we shall see new phenomena appearing, volatilization and dissolution. Another limit is passed and the atom, becoming isolated, is free henceforward from the laws of chemistry, and must now obey laws yet unknown, the task of discovering and formulating which is awaiting some worker in the realms of physics or mechanics.

The dissolution of a mineral, is it not death? Every abrupt limit of all the powers of a body is death, and all death precedes a resurrection.

As a child, which at the same moment when it opens its eyes upon the light and utters its first cry, begins already to die, so with the mineral scarcely formed, death commences. Feldspar, which constitutes in great part the soil pressed by our feet, under the influence of air and of water, of drought by day and dews by night, of the heat of summer and the cold of winter, of all agents mechanical, chemical and physical acting upon it, is reduced to its elements by a series of almost insensible transformations. Its fragments are broken to still finer bits, and when they have become dust disintegration still goes on, and gradually the silicon, the aluminum, the iron, the lime, the magnesium, and the potassium which composed them form clay. The iron oxydizes, the silicon separates itself, is dissolved by rain and carried off by the streams. Each element then enters into a new combination; sometimes it again becomes part of a stone; sometimes it helps to form the structure of a plant; sometimes that of a man. Where can birth, signifying the beginning of all existence, be placed, or where shall we find any real death? I perceive only periods of life.

Of old, naturalists made more frequent and much stronger affirmations than they do to-day. Confidence in self is the property of youth; maturity learns to doubt, which is the beginning of wisdom, provided that it does not remain content, but rather compels man to seek with increased ardor the truth which seems to fly from him. The ancients placed between the animal and the vegetable limits which in reality did not exist. Up to the present time limits of the same nature have been set between organic and inorganic life. But in proportion as we examine minerals we shall see the differences

disappear and the resemblances increase. Man is born of parents; the whole animal and vegetable worlds are perpetuated in obedience to the laws of reproduction, each after his own kind. It was this absolute identity between parent and offspring that separated distinctly the other kingdoms from the mineral; but recently a scientist has discovered that the same fixed law is established in this department of life also. M. Gernez prepared a solution consisting of octahedral borax in five equivalents of water, and rhomboidal borax in ten equivalents of water. The two bodies, excepting their proportion of water, had the same chemical composition. The liquid, treated with suitable precaution, remained perfectly limpid, and he could place in it fragments of all imaginable substances, without causing it to give rise to any remarkable phenomena. But when even an infinitely small crystal of octahedral borax was dropped into it, the temperature rose, and in a few minutes all the octahedral borax contained in the solution took the crystalline form. Meanwhile, the rhomboidal borax was held in solution, and in order to crystallize it in its turn, there was needed only the contact of a rhomboidal crystal.

The mineral was evidently born of a parent; it was identical with this parent; its symmetry was the same under the same circumstances. Similar results from numerous experiments with other substances were obtained.

... Under the influence of agents whose masters we are, molecules group themselves, following fixed laws, and arrange themselves in their relative positions. Just as soldiers off drill, and scattered throughout the camp, when the order of the commander is given, obey and fall into line, so do molecules obey the forces in command over them.

Stranger still, this crystal perfectly formed, seems sometimes to have a conception of an ideal of beauty, a perfect symmetry, the ellipsoid of the cubic system, which is a sphere; it seeks it, tries to reach it, and if it can not be attained, it falls to acting a part. It disguises itself, just as is sometimes done among men, and strives to appear the being it is not. The crystal, no more than the man, will ever assume a place in a lower rank; each seeks to appear better than he is. To attain its object the crystal will unite itself with the other crystals of the same kind; then these will gather into groups. As they can not modify their own angles they will crowd one against another. Let it cost what it may, if it is a possible thing they will have their

imperfections removed, and will improve their individual appearance, and if any measure of success is attained, the little crystals will enjoy in silence their usurped glory.

If science, with the apparent rigidity of her measures, weights and figures holds for the scholar oftentimes disagreeable surprises, she sometimes cheers him by rewards full of a strange grandeur. Azote, or nitrogen in its free state, constitutes more than three fourths of the volume of the atmosphere, and is in its appearance the type of inertia. Its presence seems to have no other rôle than to reduce the over-exciting action of the oxygen upon our organs of respiration. In order to cause it to enter into combination with other substances, it is necessary to have recourse to the most energetic forces. Among these in nature only one, electricity, lightning, is able to accomplish this result. But the union once effected, the gas is capable of undergoing a thousand variations. As passive as it was while free, so active does it become after entering into any combination. As it is found in the constitution of all animal and vegetable life, we find that without the storm-cloud no organic life could exist. The origin of all creatures is to be found in a clap of thunder.

Such examples as these show that imagination as well as science derives great profit from the intimate study of the phenomena presented by minerals. One commences their study by measuring, by weighing, by carefully analyzing; one gathers now and then slowly a little knowledge; then suddenly this apparently barren field disappears to give place to large horizons, to vast generalizations of majestic simplicity, resting upon the solid foundation of experimentation. Let us not underestimate the rôle of the imagination in scientific researches. It gives to the scholar persistence in his daily toil; it is his hope at the moment he begins an undertaking, his guide during the work, and his recompense when he has finished. What a charm in the frequent discoveries of analogies between the highest orders of beings and those which occupy the lowest rounds in the ladder of perfection!

Similarity is to be observed also in the growth of individuals in the different kingdoms. One sees at first crystal skeletons, then gradually the crystals developing into perfection. Neither the chemist with all his delicate tests, nor the physician armed with his accurate instruments can decipher

the feeblest trace of heterogeneity; the child grown has become a man; the mineral fully developed has reached also its age of virility.

Minerals may be hindered in their development, may become irregular, imperfect, deformed; upon certain of their angles new facets may appear, in other parts facets may slowly become obliterated. As soon as the obstacle causing the trouble is removed the wounds will heal over, perhaps leaving their scars, and the crystals will pursue their normal course. Sometimes an accidental circumstance, as that of too ardent a sun, or a season too wet, will cause a fissure, and a malady commences. Oxydation or hydration is produced, and the mineral begins to disintegrate; finally, as a result of the accident, the last particles are lost to sight. We think it has been destroyed. But it is dead; it has died just as a man dies. Its elements are just as imperishable as are those of man's body, which, when it is laid away in the grave are not annihilated, but, as they are resolved, enter again into new forms in the great torrent of life. Their atoms are immutable, what they have been, they are, and will be to all eternity; eternally young, eternally the same, moving without rest, unmindful of time or of combinations. The ancient symbol of the serpent with his tail in his mouth well represents the cycle of life. Periods succeed periods.

The day ends in twilight and the night is followed by a new dawn. All limits are effaced. The stone, the flower, the animal intermingle their natures. With this thought in mind all life seems like a great net-work, whose meshes are interlaced in countless ways, before which the seeker after truth stands with ardent soul. But at the moment he thinks to grasp the solution of the absorbing problem, he is only made more deeply aware of his own weakness. And looking forward over the great expanse stretching out before him to infinity, he experiences only one sentiment, that of admiration; and his desire ever increases to learn still, and to learn always.

THE MACHINERY OF OUR FOREIGN SERVICE.

Report of a lecture delivered by Hon. Eugene Schuyler in the National Museum, Washington, D. C., on Saturday, February 28th.

This topic is especially interesting from the fact that so little is known of it except by those in the service of the government whose duties are connected with the foreign service. The government of the United States, in uneventful times at least, is a despotism in the hands of five or six men, working under and through constitutional forms, and subject only to the penalty which is always exacted from very grave mistakes. These men are the President, the Secretary of State, the Secretary of the Treasury, the Speaker of the House of Representatives, the chairman of the standing Committee on Appropriations, and the chairman of the standing Committee on Ways and Means. In times of disorder, others are added to this list, both from the Senate and from the Cabinet officers. The chairmen of committees for other branches of the service also, at such times, rise into prominence. Without the consent of some one, two or three of these dignitaries no important step in public affairs can be taken. The Secretary of State and the Secretary of the Treasury are the only Cabinet officers who in ordinary times can influence, not only the policy of the government, but also the welfare of the country, without the permission of Congress; it may be, even without the knowledge of the President.

The currency question, the silver coinage, the position taken recently by the Clearing House in New York, and the state of the gold market, show how a sudden emergency may induce, if not compel, the Secretary of the Treasury to take action which might strongly affect for good or for evil the most vital interests of the nation. Nor is it otherwise with the Secretary of State, who by an intemperate or ill-timed insistence on national or individual rights, or by even a want of tact may cause irritations hard to be appeased. On the other hand, by an ignorance of precedence, an unguarded

admission or an act of good nature, he may give up rights which the nation has jealously claimed for a century, or has held in reserve for future use. However, judging by the past, I think our Secretary of State will do none of these things. This official is selected with greater care than any other public officer. He is usually a statesman of high rank or of long experience, and frequently a cautious and shrewd lawyer into the bargain. The *possibilities* of diplomatic mistakes, however, are such that it is necessary for the Secretary of State to be surrounded by thoroughly trained and skilled subordinates. This department is among the earliest of the great divisions of the administration created by Congress in 1789 for facilitating public business, and during the first forty years of our national existence was in reality, as now in rank, the leading department of the government. Years ago, indeed, our foreign policy was of far more consequence to the country than our domestic policy, although we still had to struggle, if not for our existence, at least for our position and our national rights. The Secretary of State, therefore, is the leading statesman of the party, and at one time in the nation's history was almost sure of succeeding to the presidency. The duty of the Secretary of State not only is the supervision and management of all the foreign relations of the United States, but also those duties which in other countries are generally given to the Keeper of the Seals, or to the Minister of Justice: such, for instance, as the keeping, promulgation, and publication of the laws; the custody of the great seal, and the preservation of the government archives, as well as the charge of all special relations between the general government and the several states. The first Assistant Secretary is to be considered as a political officer, in the full confidence of his chief, able to advise him, and even at times to replace him; while the second and third Assistant Secretaries have by necessity and custom become permanent officers. The affairs of the department are managed with great secrecy, not only because the officials are careful and trustworthy persons, but because the general public, as a rule, is but slightly interested in matters pertaining to our foreign relations, save when some great subject is under dispute. In England, France or Italy the case is different, since the Minister of Foreign Affairs has a place in Parliament, and can be interrogated at any time with regard to particular questions arising with foreign countries, by which means the public can not help being more or less informed on such matters, even though the progress of negotiations may be kept secret. Here the only method for obtaining such information is

by a resolution of either house of Congress, asking from the President the papers on the question in point, and making an investigation, if considered necessary, through the Committee on Foreign Affairs. These papers, however, may be refused, if thought by the President that their publication would be disadvantageous to the interests of the government. There is probably no other country, even Russia or Germany not excepted, where so little is known by the public of the negotiations carried on at any one time by the Secretary of State. This has great advantages, enabling the government to conduct with tranquility a negotiation which may be extremely necessary, and often to settle disputes which, if public opinion were excited, might result in a breach of friendly relations. On account of this quiet way of doing business, many people are of the opinion that very little work is done by the State Department. Clerks often work till late at night and all Sunday, sometimes, preparing commercial and statistical information in response to a question asked in Congress. The work of the chief clerk, in one sense, is the hardest of all, for he has to work in a public room, accessible to all, must inspect every paper that comes in or goes out, must carry the whole business of the department in all its details in his head, must see every one who calls, assist those who have legitimate business, listen to others, giving "suave answers, but no information," and withal be patient and keep his temper. During the last fiscal year the real expense of the State Department to the nation was less than \$400,000; since the total sum expended (\$1,288,355.28) was in great part met by the fees, which amounted to \$899,652.67.

The State Department has not sole authority for the administration of foreign affairs, for the consideration and approval of the Senate is required, not only regarding nominations to diplomatic and consular posts made by the President, but also regarding treaties made with foreign powers before they can be ratified. It is fortunate, however, that the Senate can only affirm or reject a treaty; but, owing to the wording of the article of our Constitution, which says that the President "with the *consent and advice* of the Senate shall conclude treaties," the Senate considers that it has the right to amend a treaty already negotiated, a practice which causes great difficulty, as frequently a Senator to whom the subject under discussion is not quite clear, insists on the addition of two or three words to an instrument, which causes a long delay and frequently protracted

negotiations. Treaties are discussed in secret session, partly because the Senate is acting as a privy council to the President, and partly because, if the debates were open, things might be said which would give offense to foreign governments. As to this latter point, I can only observe that the practice of debating a treaty in open session has not been found to work badly in those countries in which it is the habit.

A feeling of jealousy has been growing up between the House of Representatives and the Senate, and has become very evident during the last few months, the House maintaining that, as it alone was empowered to initiate measures touching the revenue, the President had no right to negotiate a commercial treaty without previously consulting that body. I do not think that this contention is supported by the Constitution, but at the same time the practice of our government has changed so much of recent years, in giving larger and larger powers to the lower House, that it is not without some reason that such a view is supported. In order to obviate such disputes, the Secretary of the State Department, before making a commercial treaty and engaging the country in a new commercial system, should, as was done in the negotiation of the Mexican treaty, ask Congress for authority to conclude it. Again, the powers allowed by the Senate to its standing committees form another obstacle to the ratification of treaties, since it is impossible, except by an actual vote of the Senate, to compel the committee to report to the full Senate a treaty which has already been referred to it for consideration. In the Senate committees are elected; in the House they are named by the Speaker. The sub-committee of three, which is in charge of the appropriations for the diplomatic and consular service, is generally named by the chairman of the Committee on Appropriations, and in nine cases out of ten is composed of persons possessing no previous acquaintance with the subject. To the sub-committee are presented the estimates made up by the Secretary of State, and a bill is then prepared. It can raise a grade here, establish a consul there, pare down a salary in one place, or abolish a mission in another. Of course some of the changes made by this sub-committee are often very excellent, and even necessary but its main idea seems to be to reduce the appropriation to the lowest limit from motives of economy; not that the nation at large cared for a saving of ten or twenty thousand dollars, but because by gaining the reputation of being economical, constituents might believe its members worthy of a new

election. The bill is next reported to the House, where party strength is drilled to support the committees. Every amendment is there voted down, for the men whose salaries are sometimes retroactively voted down, are too far away to be heard. From the House the bill is passed to the Senate. The general theory of the Senate committees is to reject every change made by the House, and to hold pretty closely to the law of the last Congress, restoring what had been omitted, and adding some appropriations for unforeseen expenses, secret service money, or as technically expressed, for "expenses in carrying out the Neutrality Act," etc. The Senate generally passes the amended bill with slight debate, except in unusual cases. The House next, on motion of the sub-committee, is wont to reject without debate all the Senate amendments, and very often suggests a committee of conference. In like manner the Senate refuses to recede from its amendments, and accepts the conference. Then a secret meeting is held of the two sub-committees, who bargain with each other, giving and taking, each yielding part, and reporting the results to their respective houses in such a technical form that it is impossible to understand it without a careful examination of all the papers. This the clerk reads hastily, and it is passed without debate, often containing new matter never before proposed in the open House. I am not blaming either body, but simply explaining a system which is becoming the habitual way of passing all appropriation bills. How can an already underpaid consul perform his duties properly and vigorously, when every few months he has to consider the chances of having his salary cut down, or when engaged in an important investigation by order of his government, he is quietly informed that his salary ceased a month or six weeks before?

The interests of our country demand that our diplomatic and consular service should be fixed by a general law, subject of course to necessary changes, to be recommended by the department, and not undergo this annual tinkering, to which no other branch of the government and no other class of officials are subjected.

Let us next consider the duties of the agents of the government under the control of the State Department, which belongs to one of two classes, those in the consular and those in the diplomatic service.

Consuls differ from diplomatic agents (by whatever name they may be known), in that while the latter are the representatives of one state or government to another, consuls are the representatives of the individuals of the nation sending them, empowered to protect individual interests, and to procure for their fellow-citizens, as far as possible, the same protection to their rights that they enjoy at home. They represent commercial interests only. They can address themselves directly to the local authorities when the rights of their fellow-citizens are infringed, but if redress be not given, they can not apply to the supreme government, except in cases specially provided for by treaty. They must refer the matter to legation or their own government. In other words they have *no* diplomatic or representative rights, powers or privileges. Formerly consuls had power as arbitrators, but gradually the legal jurisdiction over disputes was withdrawn in nearly all except non-Christian countries, although for purposes of wills or intestate property this jurisdiction has still been in some measure preserved. With regard to maritime matters the case is different; and here, for the purpose of avoiding protracted disputes in the courts of the country, the consuls are still allowed large jurisdiction. This is nowadays in most cases regulated by special treaties.

Consuls are in a certain way charged with watching over the execution of treaties, for they must protect any of their countrymen whose rights are invaded, and must immediately bring to the attention of their government any such infringement. In general, they observe the movements of naval forces of all nations on the coast near the port in which they are placed, and it is their duty also to watch over the dignity of their own country in maintaining the rights of their flag. Not only are they obliged to give aid, advice, and assistance to the ships of their commercial marine, but they should in their correspondence with their government report all events touching the navigation, the various changes in the commerce of the countries where they live, and especially anything touching the special commerce with the country which sends them. In fine, they are bound to keep pace with the state and progress of manufactures, the rise of new branches of industry, and in general, the increase or diminution of the public wealth, taking especial care to be well acquainted with all matters where other countries may gain advantage over their own. They are given a sort of police jurisdiction over the commercial vessels of their own country;

they are generally charged with the duty of investigating shipwrecks and saving property from the wrecked or stranded vessels; with all disputes between captains and sailors; with arresting deserters; and with sending back shipwrecked or discharged seamen. In time of war their duties in these respects are still more important, for they are obliged, so far as the international law, the special treaties, or the laws of the country in which they are placed will permit them, to protect at all hazard the commercial and naval interests of their country against arbitrary acts, whether committed by the country to which they are sent, or by the nation at war with it.

On the death of one of their countrymen they in general take possession of his effects, and in case of property left in the country, manage, keep, and dispose of it for the benefit of the heirs. They are charged, beside, with notarial duties of all kinds, and in most cases they are the only authorities who can validate legal instruments between citizens of their country, or others to be used at home.

In addition to the general duties of a consul various special duties are imposed on American consuls by our tariff system, which do not generally exist in the services of other countries.

It is necessary for our consuls to verify in triplicate every invoice of goods sent to the United States. Not only is he obliged to take the oaths of the manufacturer or exporter, but he is expected to have a special knowledge of the trade of the place and of the actual value of the goods, so that he can control the statements made to him; for our system does not accept the valuations of goods always at the actual price paid for them, but at the market value of the place where they are manufactured or chiefly sold. Besides keeping a number of official records, registers, and fee books, carrying on his ordinary correspondence with the Department of State, and carefully prescribed forms relating to the business of his office, and of everything of interest of a commercial nature to the government, the consul is obliged to make quarterly, semi-annual and annual returns, both to the State Department and to the Treasury. He must, for instance, at the end of each quarter give a digest of the invoices verified by him during that period; of the arrivals and departures of American vessels, a return nowadays exceedingly simple; of deceased American citizens; a record of his notarial

services, or unofficial fees; a summary of the whole consular business; and, in case the consul has extraterritorial jurisdiction, a return of the business of the consular coast, and also a record of his official fees.

Still other duties are the submitting of quarterly, semi-annual, and annual reports. The consul at Shanghai has such duties placed upon him as give him supervisory control over all consulates in China, vest him with semi-diplomatic powers, cause him to participate in the municipal government of the foreign settlement, make him a judge in civil causes, give him charge of the gaol in which American prisoners are confined, constitute him judge of a criminal court, of a court of probate and divorce, of an equity and *nisi prius* court, appoint him United States postmaster, give him the duties of a seaport consulate, and place under his control the protection of the revenue of his government.

Diplomats are agents of a higher class and with different functions. According to Caloo, who is now generally accepted as the best modern writer of international law, diplomacy is the science of the relations existing between different states, such as result from their reciprocal interests, the principles of international law, and the stipulations of treaties and conventions; or, more concisely, diplomacy is the science of relations, or simply the art of negotiations. According to Caloo, the essential nature of diplomacy is to assure the well-being of peoples, to maintain between them peace and good harmony, while guaranteeing the safety, the tranquility, and the dignity of each of them. The part played by diplomatic agents consists principally in conducting negotiations relative to these important objects, in watching over the execution of treaties which follow from them, in preventing anything which might injure the interests of their fellow-citizens in the countries where they reside, and in protecting those of them who may be obliged to ask for their assistance. According to rules adopted by the Congress in Vienna in 1815, diplomatic agents were divided into three classes: (1) Ambassadors, legates, or nuncios; these two latter being sent only by the Pope; (2) Envoys, ministers plenipotentiary, or other persons accredited to a sovereign or sovereign state; and, (3) *Chargés d’Affaires*, who are accredited only to the Minister of Foreign Affairs. According to the old custom, ambassadors represented the person of the sovereign, and accordingly enjoyed higher ceremonial honors than were paid to other diplomatic agents. They could also address themselves personally to the

sovereign or chief magistrate of the country to which they were sent for matters of business, instead of having to negotiate with the Ministers of Foreign Affairs. Nowadays ambassadors differ from other diplomatic agents only in rank and precedence. The United States having no ambassadors, and but few envoys and ministers plenipotentiary, does not always receive equal privileges of rank with some other countries. Our interests certainly demand that in every country we should be represented by agents of the highest title known or accepted there. More questions are settled by a few informal words at a dinner table than by a formal process of correspondence, although, of course, when great principles are at stake a formal mode of procedure is necessary. It is therefore evidently to be desired that diplomatic agents in a given place should be of equal rank and on a friendly footing with each other.

There are several cases in which the Minister of the United States, if he had more official authority, could manage to have matters arranged which ultimately affect our interests. At Constantinople, for instance, where there is an effort to undermine the treaty rights of all foreigners, the ambassadors have of late adopted the habit of meeting one another in an unofficial way, and of laying down rules and taking action regarding extraterritorial matters, which are then proposed to the rest of the diplomatic body. In general, the representatives of the smaller states are asked for their approval or dissent, but given no chance to suggest or argue. Three years ago, indeed, our government found it necessary to protest against this course, for it was beginning to be tacitly understood that only the ambassadors of what were called the Signatory Powers—those who were represented at the Congress in Berlin in 1878—should have any voice in matters which affected the interests of all foreigners in Turkey. Our protest had the theoretical result of bringing about occasional conferences of all foreign representatives, but the practice remains much as before.

Foreign ministers of the United States should be enabled to live in a style suitable to their rank. Nor is this simply a question of display, but for a minister to be useful he must make acquaintance with the leading persons of the country, and entertain them at his house.

The necessary qualifications for employment in the diplomatic service are a knowledge of French, and generally at least of one other language; a

good acquaintance with history, treaties and international law. It is also necessary that he be a gentleman: *i. e.*, acquainted with the ways of the world, and the usages and manners of the best society in each capital in which he is expected to move. The word “gentleman” does not necessarily imply a man of good birth, or belonging to a well known family, although the son or grandson of the President of the United States would always have more credit and influence in the place to which he was sent than one of whom nothing was known.

It is hard to create among a Christian people, enthusiasm for an infidel, however talented he may have been, or however much good he may have done; for his revelation to man, even if true, is an unwelcome and painful revelation, adding nothing to his happiness or comfort in life or in death; while the faith of the believer is an inspiring one, filling his life with the sunshine of hope, and surrounding it with a halo of imperishable glory. Most people have an instinctive dread of the man who with ruthless hand, attempts to destroy all those sacred hopes and fears which have been instilled into their minds by their nearest and dearest benefactor, their mother.—“*How to Get On in the World,*” by Robert Waters.

MADURA AND ITS PAGODA.

BY BISHOP JOHN F. HURST.

When I was buying my ticket at Tuticorin for Madura, the station agent was kind enough to say:

“Don’t you know there is cholera in Madura?”

“What, real Asiatic cholera?”

“It’s real Asiatic cholera, and nothing else,” he answered.

“I have not heard it before,” I replied. “I have only this moment landed from the steamer ‘Nerbudda,’ and have had no news of any kind. Many deaths?”

“Oh, no. Nothing compared with last year. Five thousand died during the season. Only about ten die a day just now, and we don’t consider that anything.”

I mused a moment on the mortality of ten cholera patients a day in a place of fifty thousand, and then asked: “Do you think it safe to go?”

“I can’t answer that. It all depends.”

Two facts now came to my relief. One was, that few people in India think cholera contagious. There are no separate hospitals for such cases. Cholera patients are put in the same wards with patients suffering from fever and other diseases. The other fact was, that two weeks before, when I was in Puna, there had been a cholera case in the native bazar, and yet I had a most pleasant ride through that part of the city, and had suffered no harm, and saw no alarm anywhere. The truth is, nobody thinks of cholera as any more likely to happen than any mild disease. Dr. Waugh told me only yesterday that cholera prevailed more or less in all Indian towns, but that nobody minded it. It might be next door, but it frightened no one. The only

thing is to watch its beginning, and manage it, as you can, with care and caution. Another is, to take care of one's diet. This must be said, however, that when cholera does come, and its first stage is neglected, the collapse is very sudden.

Taking all things together it did not seem much of a risk to spend my intervening day, before meeting an engagement at Bangalore, in the Mysore, in making a halt in Madura, and using my only opportunity to see the famous Pagoda there—the largest, not only in India, but in the world.

Long before reaching Madura one can see the great towers which rise above the Pagoda, and dominate not alone the city, but the whole surrounding country. In many of the Indian cities the temple is in the suburbs, and even completely alone, in the country, having been left by the drift of the population far out into other directions. But this is not the case in Madura. The Pagoda is in the very heart of the old city. The bazars lead directly toward it, and overflow into it. It is the city in miniature, with its dirt, ill odors, poverty, wealth, superstition, and infamous idolatry. All the surging tide of tradesmen drifts toward and about it. No adequate conception of an Indian temple can be formed from any European illustration of sacred places. Perhaps the Troitskoi Monastery in Russia, where many cathedrals are grouped around one central sacred place, making the whole a very Canterbury, is as near an approach to an Indian temple and its spaces as can be found anywhere west of Asia.

Madura has long been celebrated for this Pagoda. There are conflicting opinions as to its antiquity. It is probable that the place itself was regarded sacred, and was the site of a temple long before a city was built here. It is not unlikely that the temple was the first building, and that the city grew out of it, and all about it. The immense structure gives clear evidence of its own antiquity. It was built in the third century before the Christian era, by King Kula Shekhara. It is evidently a case where the city has sprung into life from religious associations, and become the capital of a large territory. Some parts of the Pagoda are modern, and were built by Nurmala Nark, in the former half of the seventeenth century, but one can easily distinguish the newer from the older. The effect, throughout, is one of great and undisturbed antiquity.

The Pagoda space is an immense parallelogram, extending 744 feet from east to west, and 847 feet from north to south. This area is enclosed by a light wall, and is flanked, at various points, by nine colossal towers. These towers are of peculiar structure, all after the same model, and so disposed toward each other as to form a symmetrical combination. Each constitutes a kind of gateway, for entrance from different sides of the wall. As you enter you find yourself passing through a great open corridor. The *gopura* is shaped like a tent, and on every side is ornamented with carvings. These represent the fabulous doings of the god Shiva and his wife, Minakshi, and ascend in lessening rows, or stories, until the apex is reached, which is sharp and curved, and reminds one of the general form of an old Roman gallery. The colors of these *gopuras* are very rich, and, in the case of several, shine like fine tiling, or even gay enamel. The blue is especially rich, and is fairly dazzling in the bright sunlight. While Shiva is the god to whom the temple is supposed to have been dedicated, the more frequent representations of his wife Minakshi prove her to be the favorite of the people.

THE SCENE IN THE MADURA TEMPLE.

Two *gopuras* constitute the great entrances. Through one of these I went, with a crowd of about fifty ill-clad beggars following me. They held high carnival as they passed around and against me, and called for alms. I noticed many sleepers in the darker corners, in various parts of the temple spaces. They lie in every position. It seems a habit of the Maduran when he gets thoroughly tired in his tent, or in the bazar, to drop into this temple and fall down for a good nap at the feet of Shiva, or some other idol, for Madura is a spot which for ages has been held strangely sacred by the Hindoo worshiper. Having passed through the *gopura*, and completed the passage of the great corridor, you see the beginnings only of this wonderful temple. There stretch out before you great reaches of passages, and halls, and still farther corridors, in all possible directions. But for my safe guide, who added to his other duties the good one of keeping off the crowd of ragged and starving and ill-smelling beggars with a stout bamboo rod, I should have lost my way at once. At your right you see an immense hall, the Hall of One Thousand Columns, which extends far away until it is lost in such

dark and distant spaces as I cared not to explore. But, beyond it—for I came back that way—there is a special temple sacred to the ruling god, Shiva. At your left are venders of images, sweetmeats, toys, and various other articles, which, for some reason, are permitted to be sold within the sacred walls. The men who sell them are squatted over the floor, on mats of palm, and their wares lie about them. Think of a seller of small wares, in a temple, sitting or standing, with his goods arranged on a counter or row of shelves! Such a thing would be preposterous beyond measure. The drift is downward. No Hindoo will stand if he can possibly drop on the floor. He doubles up his legs under him. That is his normal position. He may be talking with you this moment, and as much interested in standing or walking as any one. But a sudden change comes over him. Down he drops, and no boy ever closed the two blades of a jack-knife more quickly than the Hindoo doubles himself up, either on the temple floor, or at the side of the street, or in his own doorway. And there he can sit by the hour, nay, the whole day, and be as calm as the serene face of Buddha himself.

Perhaps these sellers in the Madura Pagoda have some ancestral claim on the favors of the authorities, by which they receive the privilege of spreading out their wares in the holy place. Over your head there flies about a flock of doves. They are sacred, and woe to the hand that would hurt a feather on their sweet heads! The worshipers feed them. It is a sacred privilege. Yonder, to your left, three sacred elephants are feeding and frisking their trunks about as if they really knew that they were picking up great wisps of straw and hay within the most holy place in all this region. Come, I must hasten, or their priestly keepers will loosen the chains of one of them in a trice, and have the mammoth dropping down on all fours, and pulling me up on his back, to take an elephant ride through this labyrinth of marvels. Imagine the absurdity of an elephant ride on a temple floor! Yet that is what you can do here, and take a long promenade, and never have him repeat his pathway. I have had two elephant rides, and want no more for a decade, at least. But by going through this first doorway I get away from the venders, and the elephants, and pass out of sight of the Hall of a Thousand Columns, and its great, interminable spaces. Here one is in a corridor nearly two hundred feet long, with pillars groaning beneath a wealth of sculptured images. Now comes a brazen door. The frame is vast

and heavy, and is entirely surrounded with brazen lamps, all of which are lighted during a festive season, perhaps the *Tailotsava*, “the oil festival.”

Monier Williams happened to visit the Madura Pagoda at the time of the “oil festival,” and thus describes the wretched scene: “A coarse image of the goddess (Minakshi), profusely decorated with jewels, and having a high head-dress of hair, was carried in the center of a long procession, on a canopied throne, borne by eight Brahmans, to a platform in the magnificent hall, opposite the temple. There the ceremony of undressing the idol, removing its ornaments, anointing its head with oil, bathing, redecorating and redressing it was gone through, and shouting, singing, beating of tom-toms, waving of lights and cowries, ringing of bells, and deafening discord from forty or fifty so-called musical instruments, each played by a man who did his best to overpower the sound of all the others combined. At the head of the procession was borne an image of Ganesa. Then followed three elephants, a long line of priests, musicians, attendants bearing cowries and umbrellas, with a troop of dancing girls bringing up the rear.

“No sight I witnessed in India made me more sick at heart than this. It presented a sad example of the utterly debasing character of the idolatry which, notwithstanding the counteracting influences of education and Christianity, still enslaves the masses of the population, deadening their intellects, corrupting their imaginations, warping their affections, perverting their consciences, and disfiguring the fair soil of a beautiful country with hideous usages and practices unsanctioned by even their own minds and works.”—“Religious Thought and Life in India.” Part I, pp. 442-443.

You are now introduced into a darker corridor, and then again into a broad and pillared space, where the columns are sculptured, being cut through and through into figures of dancing gods, like Shiva when he played his flute to the shepherds. You now look out upon a little sheet of water with a miniature temple in the middle of it. This is the Lake of the Golden Lilies. Near by it is the little chapel where Queen Mangammal’s subjects starved her to death in 1706, having placed food so near that she could see and smell it, but not taste it. We now enter another department of the temple; above there are stone images, up around the pillars, in all corners, and hanging down over you wherever you go, near walls or archways. These images are not grave and majestic, but, in the main,

grotesque, bacchanalian, in fantastic attitudes, and often combining the bodies of man and beast. They represent, for the most part, the escapades of Shiva. Every now and then one comes to a shrine, where worshipers lie prostrate before it, and remain motionless for a long time. No one knows how long it has taken these poor dusty pilgrims to reach this sacred place. Perhaps they have been three months on the journey. They come from the very base of the Himalayas, or the borders of Thibet, and now that they have reached the end of their pilgrimage, would die with a happy heart. There are several gold plated images, veiled from view, which represent the god Shiva, or his wife, in some part of their marvelous career. The representations in stone, both of men and the brute world, are frequent everywhere. Elephants, horses, cattle, and every kind of animal held sacred in the Hindoo mythology, are cut out of stone, and made to portray the supposed divine attributes of Shiva and his wife. Here, too, are the very *vehanas*, or great chariots, plated with gold, in which the god and his wife are taken out on special days in the year, to ride. Beside these there are silver litters, which serve the same divine purpose on other days.

One grows weary of the procession of splendid but gross images and idols in this vast space. Now you are out for a time in the open air, where a vacancy has been left in the roof, and the beautiful sky throws down its blessed sunlight upon this terrible picture of idolatry. But very soon you are brought again under the shadowing and lofty ceiling, and before you are aware of it, you are almost lost in a dark labyrinth of sculptured pillars, black idols in gold wrappings, dusty and absorbed pilgrims, cheerful doves, and the constant crowd of men and boys, who follow you, either to sell you their sweets, or beg for your loose coppers. All at once you come out from a corridor to the marble steps of a miniature lake. Be careful now. Only the real Hindoo dares to step down into its waters. For every drop is sacred, and must touch only the skin of Shiva's children. Over the calm surface the towers stand as gay sentinels, from century to century. Turning again, you must look carefully, or you will tread upon a sleeping form, which has dropped in from the hot air, and let fall its burden, and eaten its crust, and now rests an hour. There is a mother, with a nose-ring so large that it hangs down over her mouth, and she must eat through it, or starve. Her ankles are encircled by heavy silver anklets, cut like serpents. Her toes are glittering with jeweled rings. She has led her child up before an image of Shiva's

wife, and is explaining what it all means. Poor woman! Little she knows the truth. The One Name above all others she has never once heard. Here is a dwarf, who stands beside a shrine, and holds out his withered hand for an *anna*. Here, in a place where the statuary has given way to the wear of ages, are workmen in stone, who are making new pillars, with sculptured flutings, to take the place of the old. All the work, every stroke of mallet and chisel, must be done right here, where everything is holy, and Shiva smiles down upon the labor.

ANECDOTE OF JERROLD.—His heart was as kindly a one as ever beat in a human bosom; and his hand most liberal, and often far more liberal than his means might have justified. He was once asked by a literary acquaintance, whether he had the courage to lend him a guinea. “Oh, yes,” he replied, “I’ve got the courage; but I haven’t got the guinea.” He had always the courage to do a kind action, and when he had the guinea it was always at the command of the suffering, especially if the sufferer was an honest laborer in the field of literature.—“*Personal Traits of British Authors.*”

GEOGRAPHY OF THE HEAVENS FOR MAY.

BY PROF. M. B. GOFF,
Western University of Pennsylvania.

THE SUN.

Although at the time these lines are written the sun has not in his northern course reached the equator, and with us here in the north the ground is covered with snow, yet by the time our readers see these words in print a great change will have taken place in the face of nature; the beautiful green of the winter wheat will cover the fields, the tulips and hyacinths exhibit their brilliant colors, and our forests begin to display their refreshing foliage, and “Old Sol” himself will have completed half his journey to the tropics and have measured for us many days of the “little span” allotted to the life of man.

“Men may come and men may go,
But I go on forever.”

And thus are we ever reminded of the “flight of time.” The days grow longer and the shadows shorter; but “all too soon” the shadows begin again to lengthen and the nights increase. Of this, perhaps, we should not complain; for the many long days of summer give us ample opportunity to perform our duties during the “noble sunlight,” and we shall probably be glad of the rest that comes with the “shortening hours.”

During May our time is slow, the sun coming to the meridian about three minutes before noon, as indicated by our clocks. Sunrise occurs at 4:58, 4:42, and 4:32 a. m., on the 1st, 16th, and 30th, respectively, while sunset is at 6:55, 7:10, and 7:22 p. m. on the corresponding days. Day breaks on the 16th at 2:43 a. m., and twilight ends at 9:09 p. m., giving eighteen hours and twenty-seven minutes from “early dawn to dewy eve.” The length of day varies from

thirteen hours fifty-seven minutes to fourteen hours fifty minutes. Increase in right ascension, north $6^{\circ} 36'$.

THE MOON'S

Phases occur as follows: last quarter, on 7th, at 3:35 a. m.; new moon, 14th, at 10:09 a. m.; first quarter, 21st, 12:37 a. m.; full moon, 28th, 3:22 p. m. Rises on the 1st, at 9:16 p. m.; sets on the 16th at 9:29 p. m.; rises on the 30th at 8:49 p. m. Farthest from the earth (in apogee) on the 4th, at 5:18 a. m., and again on the 31st, at 6:54 p. m. Nearest to earth (in perigee) on the 16th, at 4:54 a. m. In latitude $41^{\circ} 30'$, least elevation on the 3d, amounting to $30^{\circ} 11' 56''$, and again on the 30th, amounting to $30^{\circ} 5'$. Greatest elevation on the 17th, equal to $66^{\circ} 51' 38''$.

MERCURY

Affords sharp-eyed early risers before and after the 25th, a few days' opportunity to get a glimpse of his countenance, as he reaches his greatest western elongation at 7:00 a. m. of the above named date. On the 11th, at 4:00 a. m., he is farthest from the sun; same date, at 2:00 p. m., stationary; on the 12th, at 10:59 p. m., $22'$ south of the moon; on 13th, at 3:00 a. m., $2^{\circ} 27'$ south of Mars, and again on the 30th, at 4:00 p. m., $2^{\circ} 56'$ south of same planet. Motion $2^{\circ} 27' 12''$ retrograde up to the 11th; and from 11th to end of the month, $14^{\circ} 54' 35''$ direct. Diameter diminishes from $12''$ on the 1st to $7.4''$ on the 31st. The times of his rising are as follows: On the 1st, 4:49 a. m.; on the 16th, 3:59 a. m.; and on the 30th, 3:36 a. m.

VENUS.

During the month the beauty of this planet is quite overshadowed by the superior light of the sun. Her times of rising and setting are nearly his own, and her diameter ranges from $9.8''$ to $10''$. On the 4th, about noon, the sun is between her and the earth (in superior conjunction). On the 11th, at 6:00 p. m., she is $1^{\circ} 15'$ north of Neptune; on the 14th, at 1:17 p. m., $3^{\circ} 47'$ north of the moon; motion direct, amounting to $39^{\circ} 15' 47''$. On the 1st, she rises at 5:05 a. m., and sets at 6:45 p. m.; and on the 16th, rises at 4:59 a. m., sets at 7:21 p. m.; on the 30th, rises at 5:03 a. m., sets at 7:53 p. m.

MARS,

Like Venus, keeps near the sun during the entire month, rising on the 1st at 4:24 a. m.; on the 16th, at 3:43 a. m., and on the 30th, at 3:25 a. m., and setting on the corresponding days at 5:22, 5:21, and 5:19 p. m. respectively. His diameter is 4.4", and his motion $22^{\circ} 14' 33.6''$ eastwardly (direct). On the 12th, at 10:55 p. m., he is $2^{\circ} 3'$ north of the moon; on the 30th, at 4:00 p. m., $2^{\circ} 56'$ north of Mercury.

JUPITER,

Now that Venus "hides her diminished head," "does himself proud," attracting the eye of the most casual observer. His proximity to the star *Alpha Leonis* (Regulus), particularly on the 30th, when he is about two thirds of a degree north of the latter, detracts nothing from his prominence; but on the other hand, rather renders him more conspicuous. On the 17th, at 10:00 a. m., he is just 90° east of the sun; and on the 20th, at 9:37 p. m., $4^{\circ} 17'$ north of the moon. His diameter decreases during the month from 37.2" to 34.2" and he makes a direct advance of $2^{\circ} 3' 51''$. On the 1st, he rises at 12:25 p. m., and sets next morning at 2:03; on the 16th, he rises at 11:30 a. m., and sets on the 17th at 1:04 a. m.; on the 30th, rises at 10:42 a. m., and sets at 12:14 a. m. on the 31st.

SATURN.

Those who wish to see in all his grandeur this planet with his rings, must not longer delay. Each day brings him nearer the sun, so that by the close of the month his time of setting is only about one hour after sunset. His diameter decreases four tenths of a second of arc, and his motion is $3^{\circ} 44'$ direct. On the 16th, at 9:35 a. m., he is $4^{\circ} 2'$ north of the moon. He rises on the 1st at 7:23 a. m. and sets at 10:05 p. m.; on the 16th, rises at 6:31 a. m., sets at 9:15 p. m.; on the 30th, rises at 5:44 a. m., sets at 8:28 p. m.

URANUS.

This planet will be an evening star, and afford a fine opportunity for observation to those who have the means at hand profitably to view it. Our limited knowledge of its physical properties make it, to the ordinary observer,

a matter of little interest. It rises on the 1st at 3:15 p. m., and sets on the 2nd at 3:21 a. m.; on the 16th, it rises at 2:15 p. m., and sets at 2:21 the next morning; on the 30th, it rises at 1:18 p. m. and sets on the 31st at 1:26 a. m. It maintains the same diameter, 3.8", throughout the month, and makes a direct motion of $2^{\circ} 13' 45''$. On the 23d, at 4:38 a. m., will be $1^{\circ} 11'$ north of the moon.

NEPTUNE.

And now we come to the "last but not least," by any means, of our planets—a planet, however, that interests us but very little, as we can only see it through a quite powerful telescope, and then only as a small, pale disk. Yet its movements are ascertained and recorded just as those are of other planets, and so far as we know them, we are just as confident of the obtained results. As much so as we are of the some two hundred and twenty small bodies that are so much nearer to us, whose orbits lie between that of Mars and that of Jupiter; more confident than we are of the orbits of those erratic bodies we call *comets*, which seem to come and go at pleasure, and were formerly the terror of all who beheld them; and of those other bodies known as *meteors*, *meteorites*, or *aerolites*, which not only terrify those who behold them, but frequently injure and destroy the beings with which they come in contact. In fact, we know that Neptune, although apparently so small, is a globe 34,500 miles in diameter, and so far away as to do us no harm, while there *may* be thousands of little invisible globes flying around our earth waiting for some favorable opportunity to break away from their restraints and hurl themselves, as those did at Stannern in 1812, or at Orgueil, in France, in 1864, upon our devoted heads or our cherished treasures. Let us, then, respect our obscure and distant friend, with whom we are definitely acquainted, and record his acts as follows: For the first part of the month he will be an evening star; from the 13th, on which date he will be in conjunction with the sun, he will be a morning star; and on the 14th, at 7:47 a. m., will be $2^{\circ} 15'$ north of the moon. His motion will be direct, and amount to $1^{\circ} 10'$; his diameter 2.5". On the 11th, at 6:00 p. m., he will be $1^{\circ} 15'$ south of Venus. On the 1st he will rise at 5:44 a. m. and set at 7:42 p. m.; on the 16th, rise at 4:48 a. m., set at 6:48 p. m.; on the 30th, rise at 3:54 a. m., set at 5:34 p. m.

THE HOMELIKE HOUSE.

BY SUSAN HAYES WARD.

CHAPTER IV.—THE BEDROOM.

“The Pilgrim they laid in a large upper chamber whose windows opened towards the sunrising; the name of the chamber was Peace, where he slept till break of day, and then he awoke and sang.”—*John Bunyan*.

It is impossible to treat of house furnishing and decoration without some allusion to what hygiene requires of the house builder. In the properly constructed house the bedroom will be light, airy, and if possible, sunny, like the pilgrim's chamber. The bedroom windows should not be so heavily hung with curtains as to obstruct the free passage of air. Thin curtains of chintz or muslin are better for sleeping rooms than heavily lined damask or cretonne, as sunlight and pure air are bedroom essentials.

The cheapest and most convenient treatment for the wall is paper hanging; but Dr. Richardson, the well known English writer of house and health papers, inveighs against wall paper upon bedroom walls, and specially against the practice of papering one layer over another, on the ground that germs of disease are liable to be cased up behind wall paper, and to remain a source of danger in after years. No doubt a painted or washable surface is best from a hygienic point of view, but with proper care paper can be risked.

Light, airy patterns are preferable, of varying tints, but the same general color as the ground, for the bedroom should never be gloomy, and the less sunshine it gets from without the more sunny should be the paper that decks its walls. Violent contrasts in color, and spotty or staring designs are a source of irritating annoyance to the sick. Let the purchaser, in selecting wall paper, stand at a distance of a dozen feet or so and look with half closed eyes, and he will get much more of the general effect, and will see more as the invalid will who may occupy the room when the paper is hung.

Then, in the matter of drainage and plumbing, there has been a great overturning in the past few years. People began to discover, about ten years ago, that their modern improvements were followed by a long train of sore throats, diphtheria, and typhoid fevers, and the wise householder was led to study the various systems of pipes and drains. Thanks to our boards of health, and to the efforts and writings of such men as Col. Waring, much has been done to improve and perfect the drainage of city houses, but in spite of the advance that has been made in this direction, modern conveniences often prove in the end to be inconvenient, if not pernicious, and the fewer set washbowls and water closets with which our houses are furnished the safer we may feel. With faucets for hot and cold water on each floor from which to replenish the water jugs, no reasonable servant could complain of the extra drudgery, much less the sensible woman who “does her own work,” and all could sleep sounder at night without fear of being haunted by any of those frightful demons of the drain pipe which were represented in a number of *Harper’s Weekly* some years ago, as issuing from a set washbowl and hovering over the innocent slumberer.

Upon this point all the writers upon house decoration are as one, and Mr. Cook, in his “House Beautiful” says: “Seeing no certain way to prevent the evil so long as drain pipes are allowed in bedrooms, many people nowadays are giving up fixed washstands altogether, and substituting the old fashioned arrangement of a movable piece of furniture, with movable apparatus, the water brought in pitchers, and the slops carried away in their native slop jars.” Whether healthier or not, I think there can be no doubt that the old way is more comfortable by far.

Setting both health and comfort to one side for a moment, there can be no doubt that the movable washstand, with its paraphernalia of bowls and pitchers, is a more sightly and decorative object in the bedroom than any set washbowl arrangement that has yet been contrived. Of course I am referring to the introduction of waste pipes into the bedroom proper, not to toilet or bath-rooms outside its walls.

In cold weather the bedroom air should be a little cooler, perhaps, than that of the living rooms of the house, but not many degrees lower.

Our fathers and mothers, when boys and girls, slept in rooms freezing cold, and broke the ice in their water pitchers in the morning; but they lived in spite of this, not because of it. There is a deal of loose thinking on this subject. Cold

air is no healthier than warm. It is impure air, warm or cold, that is unhealthy, the cold being specially pernicious; witness the church influenza, that most obstinate and unconquerable of all colds, because contracted by sitting in a chilling atmosphere after the body's vitality has been reduced through breathing air that has not been renewed since the last service held in the room.

There was a clever story called "Lizzie Wilson," published in *Littell's Living Age*, years ago, in which a clergyman's poor widow is represented as bringing up satisfactorily, through many straits, a family of young children. As their bedrooms were not heated, they had a joint dressing room, where the boy of the household first lighted the fire, and then dressed himself, his mother and sisters occupying the room later, in turn. This indulgence in the way of comfort, which might have been deemed an extravagance by others as poor as themselves, was paid for by going without dessert three days of the week; and the children, when cosily warming their backs before the dressing room fire, were pleased to call it "taking a slice of pudding." A wise household economy of this sort, less pudding and pie and more fires, would not be amiss in many American homes. To keep one room intolerably hot, and all others without any heat, is a wasteful retrenchment, which must be paid for in doctors' bills and funerals.

The question of single or double beds is also one of some hygienic importance. When a room is to be occupied by more than one person, the European custom of placing two single beds side by side has great advantage over the English double bed fashion. I have known mothers to assert that they observed a marked improvement in the health and temper of nervous, irritable children, after the little ones had been removed to single beds, where they could rest without disturbance from a bedfellow; and no one doubts that sickly or delicate people should occupy single beds.

As to color, I confess to a stout prejudice against getting up rooms all in one hue. I would banish altogether the young-ladyish dainty pink or blue room, and confine the green room to the theater. It is very hard to so manage a symphony in blue, for example, that it shall be truly symphonious. The cretonne furniture covers are apt to contain some aniline dyes that fade to forlorn and sickly hues in place of their original smartness. The blue of the wall paper will never agree with that of the carpet, and the cheap paper cambric or stouter jean that peeps through the muslin toilet cover grows paler with age, and each passing day increases the general discord.

White rooms with snowy and spotless walls, curtains and bedcovers, such as certain nun-like story-book young ladies affect, are chilling in the extreme. Their immaculate purity alone renders them endurable, and even then the obtrusiveness of their Dutch-like cleanliness is exasperating. A dingy white room is even more ugly than an ill-assorted blue one.

If the walls are plain, let the curtains be figured with various colors; if the walls are papered with figured polychrome hangings, let the curtains be plain, but harmonizing with some one color of the wall paper. That same color can be emphasized and repeated in carpet, rugs, and table or bureau cover, but no one color should be used to the exclusion of all others, as the eye wearies of neutral tints unrelieved by positive color without a large proportion of neutral tinted space.

A bedroom should look as if intended for the use of its occupants. Much millinery, quilled and ruffled muslin, and toilet tables in fine petticoats are only allowable in the room of a dainty young girl who has plenty of time to spend in renewing and freshening up her ephemeral finery, or in a guest chamber that is seldom used, and is thus made to look pretty at slight expense. Knick-knackereries of this sort provoke the righteous wrath of sturdy men, and they are quite out of taste in that most home-like of all gathering places, the mother's room. For the name of that chamber should always be Peace and Comfort. It should be of all bedrooms the most commodious, the most convenient of access, with the largest of drawers, the roomiest of closets, the most restful of chairs, and a boundless welcome to all the household.

Closet room should be struggled for in the building of a house. This is a point where the masculine intellect shows its weakness and the feminine its strength. A quick-witted woman will suggest to her architect, nook after nook of waste space to be utilized as closet room which would altogether escape his notice. No bedroom should be unfurnished in this regard. When closets are not built in, portable wardrobes should be supplied.

There is fallacy in the supposition that the most attractive portion of the house should be reserved as a "spare room" for the casual guest. The family should first be made comfortable; when that has been done, if one would use hospitality without grudging, it will be necessary to imitate the great woman of Shunem, and at least furnish a little chamber with the necessary bed, table, stool and candlestick. Moving out of one's own room and doubling up with another for a night or two does very well in the holiday season, when the spirit

of hospitality and good nature is in the air; but, ordinarily speaking, it is quite a task to empty the upper drawer of one's bureau, and leave one's own comfortable quarters.

So far as health, neatness and style are concerned, brass bedsteads are the best. They are very simple in form and construction, and so are some of the iron bedsteads, which can be kept absolutely nice and clean in any climate, and are, unlike brass, quite inexpensive. The most objectionable of all bedsteads is that

“Contrived a double debt to pay,
A bed by night, a chest of drawers by day,”

which is only to be tolerated where a parlor must serve temporarily as sleeping room. A well made bed is the essential piece of bedroom furniture, which may be hidden from view by a screen or curtains, but should not be slammed up and boxed in against the wall, or made to stand upon anything but its own merits.

Wire net springs are probably as good as can be got, and a feather bed under the mattress is an improvement to the best modern bed, if properly aired, turned and shaken daily. Mattresses should be remade and their contents pulled lightly apart before they grow matted or ridgy. Curled hair mattresses are, of course, the best, but English flock, excelsior, and straw, all make respectable beds, and can be made easier by covering them with thick comfortables or blankets, under the sheet. It is quite worth while to make slip covers for mattresses.

Sheets should have an allowance of at least three quarters of a yard for tucking in. Three yards will not be found too long for comfortable home sheets. Blankets are apt to be too short. It is better to tear a pair of blankets apart, and finish the edge with a buttonhole stitch in worsted. The old fashioned “blanket stitch,” as it was called, a long and short stitch alternating, is very pretty. This finish is better than binding, which is apt to shrink and tear off.

It seems a waste of time to make cotton patchwork when pretty quilts can be bought so cheap. In the days when cotton cloth was costly, every scrap was

worth saving, but now patchwork seems only serviceable in teaching little girls to sew overhand seams.

Hand-wrought spreads look well when pulled up over the pillows, covering the whole bed, and should be treated with respect and carefully folded and laid away at night.

Pillow shams are troublesome to keep in place, and can be discarded without regret, when a pretty covering of this sort conceals the whole bed. Of my own choice I would never make use of anything with so disreputable a name.

The fault of crazy quilts is their craziness. To be really pleasing they should have some design, like a Turkish rug which, though very irregular in detail, has yet a general plan, a distinct centerpiece, and a plainly defined border. One of the most objectionable features of the ordinary “crazy” quilt is the huddling together in the same piece of work of painting upon silk and embroidery, two widely differing sorts of decoration, which will not bear being brought heedlessly in juxtaposition.

Very pretty comfortables, to be folded like a silk quilt, and thrown over the foot of the bed, can be made of paper muslin, in dainty colors, or of cheese-cloth, lightly filled with cotton batting, and knotted with bright colored wools. Cotton comfortables are not so serviceable as blankets, but they are much cheaper, and it is well to keep a supply on hand for use in cold winter weather, or to make up an extra bed with in case of emergency. They can be folded under the sheet to soften a hard mattress, or white palliases filled with cotton can be made for the same purpose, but great care should be taken to air bedding of this sort very thoroughly.

A roomy lounge in a bed chamber is a great convenience. It affords an opportunity for an afternoon nap without disarranging the well made bed, and many a careworn woman would lie down for a few minutes upon a lounge in her bedroom who would not think of resting in the daytime upon the bed. A long, broad, pine box, with wooden castors attached, makes an admirable lounge frame, or a narrow cot bedstead could be cut down to be of suitable height for a lounge frame. This should be supplied with a good mattress, or a covering of chintz or cretonne could be drawn over it, with a frill falling nearly to the floor. From one to three square pillows, similarly covered, would perfect this lounge, which could serve readily for a bed in time of need.

Bed hangings and canopies are pretty and unnecessary, except in mosquito countries, where lace net, gathered full upon a hoop suspended horizontally from the ceiling, and falling in ample folds to the floor, will serve to keep many out, and one or two teasing marauders in, the long night through. Bed hangings proper are prettiest when made in the form of a canopy over the bed head, and should be of a material that will bear washing.

An ample supply of choice bed linen and towels, all handsomely marked, is no less a subject of pride with housekeepers than dainty table damask, and people of wealth in these days spend lavishly upon hemstitched linen or silk sheets, elegant towels, and elaborately embroidered letterings.

This fondness for well stocked linen presses is a womanly and pardonable weakness, inherited from our far away ancestresses, who strewed stalks of lavender between the sheets in their chests and presses, a custom that has not gone altogether out of date among old fashioned European housekeepers.

Other comforts of the sleeping room where bath-rooms are not attached are plenty of water and bath towels, a washstand for each occupant of the room, generous bowls, a well filled pail with which to replenish the pitchers, foot tub, a portable bath tub, capacious slop jars, a rubber or enameled leather cloth to spread upon the floor, a screen for seclusion's sake, and room to splash. If the bedroom china, pails and jars be pretty in shape and color, so much the better, but at any rate, let them be large enough.

A wooden topped washstand should be protected with a piece of enameled leather, over which a plain towel can be spread for look's sake. Fanciful fringed and colored mats are out of place on the washstand, where water should be free to spatter.

Where "splashers" are used to protect the wall, they should be simple of design and easy to wash, and mottoes, if introduced, should be appropriate to the place. "Sweet Rest in Heaven," which I have known used for this purpose, can hardly be considered suitable; nor yet the prophet's command to the leprous Syrian captain, "Wash and be clean," a too suggestive motto, wholly subversive of the theory that bathing is a luxury indulged in for refreshment's sake; nor yet again a representation of birds dipping into a stream, with the scriptural allusion to the Good Samaritan's washing and binding up of wounds, "Go and do thou likewise." These sentences might be appropriate in

the accident ward of a charity hospital, but hardly suit the wall decoration of a lady's dainty bed-chamber.

Something more suggestive of the sparkling, limpid purity of the crystal spring would be in better taste—such as:

“See the face of Laughing Water.”—*Longfellow*.

“She sprinkled bright water from the stream.”—*Shelley*.

“From a thousand petty rills
That tumble down the snowy hills.”—*Milton*.

There can be no lack of good mottoes to those who look for them.

A roomy, deep drawered bureau is best for a woman's use, a dressing table or bureau with small and large drawers for a man. There should be looking glasses suited to the needs of each, but for a lady it is convenient to have a glass so placed as to reflect her full figure, so that she may judge of the “hang” of a skirt or the looping of dress drapery.

A candle-stand by the bed, with candlestick and matches, a table or desk for writing purposes, chairs low enough for sewing or lounging in, and a big, old-fashioned stuffed chair for the solace of the sick, these are all bedroom comforts.

I have said nothing of servants' rooms, though much might be written on the thoughtless neglect which generally makes such rooms unpresentable. I can recall to memory but one house containing a model room for servants' use. That bed-chamber was as exquisitely nice in its appointments as the room of the mistress herself, though its furnishing was, of course, much less costly.

Boys' rooms, also, especially in country homes, are apt to be cheerless, neglected spots, wholly unattractive to their occupants. Boys ought not to be burdened in their rooms with the care of those little prettinesses in which their sisters delight; still they should be educated to enjoy what is truly refined and beautiful. Their bedrooms should be tasteful and comfortable, and they should be taught to keep them in order, to hang clothes tidily in the press, to lay away

neckties carefully in the drawer, and to take pride and pleasure in making their rooms attractive to themselves and their young friends. They should be encouraged to feel at home in their rooms, and if no attic or shed room can be given up for their boyish gatherings, for whittling, tinkering, kitemaking, and other important youthful manufactures, to say nothing of choice collections of sticks and stones, then banish the carpet, retaining only a warm rug before the bed, and let them make whatever clutter their legitimate pursuits involve, so long as they are rigidly required to right all disorder when the work is done. Free permission to carry on such innocent occupations within his own domain, with a kindly winking of the maternal eye at an occasional pillow fight, would tell more as a means of grace on the boy who now slips out of the house to find doubtful recreation elsewhere, than a whole barrellful of Sunday sermons. The boy who has once learned to take pleasure and pride in the appointments of his own room will want some time for enjoying them, other than the hours spent in bed, and as he of choice lives more within the walls of home, and enters more and more into the spirit of home he will be so much the more likely in his turn to be one day the master and joint possessor of a homelike house.

“CONSIDER THE LILIES.”

BY MRS. MARY N. EVANS.

Pure white the lily's cup!
Reminding thee of those whom God hath clad
In linen robings—fine, and clean, and white—
The righteousness of saints!

Perfumed, the lily's cup!
Reminding thee of incense-breathing lives
Whose fervent prayer, whose loving words and deeds
All earth with fragrance fill!

With honey stored, its cup!
Reminding thee of pure heart-chalices
Whose limpid depths are wells of heavenly love
Whence all who will may draw.

The lily's cup shall fade!
But withered flower and body death-destroyed,
Or Spring, or Angel-trump shall wake again
To resurrection, life!

This, then, the lily saith:
“Pure, perfumed, stored with sweetness be *thy* life,
On earth a benison—then, glory-crowned,
Immortal be in heaven!”

A BIRD'S-EYE VIEW OF FORESTRY.

BY THE REV. S. W. POWELL.

In this country many people have only just begun to know that there is such a thing as forestry. Very few understand clearly that it is both a science and an art, and is one of the most important subjects which can come before the mind. For convenience we will treat it under two heads:

A, the value or benefit of forests; and

B, How shall we get the greatest benefit from them?

A. We may regard forests:

I. As yielding a vast amount of products necessary for civilized people.

II. As efficient in preventing certain evils, such as:

(a) Washing soil from hillsides;

(b) Depositing this material where it does great and lasting mischief;

(c) Droughts and floods;

(d) Harm done by drying, chilling or malarious winds;

(e) The shifting of wind-driven sands which, when not held in place by forests, often cover and ruin fertile land, and even bury fences and buildings;

(f) The undue multiplication of insects harmful to vegetation.

III. As beautifying a region and affording healthful retreats for tired and sick people.

But forestry treats not only of this use and value in the woods, but also tells us:

B. How to make them of use: that is, how to manage forest property so as to make it yield the greatest benefit *in the long run*, to the individual owner, to the community, and to future generations. This involves the study of a great many questions which we may classify as follows:

I. On what kinds of soil and in what situations shall we keep or plant trees?

II. What kinds of trees shall we raise in any particular place?

III. At what age, and in what way shall we cut the trees of each kind in a given region?

IV. What are the methods of marketing forest products which will secure the greatest profit?

V. How shall we protect trees from disease, from robbery, and from fires?

Few Americans have studied forests with any other design than that of getting from them the greatest possible amount of *immediate* profit. Scarcely anywhere has care been taken to so use them that they should continue to yield their many sided benefits to succeeding generations. Neither have they been regarded as of much use in the *present* except as sources of certain products, such as lumber, timber, tan-bark, charcoal, turpentine, resin, tar, wood-pulp, etc. As a rule, no consideration has been given to the effect they have upon climate, rainfall, droughts, floods, health, or the beauty and attractiveness of a region.

The first settlers cleared off, in the quickest and cheapest way, great forests of the finest trees which, if standing now, would be worth far more than the ground on which they stood can ever be worth for farming. These splendid forests of species of timber that now bring a high price—from \$45 to \$150 per thousand feet for the best quality—were cut down, hauled together, skidded up

in piles and burned to get rid of them. And this was called *improvement* of that land!

They often cleared in this way steep hillsides which, after yielding two or three good crops by means of the rich vegetable mould that always accumulates under a forest, were almost worthless, even as pastures, and entirely so for tillage. As a result, in large regions so *improved*, springs and brooks fail in the dry season, and in a wet time floods become more and more destructive. Had these hillsides been kept as forests—that is, cut over in such a way as to ensure a new growth of equally good trees—they would have kept on affording in winter steady and remunerative employment; springs and streams would have preserved a more even and permanent flow; climate would have been more favorable for the production of all kinds of crops, and especially of fruits; men and animals would have enjoyed better health; and regions now barren, uninviting, and thinly inhabited by poverty-stricken and unambitious people would furnish a good living to large and vigorous populations, and would beside be attractive to summer visitors in search of health or recreation.

In a word, we may say that forestry—using the term as meaning the science and art of getting from the woods the greatest and most lasting benefits—has never been studied except by a very few of our people. One reason for this is that until quite recently all forest products have been abundant, and the injury to water supply, health, farming, manufacturing and navigation resulting from the destruction of the woods has only just begun to appear.

Besides, these mischievous results are not by most folks assigned to their true cause, *e. g.*, certain parts of the lower peninsula of Michigan are far less adapted for raising wheat, corn, clover and peaches than they were before, to so great an extent, the great sheltering forests of that State were cut off.

A commission appointed in 1867 by the legislature to examine the subject, reported that for forty years the winters had been growing more severe; and that thirty years before the peach had been abundant, and the crop rarely failed, frost being unknown from May to October; but that at the time of the report it was very uncertain on account of unseasonable frosts. The further statement was made that:

“The destruction of the wheat as well as the corn crop is becoming a matter of great anxiety to our farmers in many sections, and the winter-killing of the

clover in the eastern part of the State last winter, not by 'heaving,' but, apparently, by being frozen dead in the ground, as it appears black and rotten in the spring, may be another proof of climatic changes of great significance to the farmers and the dairymen."

It was estimated that the damage to winter wheat by its exposure to cold, wind and sun for want of its former usual covering of snow, caused a loss of half the crop, or 5,000,000 bushels in a single year. (United States Department of Agriculture, Report on Forestry, 1877, p. 271.)

Yet, notwithstanding this evidence of the injury done by forest destruction, it aroused no such general demand for the preservation of at least shelter-belts across the tract of injurious winds that the legislature felt obliged to interfere and secure such preservation. Since 1867 the destruction of the forests of Michigan has gone on much faster than before.

Prof. B. G. Northrop, of Connecticut, who has done so much to promote the observance of "Arbor Days," says that in visiting the regions where the floods in the Ohio river in 1883-4 did so much damage—that of 1883, it was estimated, destroyed \$60,000,000 worth of property, beside a great many lives, and that of 1884, which was five feet higher, did less harm only because that of 1883 had left less property within reach—he often met, even among the sufferers, a doubt or denial that cutting away the forests on the head-waters of the Ohio had much effect in causing the floods.

Another reason for our apathy is that people get used by degrees to changes in climate, springs and streams; in the adaptedness of a region to raise fruit, vegetables, grain or stock; or in the price and quality of forest products. In all these respects, large portions of the country are already suffering great loss, but most of it has come on gradually. Of late years, however, so much has been said in the papers and magazines that floods, drought and injurious changes of climate are more generally attributed to forest destruction. But people do not seem as yet to be very uneasy about the waste and destruction of the valuable material afforded by the woods. We had so much when we received this continent from God's hand that it never seemed as if we could suffer from lack of it. In fact, few people realize how great is the money value of what every year we draw from this bank. Most folks would be greatly surprised to learn that what we get from the woods is worth more than any other one crop. No one yield of cotton, corn, wheat or hay is worth so much in dollars and cents.

In 1880, the last census year, these products were worth *the enormous sum of \$700,000,000*, which is one and two-fifths times the value of our breadstuffs; two and one-eighth times that of the meat we raise; two and one-half times that of all the steel and iron we make; almost three times that of the woolen, and three and one-eighth times that of the cotton goods turned out by our mills; more than three and one-half times that of the boots and shoes; four and one-half times that of the sugar and molasses; *eight and one-fourth times our total outlay for public education*; ten times the output of our gold and silver mines, and three times that of the entire product of coal and ores of all sorts.

Now, this immense sum is what the raw materials afforded by the forests are worth. But these raw materials are themselves the necessary foundation of a vast number of the most important industries, such as the manufacture of furniture, wagons, agricultural implements, railroad cars, pianos, organs and other musical instruments, house-building, etc., etc. It would be a useful exercise to write out a list of the trades and occupations one can think of which must have forest products, or something more costly, as their raw material. We should find that those industries which depend directly on these products include the most important ones, while every branch of manufacture and every kind of work is indirectly dependent on them. Different branches of industry are more and more interwoven with and dependent upon each other, as civilization advances. The greater the number of parts entering into a machine, the greater is the loss from the stoppage of the whole if any one breaks down. A chain is no stronger than its weakest link.

Another reason for our indifference is that most things made from forest products are, as yet, cheap, because improved methods and machinery, together with sharp competition, have lessened the cost of finished articles. Men employed in getting these products out of the woods and carrying them to the consumer, are constantly devising more efficient methods of work. A modern saw mill, planing mill, sash and blind or furniture factory, is as much more efficient than anything known fifty years ago as an express train is better than a stage coach. Then, too, wherever the ground is moderately level, the narrow-gauge railroad often takes the place of the old fashioned sleds or trucks drawn by teams. This makes loggers independent of high water for floating their logs. A train of twenty-five cars, containing 40,000 feet of logs, is on the average loaded in seventy-five and unloaded in nine minutes, and the train will run, one day with another, 160 miles. By this means much timber is

now reached that grows so far from streams that it would not have paid to carry it to the mills by the old methods. A dollar or a day's work will, by means of these contrivances, accomplish so much more now than it used to, that under the pressure of competition, most of the finished articles made of wood in whole or in part, are sold cheaper than formerly. But really good lumber and timber in the tree or log is very much dearer, and this because our enormous consumption is exhausting the stock. But since people in general are impressed by what a finished article costs when they buy it, we are not likely to be goaded into the necessary measures by feeling the lack of forest products until the greater part of our woods have been used up. Nothing but agitation and educational work, such as that done by "Arbor Days," will arouse us in time to prevent the destruction of our forests.

Before leaving this part of our subject, which has to do with the value of the forests as sources of valuable products, it may be well to say a word about the important matter of forest fires. All experts agree that they consume at least as much as the axe and saw, and that is not less than \$300,000,000 worth a year, which is about \$5.00 for every man, woman and child in the country. But the indirect damage they do by preventing the proper care of old, and the planting of new woodland, will, quite possibly, prove greater than that which they do by destroying what we already have. The most profitable tree culture is that which produces mature and good timber, because it always will command a high price; while there is now, and probably for a long time there will be, a large supply of, and a low price for, immature and cheap timber. But to raise mature trees, we must wait longer for the profit on the time and money expended. Most of our hasty people want quick returns, and if anything makes the long investment risky, these two objections—delay and risk—will weigh more than any arguments that can be put into the other side of the scale, and it is so hard with our present laws and habits to keep fires out of the woods, that it makes it hazardous to spend time and labor in keeping and caring for trees long enough to get the best timber from them, and therefore few undertake it.

II. But we were to give a glance at the usefulness of forests as *preventing certain evils*. Taking these up in the order named, we come first to (a) *The washing of the soil from hillsides*. It was estimated by the exact and cautious George P. Marsh ("Earth as Modified by Human Action"—a book that no one can afford not to read—pp. 282-3), that during the last two thousand years there has been washed away from the portion of Italy drained by the river Po, enough soil to raise the *entire surface forty-five feet*! Had the woods been left

on the steep hillsides as they should have been, most of this havoc would never have occurred. For lack of that soil many districts once fertile are barren, and much of the material of which they have been robbed has been deposited where it is an almost unbearable nuisance. *E. g.*, it has little by little raised the bottom of the Po itself, and the dykes have been built higher and higher to keep the river from flooding the plains through which in the lower part of its course it flows, so that it runs far above the surrounding country, in a sort of aqueduct. The same thing has occurred in the lower part of the Mississippi. From the deck of a steamboat, for a long distance above New Orleans, one looks down on the plantations. This elevation, of course, makes the pressure of the water and the cost of keeping up the dykes, or levees, greater every year, and when a break occurs in time of high water, of course it is more destructive, because it pours down from a higher level.

Now, were all the steep land in the Mississippi valley kept covered with trees, as it should be, this enormous amount of sediment would not come down to raise the bottom and the dykes, and to do much other harm. In time it may break through its banks and make new channels for itself, leaving important towns high and dry, and, of course, destroying much property where its new route is cut. The vast mass of vegetable mould, ashes, etc., which will be washed down into the Hudson—if the forests of the Adirondacks are ever destroyed by fire, as there is danger that they may be—by the great floods which denudation of those mountain sides will often cause, will quite possibly ruin the navigation of that river and the harbor of New York, not to speak of the destruction of farms, factories, and towns lying where those floods can reach them.

Next in our list of evils prevented by forests come Droughts and Floods (*c*). The annual supply of water from rain and snow, if held back by woods on steep hillsides until it can soak down to the underground sources of springs, or if stored up as in a sponge by the mass of fine roots, dead leaves, decayed wood, moss, etc., which often accumulate on the surface under old forests to the depth of two or three feet—may be made to last through a whole summer. There falls from the clouds—one year with another—only a certain amount of water in any particular region. If there is nothing on the hillsides to hinder its rush down into the streams, it is lavished like the money of a spendthrift, where it does little or no good, and very likely much harm. The difference between the two is that between feverish, riotous waste and sober plenty.

“Waste not, want not,” is as good a maxim for the management of water as for that of cash.

A torrent is a stream liable to extreme and sudden increase and decrease—usually very small or quite dry in a dry time, but liable to rise suddenly to a great height, and as quickly to shrink to its former size. By the loss of its once rich forests, the Ardèche, a tributary of the Rhone, became such a torrent, its principal branch often being entirely dry. It has been known to rise sixty feet and dwindle back to almost nothing within a few days. The upper Hudson has apparently all the conditions necessary for becoming such a torrent if once its forests are exterminated. It descends some 4,000 feet in a short and steep course, from a region where there falls a great deal of rain and snow.

As the headwaters of this important river, unlike those of the Ohio, lie almost within the limits of a single state, and the control of a single legislature, great efforts have, within the last two years, been made to secure the appropriation of the great Adirondack region, which is entirely unsuited for farming, to be kept forever as a forest. It has been objected that this would cost too much, and that if such laws were enacted as would enable the State (which now holds mostly by tax title some 750,000 acres) and the permanent owners of large tracts to protect their land from fire and timber stealing, those who hold smaller lots, and do not care to keep them after cutting off the spruce and hemlock they contain, would let them pass into the possession of the State by non-payment of taxes. But a recent judicial decision renders it very doubtful whether in this way a sufficiently valid title could be secured. There are strong arguments against any effective measures. Most men who have invested money in lumbering, tanning and pulp mills, and iron works requiring charcoal, have been accustomed to carry on their work in such a way that it has made the destruction of the woods liable or almost certain to occur sooner or later; and they are not willing—indeed, they declare they are not able—to go on with their business if there is added the cost which would be involved in the changing to safer methods. They are certainly mistaken in their violent opposition to legislation which aims to protect the woods. They would, were a proper system of forestry once put in practice, find that their tracts of land would yield so much more, and so much better material, and that their losses from fires, floods, etc., would be so diminished, that in the long run they would be gainers, and at any rate the damage which would result from the denudation of those mountains would be so vast and so lasting that all the

possible cost of paying these men a fair equivalent for any loss such protective measures might occasion would be a mere trifle in the comparison.

As to the value of forests in preventing damage done by drying, chilling or malarious winds (*d*), there can be no doubt that it is very great. It is probable that all through the region between the eastern boundary of the Indian Territory, Kansas and Nebraska, and 105° west longitude, dry winds from the south and west are very detrimental to both vegetable and animal life. If any species of trees can be made to grow there—and by doing this over areas large enough to warrant the cost of irrigation and other protective measures the undertaking might succeed where it would not if attempted on a smaller scale—it is very probable that in the line of such belts of timber other species and many crops might thrive which can not now be raised.

It is certain that everywhere in the northern prairie states a grove that breaks the force of the cold winds from the north and west adds greatly to the value of a farm. And it is gratifying to learn that so much tree planting has already been done in those states. Some have even been so sanguine as to predict that when the soft and hard timber of Michigan, Wisconsin and Minnesota is gone these new prairie states may be able—at least partially—to supply the needs of those just named, whose forests were confidently asserted to be inexhaustible.

The effect of certain trees—indeed, of almost any—as fences against malarious winds has been carefully studied in France and Italy, and the verdict is that it is very great.

Marsh (“Earth as Modified by Human Action,” p. 159) says: “It is well known that the great swamps of Virginia and the Carolinas, in climates nearly similar to that of Italy, are healthy even to the white man, so long as the forests in and around them remain, but become very insalubrious when the woods are felled.” He quotes Jules Clavé, a French expert, as authority for the statement that “the flat and marshy district of the Sologne, in France, was salubrious till its woods were felled. It then became pestilential, but within the last few years its healthfulness has been restored by forest plantations.” Marsh also thinks that in Germany and India belts of trees have been found very beneficial in warding off cholera. A lumber journal recently asserted that cholera has never prevailed in pine-producing districts.

Lanisci says that in the time of Gregory VIII. (who came to the papal chair in 1572 and reformed the calendar) Rome became much more unhealthful when a pine forest lying to the south was cut down because infested by brigands. The abbey of Trois Fontaines, considered one of the worst places in the fever infested Roman *campagna*, was much improved in three years by plantations of the Eucalyptus, and this tree has been used with good effect for the same purpose in the French settlements in Algeria (Hough, U.S. Forestry Report for 1877, p. 285).

Of the service rendered by forests in preventing the drifting of sands (*e*) the most remarkable instance is afforded by the once dreary regions in the extreme southwestern part of France, where plantations of the maritime pine have, in the departments of the *Landes* and *Gironde* transformed over 4,000 square miles of poverty-stricken country into populous hives of an intelligent and thrifty population. In the lower part of the valley of the Wisconsin river, much loss and inconvenience is experienced by the drifting of the sand which, driven by the prevailing west winds, covers and ruins fields and gardens, and in many cases, even fences. A few belts of timber running across that valley would be worth many times their cost in preventing this nuisance.

Woods prevent the increase of noxious insects (*f*) in two ways: They shelter birds, nature's great insect-police, and they stop the progress of many species, such as the grasshoppers, which scourge some of the western states, and the chinch-bug, so much dreaded by wheat growers. It is said that the latter pest never traverses a belt of thick trees as much as seven or eight rods in width. So, too, it is affirmed on apparently good authority, that winds carrying the fungus called wheat-rust deposit their baleful load if they find a forest in their track.

III. Scarcely any room remains to speak of the important service which forests render in beautifying a region. Besides preventing the disfiguring ravages of wind and water, they add a positive element of beauty. No one accustomed to the palpitating glow of autumn color in the Berkshire Hills of Massachusetts; or to the more subtle attractions of the shifting half-tints in which spring drapes budding trees in the same region; or to the splendor of a wooded mountain side with the diamonds of an ice-storm glittering in the sun; or to the restful coolness of a dark hemlock grove in July heat, can ever feel quite at home in a treeless region.

B. How shall we manage forest property so as to make it yield these benefits and ward off these evils? The questions arranged under our five heads have for more than a hundred years been exhaustively studied in Germany and France. Germany thinks that before a man is put in charge of the immense interests which center around her forests, he needs from ten to fourteen years of hard work and study after he has as much education as the average graduate of an American college. The science elaborated in these schools, and in the forests under the charge of their graduates, is embodied in a large, learned and rapidly growing literature. The application of these principles to any particular region must be learned upon the spot. For this purpose, one of the first things which our national and state governments should do is to establish in different parts of the country experiment stations connected with large tracts of land.

At these stations we could work out specific answers to the questions suggested under our second main division (B). It is plainly impossible in the limits of a single article to do more than give a hint of some of these practical questions. Not even the most expert German forester could give adequate answers until he had before him the results of years of experiment conducted in different parts of the country by able men. Even then such answers would fill many books.

The end aimed at in this paper has been reached, if the numerous readers of *THE CHAUTAUQUAN* get from it some distinct impression of the magnitude of the interests with which the science of forestry has to do, and of the pressing need that the American people begin at once, and in earnest, to protect, to improve, and to extend the forest estate which up to the present we have been so heedlessly wasting.

GOVERNMENT EMPLOYMENT FOR WOMEN.

BY MRS. PATTIE L. COLLINS.

Since the appearance of my first article on this subject, in the October number of THE CHAUTAUQUAN, I have received letters, asking further information, from New York, Connecticut, Massachusetts, Virginia, Maryland, Iowa, Indiana, Missouri, Ohio, Pennsylvania and Wisconsin. I have tried to answer all of them, but to none could I return more than a brief, and I fear unsatisfactory reply. I therefore resume the subject in this public way, with the hope of furnishing full information to all who are interested.

My former article was much criticised by my friends, scarcely one of whom agreed with me entirely. Some said that my view of the Civil Service was an ideal one—as it should be, not as it is—and others that I wrote from the standpoint of a successful (!) woman. A person high in authority gave me food for reflection in the following inquiry: “Do you not think that in justice to those who wish to apply for examination, you ought to say how many have already successfully passed, for whom places can not be found?”

This was a new phase of the question, and quite staggered me. With my accumulating correspondence around me, I began to feel as if I had ignorantly committed a serious blunder which I must lose no time in explaining. The fact that the Commission already had on file the names of many women, with a bare possibility of the list being exhausted at a remote period, was an aspect of the case which I had not adequately considered.

The reason for the great excess in the number of women’s names over those of men left on file, is officially stated to be that, whenever vacancies occur in any of the departments a demand is usually made for a male instead of a female clerk. I think that I am not mistaken in saying that appointments have hitherto been made in the ratio of six to one. The Commission has been allowed no discretion. When a demand was made for a male clerk, a female could not be supplied. With my observation and experience among department clerks, this appears to me an unjust discrimination. I have often asked the

reason for this of those who ought to be able to furnish intelligent opinions upon the subject, but have never succeeded in eliciting anything satisfactory. One general argument used is that it is more difficult to enforce discipline among women. This, however, is such a weak statement, though oft repeated, that I am not inclined to discuss its merits or demerits. A really strong point is made in their average loss of time. It is claimed by some of the aggrieved class that this would be rectified by a just apportionment of salary as it would relieve them of much additional labor after office hours. I am not prepared to suggest the remedy, I only desire to speak of the service as it exists. Whenever a proposition is made to introduce a woman clerk into one of the cosy, well-furnished apartments occupied by three, four or six clerks of the opposite sex, it is met by a prompt and indignant protest.

Why? Well, this is a question I will answer myself, and if it be treason, I am absolutely indifferent to the consequences. Pass along the corridors and glance through the half-open doors into these pleasant offices; observe the occupants engaged with fresh newspapers, their chairs tipped back, barely preserving a perilous balance by making footstools of the desks; observe also the clouds of fragrant smoke; if the weather be warm, graceful negligé costumes will not escape you; a vacant chair, perhaps many vacant chairs will be sadly suggestive to the unenlightened, but if you inquire, the answer will be reassuring: "He has only stepped out" (for an hour or two). Paradoxical as it may appear, the affairs of an entire department may be administered while the \$8,000 head, accompanied by his bureau officers, is fishing in the Adirondacks, making an excursion to Florida, crossing the plains, or summering at Mount Desert; and all along the line there are delicious masculine privileges which the ambitious female must not even cherish a profane desire to possess. In the light of these exclusive dispensations I have often seriously considered the tenets of the Arcadian settlement upon the bayou Têche. No woman is allowed to enter its peaceful borders who can read and write, consequently they are lovely cooks and have no wrinkles. The difficulty seems to begin when they are allowed to master the alphabet—after that there is no stopping them. In a word, the intelligent, deft, skillful, conscientious women workers of the departments have trespassed alarmingly upon the ancient, solidified rights of the original "lords of the territory," and that, too, just as they were most aggressively and menacingly arrayed. The passage of the Civil Service act was "a trumpet pealing news of battle above the unrisen morrow"—to women. Visions of the beautiful "equal work, equal pay" doctrine rose thick and fast beyond the dark limit line of a hitherto

narrow horizon; visions of regular promotions for well performed work. Alas, instead of this, the departments are permitted to discriminate in favor of male clerks, and no laws have yet been enacted in reference to promotion. In view of these two truths perhaps I ought to take back all that my former article contained. If the operation of the law is to practically exclude women, while it offers no opportunities of advancement to those already within the gates, certainly I owe it to my readers to admit it. But this admission—even with the six to one, or even greater ratio confronting me, I am not prepared to make. I have shadowed forth the reasons for this state of things, and I shall continue to hope and to believe that wise legislation will regulate the system, and time will adjust it to harmonious working order.

In reading the complete and exhaustive work upon the Civil Service of Great Britain, by the Hon. Dorman B. Eaton, chairman of our own Commission, I find that when women were first employed in that enlightened land they were considered too dishonest to be trusted with letters that had not been first cut open by men clerks, and all valuables removed! They were also under the constant surveillance of a “matron.” Now they fill numberless places of trust and responsibility, and a large class of earnest, self-reliant, independent women has been the outgrowth of an experiment which had this puny, almost despicable beginning. Therefore, despite facts and figures, we need not be discouraged in this country. It ought to be comforting to those clamoring upon the outside that there are friends and allies within. It is said that there are twelve hundred women in the Treasury alone, while those in the other departments must aggregate many hundreds more. A certain historian affirms that the Bourbons mainly owed their restoration to the throne of France to the following message sent by Prince Talleyrand to the allied sovereigns and their generals: “You may do everything and you dare nothing; for once, then, be daring.” I repeat the warlike sentiment to women desiring work here. If you do not ask for it, assuredly you will not receive it. If you do ask—you, or you, my sister—from the North, East, South or West, may be successful. Some of you certainly. In direct response to questions which have been asked me by numerous correspondents, I make the following statements: Each state and territory is entitled to a quota of appointments based upon its population, without reference to present incumbents who were appointed under the old system. Examinations are held in various places, thereby affording all who desire an opportunity to attend. A letter addressed “Civil Service Commission, Washington, D. C.,” containing a request for information, will obtain a printed form containing full particulars as to every

step of procedure. Age, moral character, previous employment, where educated, etc., must be testified to under oath, and each applicant must be endorsed by not less than three nor more than five reputable persons to whom he or she must be personally known. All applicants will be informed when and where the first examination most convenient to them will be held. A thorough, rudimentary education is all that is necessary for the \$900 examination. For the \$1,200 places something more is required; a knowledge of book-keeping, and our civil government, and wider historical and geographical information. There are special and technical examinations for linguists, mechanical experts, etc. The ordinary work—by which I mean all that is usually performed by clerks of the lower grades—is never of an intricate character. It is easily learned, whether it be keeping books of records, copying or briefing letters, counting money or reckoning percentages. I do not wish to convey the impression that there is any child's play in the matter—far from it. There is an abundant field in which industry and intelligence can exhibit themselves advantageously. The extracts appended are from the first annual report of the Commission, submitted to the President in February, 1884. I have endeavored to outline the difficulties that lie all along the way, and to show that long waiting may follow even a successful examination. I am not conscious of being even at heart a strong partisan, and think that I have presented a calm and impartial statement. In conclusion, I may add that although good penmanship may be considered as a rather low accomplishment, it is an important factor of success in this special direction.

APPENDIX No. 5.

The questions below are an example of those used in the grades which fall under the first and fourth clauses of Rule 7, known respectively as the general and limited examinations. They are a fair sample of all those used in those grades. It is at one or the other of those grades that fully 95 out of every 100 applicants have been examined under the rules. The questions are frequently varied, indeed almost at every examination, without materially changing their grade, and there are special adaptations of them to various places in the postal and customs offices. For the sake of brevity the ample spaces for the answers on the examination papers are omitted.

GENERAL EXAMINATION UNDER CLAUSE 1 OF RULE 7.—FIRST SUBJECT.

Question 1.—One of the examiners will distinctly read (at a rate reasonable for copying) fifteen lines from the Civil Service Law or Rules, and each applicant will copy the same below from the reading as it proceeds.

Question 2.—Write below, at length, the names of fifteen states and fifteen cities of the Union.

Question 3.—Copy the following, which is section five of the Civil Service act, in the blank below:

SEC. 5. That any said commissioner, examiner, copyist, or messenger, or any person in the public service who shall willfully and corruptly, by himself or in coöperation with one or more other persons, defeat, deceive, or obstruct any person in respect of his or her right of examination according to any such rules or regulations, or who shall willfully, corruptly, and falsely mark, grade, estimate or report upon the examination or proper standing of any person examined hereunder, or aid in so doing, or who shall willfully and corruptly make any false representations concerning the same or concerning the person examined, or who shall willfully and corruptly furnish to any person any special or secret information for the purpose of either improving or injuring the prospects or chances of any person so examined, or to be examined, being appointed, employed, or promoted, shall for each such offense be deemed guilty of a misdemeanor, and upon conviction thereof, shall be punished by a fine of not less than one hundred dollars, nor more than one thousand dollars, or by imprisonment not less than ten days, nor more than one year, or by both such fine and imprisonment.

SECOND SUBJECT.

Question 1.—Multiply 307968 by 490875 and divide the product by 307968. Write in full the operation.

Question 2.—Divide three fourths of eight ninths by one seventh of three fifths and subtract one seventh from the quotient.

Question 3.—Divide one thousand and eight and three one-thousandths by three and eight one-hundredths, expressing the process in decimal fractions.

Question 4.—The compensation of a clerk, beginning June 30th, was \$133.33 a calendar month. On the first of October his salary was increased 15 per cent., and so remained until June 1st, when it was increased a further amount of three per cent. on the original salary. What was the whole amount payable to the clerk for the year?

Question 5.—A commissary suddenly forced to change quarters had on hand 980 bushels of wheat which cost 80 cents per bushel. He sold six per cent. of it at a loss of four per cent. and four per cent. of it at a loss of three per cent. How much was the whole loss incurred by the sale?

THIRD SUBJECT.

Question 1.—A note for \$2,647.34 is payable eleven months from date with interest at $3\frac{1}{2}$ per cent. What will be the amount due on the note at maturity? Give all the figures in the operation.

Question 2.—A disbursing agent failed, owing the government one item of \$308.45, another of \$2,901.02. The government agreed to make a discount of 13 per cent. on the first item and $11\frac{1}{2}$ per cent. on the second. How much was payable under the agreement?

Question 3.—June 30, 1880, A gave B a note for \$1,005 payable July 4, 1882, with interest at 4 per cent. May 1, 1882, A paid \$235. What was the amount of principal and interest due B when the note matured?

Question 1.—A contractor furnished the government articles as follows: June 8, 1880, 300 barrels of flour at \$4.50 a barrel, and July 6, 1880, 187 yards of carpet at \$1 a yard. August 4, 1880, 1,000 yards of carpet at 87 cents a yard. The government paid on account as follows: June 12, 1880, \$1,000; July 10, 1880, \$100; August 4, 1880, \$500. State the dealings between the parties in the form of a debit and credit account, showing the balance due.

FOURTH SUBJECT.

Question 1.—Give a definition as full as the space will allow of (1) a verb; (2) a noun; (3) an adverb; (4) an adjective; (5) a preposition; (6) a conjunction; and of (7) the phrase, “the grammar of the English language.”

Question 2.—Write a letter, addressing it to the President and giving your views, as far as you are willing to express them, in regard to the duties and responsibilities of an officer in the public service which you seek to enter. Let it fill, as nearly as may be, the following space.

FIFTH SUBJECT.

Question 1.—Which States extend to or border on the sea or tide-water? What is the capital of each of said States?

Question 2.—What is meant in our history, (1) by the Colonial period; (2) by the Continental Congress; (3) by the Declaration of Independence; (4) by the Emancipation Proclamation? Let your answers, as nearly as may be, fill this blank.

Question 3.—State in general terms, but as particularly as the space below will permit, what are the authority and functions of (1) the Congress of the United States; of (2) the Supreme Court of the United States; of (3) the President of the United States; and give the names of each of the executive departments at Washington.

LIMITED EXAMINATION UNDER CLAUSE 4 OF RULE 7.

First subject same as in examination under Clause 1.

SECOND SUBJECT.

DIRECTION.—In case the examiners think that any of the following examples may have been seen by the applicants, they can in the first strike out a line of the figures, and, in the others, change some of the figures without altering the grade of the question.

Question 1.—Add the following:

64379582
28597346
91731625
52613719
26598421
53679713

$$\begin{array}{r}
 83576532 \\
 62985274 \\
 \hline
 79365497
 \end{array}$$

Question 2.—Find the difference between the following numbers:

$$\begin{array}{r}
 905127038624 \\
 605138759928 \\
 \hline
 \end{array}$$

Question 3.—Subtract ten thousand one hundred dollars and six cents from one hundred thousand and seven dollars and five cents, giving all the figures required in the operation.

Question 4.—Multiply 7089 by 983.

Question 5.—Divide 368506 by 375.

Question 6.—When board costs three dollars and seventy-six cents per week what will it cost from March 15th to July 4th?

Question 7.—How many times is 17 cents contained in ten thousand dollars and ten cents?

Question 8.—There are seven hundred and three dollars to be divided between nine men and three boys. The boys are to have twenty-five dollars and five cents each, the residue is to be equally divided among the men, what is each man's share? Give all the figures involved in the solution.



APPENDIX No. 6.

The following tables show the statistics of the examinations in the three branches of the classified service. These considerations should be borne in mind in considering them:

1. That the ratio of those who fail to those who succeed is likely to be much less when the grade of questions shall be better understood; for the more incompetent will see they have little chance of succeeding. Besides, a better class has appeared at each succeeding examination.

2. It was necessary in the outset to examine a large number to make sure of having those competent to fill every variety of vacancy. Many appointments may be now made without further examinations. The excessive number examined from the District of Columbia was the result of conforming to a rule having an unanticipated effect, which has been since amended.

3. In regard to education, the records of the Commission are defective in not showing how long those who have been at an academy or college have remained at either, nor how many are graduates. If a person has been but a month at an academy or college, he is put under the head of those institutions. The habit of calling so many schools academies, and so many academies colleges, helps to make this unavoidable classification the more misleading.

TABLE

Showing numbers of examinations, number of those examined, passed, appointed, age, education, etc., in the Department Service, Washington.

STATES, TERRITORIES, AND DIST. OF COLUMBIA.	Number examined.	Male.	Female.	Average age.	Education.			Number passed at 65 per cent. or over.	Number appointed.
					Common School.	Academy.	College.		
Alabama	4	2	2	42	2	2	..	2	..
Arizona Territory	1	..	1	33	..	1	..	1	..
California	7	6	1	30	2	2	3	5	1
Colorado	4	2	2	37	1	3	..	2	..
Connecticut	9	3	6	29	4	2	..	3	1
Dakota Territory	2	1	1	29	1	1	..	1	..
District of Columbia	125	54	71	25	48	53	24	74	3
Delaware	1	..	1	25	1	..	..	1	..
Florida	2	..	2	36	..	1	1	1	..
Georgia	3	2	1	25	..	1	2	..	..
Illinois	24	16	8	31	4	6	14	15	4
Indiana	40	29	11	26	15	12	13	18	2
Indian Territory	1	1	..	30	..	1	..	..	..
Iowa	3	2	1	23	1	1	1	3	1
Kansas	15	13	2	32	7	2	6	9	2
Kentucky	21	16	5	28	4	7	10	13	2
Louisiana	6	3	3	34	2	4	..	3	..
Maine	14	10	4	26	2	11	1	11	2
Maryland	66	40	26	26	13	35	18	44	3
Massachusetts	36	27	9	30	6	17	13	26	1

Michigan	18	12	6	28	3	10	5	10	3
Minnesota	7	5	2	36	3	..	4	4	..
Mississippi	4	3	1	30	2	1	1	1	..
Missouri	15	11	4	34	10	4	1	7	2
Nebraska	1	..	1	25	..	1	..	..	..
New Hampshire	6	2	4	35	..	6	..	3	1
New Jersey	16	7	9	28	5	9	2	10	2
New York	94	65	29	26	20	54	20	50	5
North Carolina	38	26	12	27	2	24	12	19	1
Ohio	64	45	19	32	20	27	17	42	4
Pennsylvania	42	30	12	30	10	22	10	22	5
Rhode Island	6	4	2	42	5	1	..	1	1
South Carolina	13	11	2	24	1	6	6	9	1
Tennessee	4	1	3	31	2	2	..	2	..
Texas	3	1	2	36	..	3	..	3	..
Vermont	5	1	4	28	2	3	..	4	1
Virginia	37	21	16	32	9	22	6	24	2
Washington Territory	1	..	1	40	..	..	1	1	..
West Virginia	19	13	6	30	7	6	6	10	1
Wisconsin	7	6	1	32	3	3	1	5	2
Total	784	491	293	32	217	366	201	459	53

Time was when Nature's every mystic mood
 Poured round my heart a flood of eager joy;
 When pageantry of sunsets moved the boy
 More than high ventures of the great and good;
 When trellised shadows in the vernal wood,
 And little peeping flowers, so sweet and coy,
 Were simple happiness without alloy,
 And whispered to me things I understood.
 But now the strange sad weight of human woe,
 And all the bitterness of human wrong,
 Press on my saddened spirit as I go,
 And stir the pulsings of a graver song:
 Dread mysteries of life and death I scan,
 And all my soul is only full of man.—*W. W. Bedford.*

THE ART OF FISH CULTURE.

BY PROF. G. BROWN GOODE.

PART II.

There are two forms of fish culture. One of these, which has been practiced for many centuries in China, and perhaps quite as long in Europe, consisted in the transportation of living fish from waters in which they were abundant, to other waters, depleted or naturally deficient in fish life. The carp and the goldfish have been so long domesticated that they have become modified, like domestic fowls and cattle. The goldfish was introduced into all parts of the world from China, centuries ago. The introduction of the carp into the United States by the efforts of our Commissioner of Fisheries has been one of the most extensive operations in fish culture ever attempted. In 1878 carp were brought from Bavaria, and from this stock, planted in Babcock Lake, over 300,000 young fish have been distributed, in lots of ten to twenty, to every part of the country, so that almost every county is now stocked with this valuable food fish. As early as 1770 some experiments in transplanting fish were attempted, at the suggestion of Benjamin Franklin and others. In 1854, the black bass, now so abundant in the Potomac, were introduced by an engineer on the B. & O. R. R., who brought them over the Alleghenies in the water tank of his engine. This fish has also been sent to England and France, where it bids fair to become a favorite. In 1873, a car was freighted with eastern fish designed for introduction into the waters of California. The car ran off the track in Nebraska, and the rivers in that region are now stocked with our best fishes.

Far more important than fish transportation and the acclimation of foreign species, is the art of fish breeding, by which it is possible to keep up the supply of fishes in waters into which they have been successfully introduced. It was in the year 1741 that Stephen Ludwig Jacobi, a wealthy landed proprietor and civil engineer of northwestern Germany, discovered the method of artificially fertilizing the eggs of fish for the purpose of restocking ponds

and streams, and began a series of painstaking experiments with that end in view. He first conceived the idea in 1725, when a youth of seventeen years, and was successful after laboring for sixteen years. His discovery was not announced till 1763. Although his discovery was thought to be of interest, and was used by physiologists and students of embryology, it was not until the French government resolved to make a grand experiment in stocking the waters of France with fish that modern industrial fish culture was born.

The establishment in 1850 at Huningen, in Alsace, by the French government, of the first fish-breeding station, or “piscifactory,” as it was named by Prof. Coste, is of great significance, since it marks the initiation of public fish culture. To this establishment the world is indebted for some practical hints, but most of all for its influence upon the policy of governments. The fortunes of war and conquest have now thrown Huningen into the hands of the German government. The art discovered in Germany was practiced in Italy as early as 1791 by Bufalini, in France in 1820, in Bohemia in 1824, in Great Britain in 1837, in Switzerland in 1842, in Norway, under government patronage in 1850, in Finland in 1852, in the United States in 1853, in Belgium, Holland, and Russia in 1854, in Canada about 1863, in Austria in 1865, in Australasia, by the inhabitants of English Salem in 1862, and in Japan in 1877.

The history of fish culture in this country is so familiar to every one who has the slightest interest in the subject that it seems unnecessary to refer to it in this place, except to show that it was largely to the growth of popular interest in the subject that the Fish Commission has owed its original and since increasing support.

The establishment of the United States Commission of Fish and Fisheries in 1871 marked the beginning of a period of great activity and great progress in fish culture, which has been quite without parallel elsewhere. The duties of the Commissioner were thus defined: “To prosecute investigations on the subject (of the diminution of valuable fishes), with the view of ascertaining whether any and what diminution in the number of food fishes of the coast and the lakes of the United States has taken place; and, if so, to what causes the same is due; and also whether any, and what protection, prohibitory or precautionary measures should be adopted in the premises, and to report upon the same to Congress.”

I think I may truthfully assert that very much of the improvement in the condition of our fisheries has been due to the wise and energetic management of our Commissioner, Prof. Spencer F. Baird. Himself an eminent man of science, for forty years in the front rank of biological investigation, the author of several hundred scientific memoirs, no one could realize more thoroughly the importance of a scientific foundation for the proposed work.

His position as the head of that influential scientific organization, given by an Englishman to the United States, “for the increase and diffusion of useful knowledge among men,” enabled him to secure at once the aid of a body of trained specialists.

I wish to emphasize the idea that *the work of the Fish Commission owes its value solely and entirely to the fact of its being based upon an extensive and long continued system of scientific investigations*, for the purpose of discovering unknown facts, the knowledge of which is essential to the welfare of the fisheries, the economical management of the national fishery resources, the success of fish culture, and the intelligent framing of fishery laws.

The resolution establishing the Commission requires that its head shall be a civil officer of the government, of proved scientific and practical acquaintance with the fishes of the coast—thus formally fixing its scientific character.

The work of the Commission is and always has been under the direction of eminent and representative scientific specialists, acting as heads of its several divisions, and the employes, with the exception of a very limited number of clerks, are trained experts, usually scientific students—so exact and special is the training required even for subordinate positions, that in a majority of cases each man employed is the only man in the country who understands and can perform his own individual work.

Pure and applied science have labored together always in the service of the Fish Commission, their representatives working side by side in the same laboratories; indeed, much of the best work in the investigation of the fisheries and in the artificial culture of fishes has been performed by men eminent as zoölogists.

The work of the Fish Commission is naturally divided into three sections:

1. The systematic investigation of the waters of the United States, and the biological and physical problems which they present. The scientific studies of

the Commission are based upon a liberal and philosophical interpretation of the law. In making his original plans the Commissioner insisted that to study only the food fishes would be of little importance, and that useful conclusions must needs rest upon a broad foundation of investigations purely scientific in character. The life history of species of economic value should be understood from beginning to end, but no less requisite is it to know the histories of the animals and plants upon which they feed, or upon which their food is nourished; the histories of their enemies and friends, and the friends and foes of their enemies and friends, as well as the currents, temperature and other physical phenomena of the waters in relation to migration, reproduction and growth. A necessary accompaniment to this division is the amassing of material for research to be stored in the National and other museums for future use.

2. The investigation of the methods of fisheries, past and present, and the statistics of production and commerce of fishery products. Man being one of the chief destroyers of fish, his influence upon their abundance must be studied. Fishery methods and apparatus must be examined and compared with those of other lands, that the use of those which threaten the destruction of useful fishes may be discouraged, and that those which are inefficient may be replaced by others more serviceable. Statistics of industry and trade must be secured for the use of Congress in making treaties or imposing tariffs, to show to producers the best markets, and to consumers where and with what their needs may be supplied.

3. The introduction and multiplication of useful food fishes throughout the country, especially in waters under the jurisdiction of the general government, or those common to several states, none of which might feel willing to make expenditures for the benefit of the others. This work, which was not contemplated when the Commission was established, was first undertaken at the instance of the American Fish Cultural Association, whose representatives induced Congress to make a special appropriation for the purpose. This appropriation has since been renewed every year on an increasingly bountiful scale, and the propagation of fish is at present by far the most extensive branch of the work of the Commission, both in respect to number of men employed and quantity of money expended.

The limits of this article do not permit the discussion of work in connection with the fisheries, or of the scientific investigations which form the bed for the

whole current of its activity.

The principal activity of the Commission has properly been directed to the wholesale replenishment of our depleted waters, as is shown by the fact that from seventy-five to eighty-five per cent. of the appropriations have been directed into this channel.

For fifteen or twenty years prior to the establishment of the Commission, popular interest in the fisheries, and a desire for their maintenance had been on the increase, the state of public opinion being doubtless under stimulation from the action of the French government in fostering the still infant art of fish culture.

The publications and experiments of Garlick, Fry, Atwood, Lyman, Green, Stone, Ainsworth, Roosevelt, Atkins, Stady, and others, awakened everywhere a sense of the fact that our rivers and streams were being rapidly cleared out, and the feeling that a similar state of affairs was probably existing in the adjoining ocean. Measures were set on foot for restoration and protection as early as 1605, when Massachusetts appointed the first commission, and prior to 1870 this example was followed by several other states. Nearly all the states and territories now have similar organizations. The United States has distanced all its competitors, as was evinced by the manner in which the prizes were distributed at the recent fishery exhibitions in Berlin and London.

The fertilization of the fish egg is the simplest of processes, consisting, as every one knows, in simply pressing the ripe ova from the female fish into a shallow receptacle, and then squeezing out the milt of the male upon them. Formerly a great deal of water was placed in the pan, now the "dry method," with only a little water, discovered by the Russian Vrasski, in 1854, is preferred. The eggs having been fertilized, the most difficult part of the task remains, namely, the care of the eggs until they are hatched, and the care of the young fry until they are able to care for themselves.

The apparatus employed is various in principle, to correspond to the physical peculiarities of the eggs. Fish culturists divide eggs into four classes, viz.: (1) heavy eggs, non-adhesive, whose specific gravity is so great that they will not float, such as the eggs of the salmon and trout; (2) heavy adhesive eggs, such as those of the herring, smelt and perch; (3) semi-buoyant eggs, like those of the shad and whitefish (*Coregonus*), and (4) buoyant eggs, like those of the cod and mackerel.

Heavy, non-adhesive eggs are placed in thin layers, either upon gravel, grilles of glass, or sheets of wire cloth, in receptacles through which a current of water is constantly passing. There are numerous forms of apparatus for eggs of this class, but the most effective are those in which a number of trays of wire cloth, just deep enough to carry single layers of eggs, are placed one upon the other in a box or jar, into which the water enters from below, passing out at the top.

Heavy, adhesive eggs, are received upon bunches of twigs, or frames of glass plates, to which they adhere, and which are placed in receptacles through which water is passing.

Semi-buoyant eggs, or those whose specific gravity is but slightly greater than that of the water, require altogether other treatment. They are necessarily placed together in large numbers, and to prevent their settling upon the bottom of the receptacle, it is necessary to introduce a gentle current from below. For many years these eggs could be hatched only in floating receptacles placed in a river, with wire cloth bottoms, placed at an angle, the motion of which was utilized to keep the eggs in suspension. Later, an arrangement of plunging buckets was invented, cylindrical receptacles with tops and bottoms of wire cloth, which were worked up and down at the surface of the water by machinery. The eggs in the cylinders were suspended in rows from beams which were worked up and down at the surface of the water by machinery. The eggs in the cylinders were thus kept constantly in motion. Finally, the device now most in favor was perfected; this is a receptacle, conical, or at least with a constricted termination, placed with its apex downward, through which passes from below a strong current, keeping the eggs constantly suspended and in motion. This form of apparatus, of which the McDonald and Clark hatching jars are the most perfect developments, may be worked in connection with any common hydrant.[\[A\]](#)

Floating eggs have been hatched only by means of rude contrivances for sustaining a lateral circular eddy, or swirl of water in the receptacle.

The use of refrigerators, to retard the development of the egg until such time as it is most convenient to take care of the fry, is now extensively practiced in the United States, and has been experimented upon in Germany.

In the discussion of fish-cultural economy, the distinction between *private fish culture* and *public fish culture* must be carefully observed, and it must also

be borne in mind that by *public fish culture*, or *modern fish culture*, I mean fish culture carried on at public expense, and for the public good. Public fish culture, to be effective, must be conducted by men trained in scientific methods of thought and work.

The distinction between private and public fish culture must be carefully observed. The maintenance of ponds for carp, trout, and other domesticated species, is an industry to be classed with poultry raising and bee-keeping, and its interest to the political economist is but slight.

The proper function of fish culture is the stocking of the public waters with fish in which no individual can claim the right of property.

The comparative insignificance of the private fish-culture of Europe is, perhaps, what has led to the recent savage attack upon fish culture in general by Malmgren of the University of Helsingfors. European fish culturists have always operated with small numbers of eggs. The establishment of Sir James Maitland at Howieton, near Stirling, Scotland, is the finest and largest private establishment in the world, and yields a handsome addition to the revenues of its proprietor. A description of this hatchery is published as one of the conference papers of the International Fisheries Exhibition, and that the distinction between public and private enterprise in fish-culture may be understood, it should be compared with the following statement by Mr. Livingston Stone, the superintendent of one of the seventeen hatcheries supported by the United States Fish Commission—that on the McCloud River in California.

“In the eleven years since the salmon-breeding station has been in operation, 67,000,000 eggs have been taken, most of which have been distributed in the various states of the Union. Several million, however, have been sent to foreign countries, including Germany, France, Great Britain, Denmark, Russia, Belgium, Holland, Canada, New Zealand, Australia, and the Sandwich Islands.

“About 15,000,000 have been hatched at the station, and the young placed in the McCloud and other tributaries of the Sacramento River. So great have been the benefits of this restocking of the Sacramento, that the statistics of the salmon fisheries on the Sacramento show that the annual salmon catch of the river has increased 5,000,000 pounds each year during the last few years.”

In the two government hatcheries at Alpena and Northville, Michigan, in the winter of 1883-84, there were produced over 100,000,000 eggs of whitefish, *Coregonus clupeiformis*, and the total number of young fish to be placed in the Great Lakes this year by these and the various state hatcheries will exceed 225,000,000. The fishermen of the Great Lakes admit that but for public fish culture half of them would be obliged to abandon their calling.

Instances of great improvement might be cited in connection with nearly every shad river in the United States. In the Potomac alone the annual yield has been brought up by the operations of the Fish Commission from 668,000 pounds in 1877 to an average of more than 1,600,000 in recent years.

In 1882, carp bred in the Fish Commission ponds in Washington was distributed in lots of from 20 to 10,000 applicants in every State and Territory, at an average distance of more than 900 miles, the total mileage of the shipments being about 9,000,000 miles, and the actual distance traversed by the transportation cars 34,000 miles.

Public fish culture is only useful when conducted upon a gigantic scale—its statistical tables must be footed up in hundreds of millions. To count young fish by the thousand is the task of the private propagator. The use of steamships and steam machinery, the construction of refrigerating transportation cars, and the maintenance of permanent hatching stations, seventeen in number, in different parts of the continent, are forms of activity only attainable by government aid.

Equally unattainable by private effort would be the enormous experiments in transplanting and acclimating fish in new waters—California salmon in the rivers of the east; landlocked salmon and smelt in the lakes of the interior; the planting of shad in California and the Mississippi valley; and German carp in thirty thousand separate bodies of water distributed through all the states and territories of the Union. The two last named experiments, carried out within a period of three years, have met with success beyond doubt, and are of the greatest importance to the country; the others have been more or less successful, though their results are not yet fully realized.

It has been demonstrated, however, beyond possibility of challenge, that the great river fisheries of the United States, which produced in 1880 48,000,000 pounds of alewives, 18,000,000 pounds of shad, 52,000,000 pounds of salmon, besides bass, sturgeon, and smelt, and worth “at first

hands” between \$4,000,000 and \$6,000,000, are entirely under control of the fish culturist to sustain or destroy, and are capable of immense extension.

Having now attempted to define the field of modern fish culture, and to show what it has already accomplished, it remains to be said what appear to be its legitimate aims and limitations. Its aims, as I understand them, are:

1. To arrive at a thorough knowledge of the life history from beginning to end, of every species of economic value; the histories of the animals and plants upon which they feed or upon which their food is nourished; the histories of their enemies and friends, and the friends and foes of their enemies and friends, as well as the currents, temperature and other physical phenomena of the waters in relation to migration, reproduction and growth.

2. To apply this knowledge in such a practical manner that every form of fish shall be at least as thoroughly under control as are now the salmon, the shad, the alewives, the carp, and the whitefish.

Its limitations are precisely those of scientific agriculture and animal rearing, since, although certain physical conditions may constantly intervene to thwart man’s efforts in any given direction, it is quite within the bounds of reasonable expectation to be able to understand what these are and how their effects are produced.

An important consideration concerning the limitations of fish culture must always be kept in mind in weighing the arguments for and against its success. It is simply this: *that effort toward the acclimation of fishes in new waters is not fish culture, but is simply one of the necessary experiments upon which fish culture may be based.* The introduction of carp from Germany was not fish culture, it was an experiment: the experiment has succeeded, and fish culture is now one of its results. The introduction of California salmon to the Atlantic slope was an experiment. It has not succeeded. Its failure has nothing to do with the success of fish culture. If any one wants to see successful fish culture in connection with this fish, let him go to the Sacramento River. The introduction of shad to the Pacific coast was an experiment. It succeeded. Shad culture can now be carried on without fear of failure by the fish commissions of the Pacific states.

Shad culture is an established success, so is whitefish culture in the Great Lakes. The experiments with cod and Spanish mackerel were not fish culture,

though there is reason to hope that they may yet lead up to it.

Public fish culture, then, scarcely exists except in America, though in Europe many eminent men of science appreciate its importance and are striving to educate the people to the point of supporting it. Germany is at present in the vanguard, and the powerful *Deutscher Fischerei Verein* is doing all in its power to advance the interests of fish culture.

[A] Trans. Amer. Fish Cultural Association, 1883.

If we could take all things as ordained and for the best, we should indeed be conquerors of the world. Nothing has ever happened to man so bad as he has anticipated it to be. If we should be quiet under our troubles they would not be so painful to bear. I can not separate the existence of a God from his pre-ordination and direction of all things, good and evil; the latter he permits, but still controls.—*Chinese Gordon*.

HONESTY IN THE C. L. S. C.

BY CHANCELLOR J. H. VINCENT, D.D.

The system of examinations which prevails in secular schools has its advantages, but is not an unmixed benefit. It is an incentive to study. It aids the teacher in determining the proper time and degree of the promotion sought by the pupil. It is an approximate test of the place which a candidate may be able to command in the advanced grades.

But the examinations may become an end instead of a means. Pupils may work for success in a process rather than for the possession of power to think on any subject at any time. Examinations may give advantage to an inferior type of mind, rewarding mere memory and facility of expression, thus putting to a disadvantage the steadier, calmer, slower movements of a thorough student.

While in the day school, which deals with youth, we think the examination in some form or other is indispensable, in our Circle it is entirely impracticable. It is equally undesirable and unnecessary. We aim at reading and not study, except as reading by mature minds, eager to know, must necessarily induce the most fruitful kind of study. This, our aim, is the highest and wisest. People join the C. L. S. C., not for degrees in college, not for recognition as competitors in departments of exact scholarship, but for direction in useful reading, and for the pleasures of association in literary pursuits. It gives no pecuniary reward in the shape of prizes or professional diplomas. It would seem to present no inducement to dishonesty on the part of its members. They read for personal profit. Their compensation lies chiefly, if not wholly, in the joy of knowing, in the sense of increasing taste and power, and in the delights of high and honorable companionship.

In our Circle moral worth is assumed. The men and women who join us have long since learned that knowledge without character is not only worthless, but is a curse. They come, through faith in the highest ends of life, to improve their intellectual faculties. The gate by which they enter the C. L.

S. C. bears this legend: "We study the Word and the Works of God." On our holiest altar they read: "Let us keep our Heavenly Father in the midst." The most fervent appeals which fall from the lips of our leaders are based upon the lofty religious standards which are lifted by the institution. The memorial days, the Sunday vesper hour, the sacred songs, all bear testimony to the religious character of the Circle. There would seem to be no inducement to dishonest souls to knock at our doors, or record their names on our lists.

A fact or two, not widely representative, justifies this word of warning. I am glad to believe that to but few members in the Circle can it be necessary or appropriate. We should all be watchful where temptation is not excluded, and we may as well recall the fact that in the old story from a very old book, there was a lurking serpent in a garden of innocence and delight.

A C. L. S. C. diploma, though radiant with thirty-one seals—shields, stars, octagons—would not stand for much in Heidelberg, Oxford, or Harvard. As an American curiosity it would not attract a moment's notice, save as thoughtful men might come to measure its real significance. But even then it would be respected, not as conferring honor upon its holder, but as indicating a popular movement in favor of higher education. No wearer of the badge of the "S. H. G." or of the "Guild of the Seven Seals," would thereby stand any chance for appointment from any of these institutions, to wear an honorary degree, or take a professor's chair.

At Chautauqua in the season, and at local circle receptions, and recognitions, the C. L. S. C. badge and diploma are not thus impotent. The color and stamp assign the holder to the place of honor. It is something even to our learned and honored Dr. Eaton to be the first member of the "Guild of the Seven Seals," and the only member of its highest degree. It is something to be able to linger to the last in the Hall of Philosophy at Chautauqua at the sunset, as the successive societies are requested to remain—"S. H. G.," "O. W. S.," "L. R. T.," and the "G. S. S." Members of our Circle appreciate the distinction, and it is a distinction with meaning in it, and with genuine pleasure accompanying it. It is something to have a high place in the Chautauqua procession, and to frame a diploma at home with increasing luster as new seals flash out upon it as stars in the evening sky.

But along these lines of promotion lie the perils indicated. The recognition given to the members of the C. L. S. C. graduates and members of its advanced societies may prove a temptation to unguarded souls, and in an evil

moment reading may be reported that has not been done, and seals solicited on false representations.

An anonymous note (which none but contemptible people ever write) called my attention to a possibility in a particular case, and a careful investigation was made. The idle boast of a thoughtless woman was reported, and an official examination of her report papers seemed to corroborate the ungracious charge. Later investigations vindicate our member and relieve her from the implied condemnation. But the subject has weighed heavily upon my mind, so that I call the attention of all to a possible peril.

Since the organization of the Circle I have been greatly pleased with the conscientiousness of its members. Many of them were afraid, when we required a report of the time spent each day in reading, that they would not keep an exact account. They were afraid that if they could not recall the contents of the chapter, or book, as students at school would be required to do, that they could not report that chapter or book as thoroughly read. Many persons refused to join the Circle lest they should not be able to complete the four years' course, believing as they did that members were pledged to such completed work.

While this conscientiousness was gratifying it was excessive, and was based on false views of the aims of the Circle. I have endeavored to correct these views, to modify details of working, and to impress all members with the simple aim of the Circle, to promote the reading of certain books, leaving every person free to decide how superficially or thoroughly the reading should be done.

Our only aim is to promote reading. If we enlist people in the reading of good books on a wide range of subjects we shall at some point strike their taste, and thus promote the culture that comes from the use of one's faculties in the line of his inclination and opportunity.

This being the modest standard of the Circle, we have a right to expect that every member will honorably discharge his duty, reporting the books he has read and none else, filling out his memoranda (when he undertakes to do it at all) by his own hand, or by dictation, not by proxy, winning the honors he seeks in our Circle by the honesty which will render his recognition a pleasure to himself and a credit to the management.

If any member feels that his conscience would be quieted by re-reading portions of the required books, let him do it.

If any member expects to gain distinction or place among us by unfairness, let him remember that self-contempt is the severest penalty we care to predict.

Let us live honestly.

OUTLINE AND PROGRAMS.

OUTLINE OF REQUIRED READINGS FOR MAY.

First Week (ending May 8).—1. “Easy Lessons in Animal Biology,” in THE CHAUTAUQUAN.

2. Sunday Readings for May 3, in THE CHAUTAUQUAN.

Second Week (ending May 16).—1. “English as a Universal Language,” in THE CHAUTAUQUAN.

2. Sunday Readings for May 10, in THE CHAUTAUQUAN.

Third Week (ending May 23).—1. “Home Studies in Chemistry,” in THE CHAUTAUQUAN.

2. Sunday Readings for May 17, in THE CHAUTAUQUAN.

Fourth Week (ending May 31).—1. “The Eyes Busy on Things About Us,” in THE CHAUTAUQUAN.

2. Sunday Readings for May 24 and May 31, in THE CHAUTAUQUAN.

PROGRAMS FOR LOCAL CIRCLE WORK.

FIRST WEEK IN MAY.—ADDISON’S DAY.

“Give days and nights, sir, to the study of Addison if you mean to be a good writer, and, what is more worth, an honest man.”—*Samuel Johnson.*

Music.

1. Roll call—Responses consisting of the name and distinguishing trait of some character in Addison's writings.

2. A Paper on the Political History of England in Addison's Time.

3. A Brief Sketch of Addison's Life and Travels.

Music.

4. Selection—"The Transmigration of Souls—A Letter from a Monkey."

5. A History of the Newspapers with which Addison was connected.

6. A Paper on two of Addison's Works—"The Campaign" and "The Tragedy of Cato."

Music.

7. Selection—"Reflections on the Delights of Spring."

8. Essay—Addison's Delineation of Woman's Character.

Music.

A delightful entertainment for an evening can be given by preparing a banquet at which the guests are to personate the characters introduced in Addison's "Vision of the Table of Fame." These characters can be studied from other sources, so that each person may be enabled fittingly to carry on the representation during the time spent at the table.

The "Exercise of the Fan" can be prepared by a little practice so as to afford much amusement.

For books of reference see Thackeray's "English Humorists," Aiken's "Memorials of Addison," Macaulay's "Life and Writings of Addison," and the books on English Literature.



SECOND WEEK IN MAY.

1. Essay—The Aryan Race.

2. A Review Lesson—Questions from Chautauqua Text-Book, No. 5; Greek History.

3. Selection—"Orpheus and Eurydice." By J. G. Saxe.

4. A Trip on Paper through the Soudan.

Music.

5. Story—"Circe's Palace." From Hawthorne's "Tangle-Wood Tales."

6. Book Review—"The Life of George Eliot." By J. W. Cross.

7. A General Talk on the Mohammedan Power of To-day.

8. Question Box.



THIRD WEEK IN MAY.

1. A Paper on the Introduction of Temperance Text-Books into the Public Schools.

2. Selections—"Prometheus" and "Epimetheus." By Longfellow. [The two read by different members.]

3. Brief Sketches of Literary Women who have assumed Masculine Pseudonyms.

Music.

4. Essay—What is the Oklahoma Boom?

5. A General Talk on the practical Home Use of the Study of Chemistry.

6. A Pronunciation Match—The circle chooses sides, the leader spells the words, and the class pronounces.

7. Critic's Report.



FOURTH WEEK IN MAY.

1. "Questions and Answers," in review.

2. Essay—May-day as Observed in Olden Times. [It might be well to suggest that a May-day suitably arranged for modern times would be fully as

enjoyable as it used to be.]

3. Recitation—"Phaëton," by J. G. Saxe. Compare this with "The Story of Phaëton," by Addison.

4. Map Exercise—Locate all the most important battle fields of Greek history.

Music.

5. A Paper on the Foreign Service of the United States.

6. Essay—What Remains of Greek Art.

7. *Conversazione*—The Wrongs of the Indians as portrayed in Mrs. Jackson's "Ramona."

Knowing there are times and places in which every little helps, we offer a few suggestions for Special Sunday. If they only serve as index fingers, pointing out the way to fields where each can glean for himself much more successfully and satisfactorily than to take what others have gathered, they will accomplish a good purpose. To hopeful, earnest, self-reliant workers with "eyes busy on things about us," more help is not needed, and perhaps not even this much. In addition to the vesper service prepared, selections, essays or papers, and Bible studies can be very profitably used.

From "The Literature of the Age of Elizabeth," by William Hazlitt, the part referring to the translation of the Bible is very fine. Also, "Christianity the Great Remedy," by Robert Charles Winthrop, LL.D. These selections can be found in Allibone's "Great Authors of All Ages." In THE CHAUTAUQUAN for January, 1885, "The Inner Chautauqua" is a good reading. In *The Chautauqua Assembly Herald* for August 9, 1884, "Mrs. Pickett's Missionary Box" Miss P. J. Walden, 36 Bromfield Street, Boston, will send for three cents "Thanksgiving Ann." Nothing could be better suited for such a service than selections from Miss Havergal's writings.

Themes for Essays are: "Personal Culture a Christian Duty." "How Best can I Help my Neighbor?" "Work in the Home Missionary Field." "How to Make the Sabbath a Beautiful Day." Papers can be prepared on Bible customs and manners, Bible lands, and Bible characters. With the aid simply of a Concordance and a Reference Bible, interesting Bible studies on any desired

topic can be prepared. Hitchcock's "Analysis of the Bible" would afford great help in arranging work of this kind.

LOCAL CIRCLES.

C. L. S. C. MOTTOES.

“We Study the Word and the Works of God.”—“Let us keep our Heavenly Father in the Midst.”—“Never be Discouraged.”

C. L. S. C. MEMORIAL DAYS.

1. OPENING DAY—October 1.
2. BRYANT DAY—November 3.
3. SPECIAL SUNDAY—November, second Sunday.
4. MILTON DAY—December 9.
5. COLLEGE DAY—January, last Thursday.
6. SPECIAL SUNDAY—February, second Sunday.
7. FOUNDER’S DAY—February 23.
8. LONGFELLOW DAY—February 27.
9. SHAKSPERE DAY—April 23.
10. ADDISON DAY—May 1.
11. SPECIAL SUNDAY—May, second Sunday.
12. SPECIAL SUNDAY—July, second Sunday.
13. INAUGURATION DAY—August, first Saturday after first Tuesday; anniversary of C. L. S. C. at Chautauqua.
14. ST. PAUL’S DAY—August, second Saturday after first Tuesday; anniversary of the dedication of St. Paul’s Grove at Chautauqua.

15. COMMENCEMENT DAY—August, third Tuesday.

16. GARFIELD DAY—September 19.

The sensible, social way in which the February memorial days were observed has brought to our mind a comment on Madame Mohl and her methods of entertaining, how, “beyond ordering a good and abundant meal, she gave little thought to the mere material details of her entertainments; but she took great pains with the intellectual *menu*. She would give time and thought and personal trouble to provide for each guest intellectually what he would most enjoy, and would carefully consider whether this person would like to meet the other, and to sit next So-and-So. Her great preoccupation was the combining of congenial elements for all in general and particular.” We feel very much as if our circles’ friends have learned Madame Mohl’s wisdom. As if the long desired reform in the methods of social entertainment was beginning in our own C. L. S. C. family. To give entertainments where wit and wisdom and social freedom prevail, where thoughts are more desired than feasts, and music and art take the place of supper tables is, it may be, the Quixote and blue-stockings way to-day—but it is the true social method. Any one who will take a glance with us over the receptions, “socials,” “at homes,” and public meetings which our circles held in February will, we believe, conclude that it certainly is the C. L. S. C. idea of “society,” and of a “good time.” Such delightful programs are rare to find.

Founder’s day is a new and very welcome occasion for observance, and very many circles made it the time of a special or public meeting. Some prepared an extra program, invited a few friends and spent a quiet evening in pleasant, friendly talk and merriment; others prepared a public meeting and strove to celebrate the day by increasing the interest in the work. At FRANKLIN, NEW HAMPSHIRE, the “Webster” C. L. S. C. had charge of a joint meeting. The program they carried out was admirable. Two other Chautauqua societies assisted the “Websters” in the entertainment; the “Pemigewassett” C. L. S. C., and the “Crystal” C. Y. F. R. U.—February 27th was so close in the wake of February 23d that the celebration of the two days was united in several places, with excellent results, too, we should judge. At MILFORD, MASS., such a union meeting was held. The circle numbers twenty-eight, and as each member was allowed to invite one guest it made a goodly company. Vine wreathed portraits of the two heroes of the evening decorated the tables of the parlor where the circle met, and the program, divided into two portions, one devoted to

Chancellor Vincent, the other to Longfellow, was happily arranged. We are glad that they have found out Lowell's tribute to Longfellow; it makes a very appropriate number.—At FOXBORO, MASS., the "Star" circle quite distinguished itself by its celebration of Founder's day. There were present several out-of-town circles, among them those from Franklin, Medfield, and Mansfield. In the notice which a local paper gave of this affair, we find some comments which are particularly encouraging: "Many have expressed their great delight at the manner in which the whole entertainment was carried out, which shows that these seasons are becoming more and more popular. Says a lady somewhere in the fifties, 'How I wish there had been such an organization when I was young. My advantages for gaining an education were limited. If my memory was not so poor I should be tempted to join, even at this late day.' Another says: 'I have enjoyed the whole program very much, and have got a better idea of the work of the Circle from this evening's entertainment than from any other source.' Still another: 'I enjoyed the whole of it very much indeed. The program was nicely carried out,' and asks 'Why don't the circle give such entertainments oftener, so that people can better understand the object of this organization?' Says a gentleman, a graduate of one of the higher schools, 'I enjoyed it immensely. The exercises from the commencement to the close were very interesting.' Another, 'It carried me back to school-days and spelling schools, and I thoroughly enjoyed it, supper included.'"—At PROVIDENCE, RHODE ISLAND, the plan of the Milford circle was followed by the "Milton" circle. A paper on "The Chautauqua Movement" was one feature of the evening. It seems to have been an evening of practical work as well as pleasure, for good results are promised us as an outcome of the meeting. The "Miltons" now number twenty-two, an increase of seven over last year. Indeed, we can hardly see how a meeting at all suitable for Founder's day could do anything else but convert people. It would necessarily be brimming over with such sparkling ideas, such enticing plans, that the fortunate guests at such an entertainment would very naturally want to join the company.—Another circle, that at NORTHFIELD, OHIO, adopted this theory, and combined the two memorial days, making their celebration a public meeting, to which about sixty guests were invited. The program was very skillfully arranged, including some excellent subjects for essays. The Northfield circle was organized last year by ten "Pansies," and has been recruited this year by two of the class of '88. A good idea of their program is that they begin each evening's work with the vesper service. The "quiz" is a prominent feature of the evening, and as "discussion" sends them home alert,

interested, and sorry that the evening is over. A discussion on a live subject, we would whisper to leaders, is one of the best methods of making your circle sorry that it is time to stop, a result which is the best possible proof of an enjoyable evening.—At SILVER CREEK, NEW YORK, Founder's day was celebrated with much enthusiasm by the circle; an excellent program of the evening was prepared and published, the week previous, in the local paper. The program was divided into three parts—the first consisting of Chautauqua songs, mottoes, selections from Founder's writings or sayings, a sketch of his life, and appropriate recitations; then a *petit souper*, and, on the principle, perhaps, that the best should come last, part third consisted of the reading of a letter from the Founder himself, sent in reply to a request from the circle for only a few words, and of a poem from the poetess of the class of 1886, Mrs. Cleveland.—The three Chautauqua circles of NEW ALBANY, INDIANA, have now about seventy-five members, and all memorial days are observed by them jointly. Among the several pleasant meetings of the year, none have been so successful as Founder's day. On the program of exercises we notice that roll call was responded to by giving quotations on the "Companionship of Books." These were collected and printed in a local paper for preservation in the circle's scrap books. Prof. R. A. Ogg, of the class of '84, presented "The Founder and his Chautauqua Idea" in his happiest manner. "The Founder at Chautauqua" was vividly pictured by Rev. W. S. Austin, secretary of the class of '86. Every Chautauquan present left this memorial meeting with the expression upon his lips, "The best of all our union meetings," and the public were loud in their praises of it.—The *sixth* circle of MILWAUKEE, WISCONSIN, young, enthusiastic '88s, all ladies, and hard at work for their own improvement and for the advancement of their circle, make a right jubilant report of the pleasant time they had with their friends of the "Beta" circle on Founder's day. They held their meeting in the afternoon, and carried out the program very nearly as THE CHAUTAUQUAN prescribed. "To some of us," a member writes, "our second *alma mater* bids fair to become even dearer than the one of earlier remembrance."—Right in tune with this glad greeting is a message which comes from the circle at INDIANOLA, IOWA, another circle that celebrated Founder's day. "We have received so many helpful hints and useful suggestions through your local circle column, that we take courage to say: 'May the Chautauqua work we love so well go on until it spreads from east to west, from north to south.'" Our Indianola friends were organized six years ago. Two of their number were members of the first graduating class. A method of leadership, which we believe to be very effective in a small circle,

is pursued by them. It is that each member take turns in the leadership of the circle.

The 27th of February is the memorial day most universally celebrated, and this because, perhaps, our memories of Longfellow are so peculiarly near and tender. This season the memorial services celebrated by the C. L. S. C. were particularly prominent. To begin "at home," the "Willis" circle of PORTLAND, ME., gave a charming entertainment. From a Portland paper we clip the following description: "The exercises were conducted in a very unique and pleasing manner. One of the double parlors was filled with about fifty invited guests, while the other, being fitted up in representation of the Wayside Inn, was occupied by members of the circle, each one representing in costume some one of Longfellow's characters. Each of the participants in the exercises was introduced with appropriate selections by 'John Alden,' while by his side sat 'Priscilla, the Puritan maiden.' Other characters represented were the 'Landlord and Daughter,' 'Evangeline,' 'Rabbi Ben Levi,' 'Astred, the Abbess,' 'Precissa, the Gipsy,' 'Spanish Lady,' 'Hiawatha,' 'Minnehaha,' 'Young Musician,' 'Paul Revere,' 'Theodore,' and 'Lady Wentworth.' On the whole, the entertainment was decidedly a novel affair and will be long remembered by the members of the circle and their many friends." This "Willis" circle was organized last September, and numbers eighteen members. They are blessed with the best of recommendations. They declare that willingness and good nature are their prominent characteristics. Certainly, with plenty of the sunlight and fresh air of society theirs, it is not strange that they "look forward from one meeting to the next, anticipating much pleasure and profit." The "Dorionic" circle of BIDDEFORD, MAINE, laid aside its studies for one evening and held special exercises in honor of Longfellow. This circle has been doing exceptionally thorough work in chemistry this winter. A full course of lectures on the subject has been delivered in connection with the study of the text-book on chemistry.—One of the most novel programs with which we have been favored is from RICHMOND, ME. The "Merry Meeting" circle send it. A gay Japanese napkin, on which the title page, the committees and the exercises all find place. As in several of the programs, we find that tableaux take a prominent position. No better interpretation of striking scenes is possible than by *tableaux vivants*, and a strictly literary program can be readily enlivened by a well selected scene.—The Longfellow memorial day was especially and appropriately observed at OLD TOWN. The pleasant and commodious vestry of the Congregational Church was filled with the members and invited guests to the number of a hundred or more. The program

—an excellent one—contained beside its essays, readings and music, extracts with tableaux from “Evangeline,” “Miles Standish” and “Excelsior.” The “Old Town” circle is one of our new friends, having been organized last October; young yet, but vigorous, for it numbers already thirty-five members.—At CASTLETON, VERMONT, the “Lone Pine” circle is doing very thorough work, and rejoicing in a good organization. Their Longfellow memorial program was very complete; though they have but fourteen members, they seemed to have no difficulty in securing the music, essays and readings for a full program. The closing feature of the evening was an informal *conversazione* over their ice cream and cake, on Longfellow. With such an entertaining “something to talk about” the circle must have gone home full of ideas and happy thoughts.—A very energetic effort was made recently by the “Berkeley” circle, of BOSTON, MASS., to bring the Chautauquans of Boston and vicinity into more intimate relations. A union meeting was the means chosen, and Longfellow’s day was the time. Fifteen circles in all participated in the exercises; among them were: “Phillips,” of South Boston; “Hurlbut,” of East Boston; “People’s Church,” of Boston; “Parker Hill,” of Roxbury; “Floral Society,” of Tremont Temple; “Longfellow,” of Cambridge; “Pericles,” of Brighton; “Clark,” of Jamaica Plain; “Sherwin,” of Dorchester; “Henry M. King,” of Roxbury; “The Pilgrims,” of Dorchester; “Bromfield Street Church,” of Boston; “Berkeley,” of Boston. About fifty persons were present. Letters were received from the Rev. B. P. Snow, the president of the class of ’86, and Prof. W. F. Sherwin, who were unable to attend. This brave effort to strengthen the bonds of fellow-feeling will not be fruitless, we are sure. “Berkeley” circle, in undertaking such a reunion, has instituted one of the most practical and useful ways of increasing the breadth and strength of the C. L. S. C. It is to be hoped that it will be made an annual feature of the Chautauqua work of Boston.—The “Mount Hope” circle, of BRISTOL, R. I. believes in the liberal use of printer’s ink. Accordingly, all of their meetings have been reported in one or both of the papers most widely circulated in the town. Three of the reviews which have been read before the circle this year have been printed, and an essay read at their recent Longfellow memorial also appeared. This “Memorial” was an unusually pleasant affair.—The circle at GOUVERNEUR, N. Y., is the outgrowth of Mr. Hurlbut’s teachings at Thousand Islands Park, one year ago last summer. It has been successful beyond the hope of its organizers. At present its membership is about twenty-five, and these are all hard workers. Some two hundred of the circle’s friends met with them on February 27th to celebrate the day. There was music, recitations in costume, and an essay on a

splendid subject—"Acadia"—and finally, a pleasant hour of social life. This circle at Gouverneur has done great good in the community and the members seem to grow more enthusiastic the longer their connection with the circle lasts. Two features of their meetings which they find very interesting are the review contest and conversation on a certain given subject.—A flourishing C. L. S. C. exists at AMSTERDAM, N. Y. It is composed of forty members—double the number of last year. The circle recently celebrated Longfellow's memorial day in a pleasant manner. The program consisted of an essay on the poet and his works; music, songs and readings, selected from Longfellow; "The Black Knight" and "Nun of Nidaros" were read and illustrated by tableaux. The program closed with a series of tableaux, taken from "Evangeline," portions of which were read.—The "Courtship of Miles Standish" was dramatized for Longfellow's day by the circle at JOHNSTOWN, N. Y., and with music and a few additional numbers, made a very interesting program. This is the first memorial day observed by the circle, and it brought together many old Chautauquans, who professed themselves highly pleased with the vigor of the circle. We hope that if any of these "old Chautauquans" are not lending to the vigor of the circle, they will hasten after this happy evening to renew their allegiance.—The program of a literary and musical entertainment given on this chosen day by the circle at EAST NORWICH, N. Y., has reached our table, in company with a genial letter about the C. L. S. C. life of that town. "We have been sarcastically spoken of," our friend writes, "as that *great* Chautauqua Circle, and no doubt we have rather bored the people by our enthusiasm. Hawthorne compared religion to a painted window in a cathedral; seen from the outside it is not admirable, and one wonders that they can be so much praised; for it must be viewed from the inside to see its full beauty. The C. L. S. C., it seems to me, could be fittingly compared to the same thing. You see, I know, for it is not very long since I was outside myself. Our entertainment was a great success. We have considerable talent in our circle, both elocutionary and musical."—Another delightful entertainment was the social given by the "Alyssum" to the "Argonaut" circle, of BUFFALO, N. Y. The program was brightened by an excellent variety of tableaux, refreshments were bountifully served, and the delighted guests departed after a hearty vote of appreciation of the pleasant evening with Longfellow.—The program carried out by the "Allegheny" circle of PITTSBURGH was characteristic of the circle—that is, very good. This circle always does something good.—The "Pansy" circle, of CHESTER, PA., observed the "Longfellow Memorial" in a very appropriate and spirited manner, by a

program which included the singing of Chautauqua songs, sketches and recitations. A pretty feature was the reading of "The Nun of Nidaros," with organ accompaniment and tableaux. The greater portion of "The Courtship of Miles Standish," with numerous tableaux, was given. The evening's program ended with choruses from the songs. This "Pansy" circle was organized last October and its membership is divided between "Pansies" and "Plymouth Rocks."——Another Pennsylvania circle formed last fall is the "Longfellow," of PHILADELPHIA. The circle is made up of eight members. Their regular order of exercises is capitally arranged to cover all points of the readings. They invited the "Sappho" circle to join with them in commemorating Longfellow's day, the members of both circles to take equal part in the entertainment. They spent such a very pleasant and profitable evening, that it has been decided to keep all the memorial days in the same manner.——In a recent letter from a WASHINGTON, D. C., friend, we find the following interesting points about one of the circles of that city: "The weekly meetings of 'Parker C. L. S. C.' during the past winter have been, for the most part, very interesting. The annual celebration of Longfellow's day was no exception. An appropriate program was carried out, one of the best features of which was the account of the poet's life. The whole time was divided into periods and assigned to different members. After the exercises, Mr. Lowe, Engineer U. S. N., at whose home the circle met, kindly gave a most interesting and instructive account of the Greely Relief Expedition, by previous request, he having been one of those brave rescuers. He was listened to with attention, and the scenes he pictured were rendered more vivid by the exhibition of various articles of fur clothing worn in that region, by photographs of the relief ships, natives, etc., and also by some of the identical food upon which Greely's party were subsisting when found."——The "Crescent" circle, organized at FREMONT, IND., three years ago, would like to extend greeting to their many fellow students, and claim a place in the great family of workers who are reading the same books and thinking the same thoughts. The circle is not large, but has abundant hope and ambition to make the work enjoyable. Every meeting is a treat to them, they say. They, too, observed Longfellow's day and carried out an excellent program, in which some of the "little folks" carried off high honors for their share in tableaux and charades.——At HAMPSHIRE, ILL., the most interesting memorial service of the year was that celebrated on Longfellow's day. A large audience collected to listen to the exercises and went away seemingly well satisfied with the ability of Chautauquans to furnish an evening which should be both literary and social. This circle is doing good work in Hampshire. The

members are more active than ever, and there is a prospect of an increase in numbers.—Another delightful parlor entertainment was that at ELK HORN, WISCONSIN. The circle, bubbling over with C. L. S. C. devotion, mingled an occasional purely Chautauqua subject with the *Longfellowana*, not at all to the detriment of the program, so we think. At the close of the literary exercises, the guests, numbering about forty, were invited to a lunch, in quality and quantity “fit for a king.” Then came the “good-nights,” and each guest left with the wish that C. L. S. C. might long continue to flourish, and that such evenings might be in the ascendant among the diversions of the town.—A dainty invitation has come to us to be present at the Longfellow memorial exercises and social reunion held March 2, at the Grand Avenue Congregational Church in MILWAUKEE, WIS. The program which accompanied it has an essay subject which we hope our friends will tuck away in their memories for next year’s use. It is “The Women of Longfellow’s Writings.”—A houseful of Chautauquans and their friends gathered on the evening of Longfellow’s day at one of the delightful homes of MARSHALLTOWN, IOWA. The “Vincent” and “Alden” circles held a joint meeting which was a source of great pleasure to both the Chautauquans and guests present. The social notes of the local paper of Marshalltown contain a very complimentary reference to this pleasant affair.—Along with the notice of a Longfellow dinner given by the secretary of the circle at MAPLE HILL, KANSAS, comes a sparkling letter of the birth and growth of that same circle. Perhaps it will be more suggestive than even the pleasant exercises of the dinner would be. Our correspondent writes: “It would be too long a story to tell of the first infection of the secretary of this circle by visiting a *live* Chautauquan in Topeka during the fall of 1883. Enough to say, she ‘caught’ the fever, as the diagnosis plainly showed. The first pronounced symptom, enthusiasm, was increased by the purchase of Pansy’s ‘Hall in the Grove,’ and finally culminated in the Chautauqua brain fever. She went home and showed a ‘method in her madness’ by inoculating her friends through the loan of that same ‘Hall in the Grove,’ and she was delighted to see the usual symptoms develop in due course of time. This same secretary was dubbed the ‘She-Talker’ by her friends, but all to no purpose, so far as discouragements go, for she had the satisfaction of forming a class in November of 1883, consisting of four members. Now be it known, Maple Hill is a sparsely settled farming town, made up mostly of large farms and ranches, and this makes it the more difficult to carry on such an enterprise. We, however, read on during the winter, but were disappointed when the fall of 1884 came round, to find that

one of our most enthusiastic members had 'taken a school' some twelve miles away, and would have to read on alone, 'probably,' but before our first month had passed we had taken five new names, and had adopted two honorary members, who, although they were fully in sympathy with the movement, could not this year take up the regular line of work. So we number eight regular members, making a class of ten. It would make my letter too long to tell of all our work. Our readings have been not only profitable, but exceeding pleasant this 'long and dreary winter,' and this united class of '87-88 extend the right hand of fellowship to their comrades all over our goodly land."——

At CHANUTE, KANSAS, the memorial service was equally pleasant. The circle there is composed of seventeen members, and they all contributed their best to make the program bright and taking. The success which attended their efforts is peculiarly gratifying, when we remember that for all save one of the members this is the first year of C. L. S. C. work.——Longfellow's day was appropriately observed at CLINTON, MISSOURI, by the "Excelsior" circle, with a program modeled on that published in THE CHAUTAUQUAN. This circle, organized with four members, now numbers nineteen, all ladies. There is a growing interest in the work at CLINTON which insures future prosperity and increase of strength.——"Out among the Rockies," at BOULDER, COLORADO, Longfellow's day was appropriately observed by the circle. The meeting was made doubly pleasant by the fact that February 27th is the birthday of the hostess of the occasion.——"Central" circle of SAN FRANCISCO, CALIFORNIA, has never yet been reported to THE CHAUTAUQUAN. It was organized in 1883, and reorganized in 1884, with a membership of twenty-three, fourteen of whom belong to the "Plymouth Rocks," '88. Their plan of work is to have a special leader for each study, and a paper or select reading has a place in nearly every program. Longfellow's day was observed in a pleasing manner. In addition to a biography of the poet, each chapter of which was written by a different member, "The Bridge," and "The Day is Done" were sung, and "The Builders" given as a recitation. At roll call responses were given from Longfellow's most beautiful thoughts. A good degree of interest is manifested, and young and old find places in their ranks. Two of the members expect to graduate this year, but, writes one of them, "I shall be none the less a 'Chautauquan,' for that I intend to be through life."

A large number of new circles are reported this month, with a few which, though organized for some time, have never before reported to us. At AUGUSTA, MAINE, a circle of fourteen members was formed in October last. The circle has been so interesting that the numbers usually swell beyond the

actual fourteen—a good sign of the manner in which the circle conducts its meetings. They have observed the memorial days, and send us an excellent program of the Longfellow exercises. One of the numbers on this occasion was an original poem—from the pen of a “Pansy,” we suspect—one stanza of which we quote:

“Take whatever God sends,
As the blossoming pansies do;
He clothes them with royal grace;
Shall he not take thought for you?
Trust—for the trustful heart
Knoweth the tenderest leading,
Knoweth how certainly God
Our need and our craving is heeding.”

The “Garfield” circle, at LEWISTON, MAINE, a new circle of seven members, gives us a delightful glimpse of their C. L. S. C. hour: “Our president is a dressmaker, and ‘we girls,’ or at least four of us, work for her. We have reading in the shop nearly every day, forty minutes or more, and then talk of what we read. Almost a Socratic school in a dressmaker’s shop! Friday evening of each week the shop takes on another look. The work is put away, the table drawn out, the bright cloth laid, the lamps trimmed and burning; the members take their seats and place at the ‘table square,’ and for two or three hours we spend a refreshing and enjoyable evening. We find the programs in THE CHAUTAUQUAN very useful, but always have to add to and rearrange the parts, for we all want to do something for the next meeting. Arrangements were made by the presidents of the ‘Garfield’ and ‘Scott’ circles to have a union meeting on Longfellow day. We spent a delightful evening. The work was divided between the two circles, and we all felt much benefited by the meeting. We heartily recommend the occasional union meeting.”

There are over twenty-one regular members in the “Alpha” circle of MELROSE, MASS., though it was started only last October, and any amount of enterprise. The secretary writes many appreciating words of the C. L. S. C.: “This is my fourth year,” she says, “but I can echo the sentiment expressed by some one in the last CHAUTAUQUAN—‘Once a Chautauquan, always a Chautauquan’—and rejoice to think that it is by no means my last year. I hope to send you annual greetings from our circle, for we anticipate a future for it.”——A share of the honor which is bestowed upon the circles of ’88 certainly belongs to the “Hestia” circle, of LEOMINSTER, MASS. Their motto, *Festina lente*, they are faithfully carrying out. In addition to the Chautauqua

course, they are taking a systematic course in botany, which they expect to enjoy very much this coming summer. One of their number is a zealous student in botany and chemistry, and is a great help to them in these branches, performing all the experiments, and explaining the difficult points.

The “Gardner” circle was organized in PASCOAG, R. I., last November, with a membership of seven, which rapidly grew to its present number of twenty-two, all, with the exception of two, “Pansies,” of the class of ’88. The circle was named in honor of Mr. E. P. Gardner, of Norwich, Conn., to whose inspiring words it owes its formation. The interest of the members is steadily increasing, and although few of the number are persons of leisure, yet the earnest work accomplished by this circle, we are confident, would cheer the hearts of those who love the C. L. S. C.

A second circle in PLAINFIELD, N. J., has met with a cordial recommendation from Dr. Hurlbut. There is no lack of enthusiasm in the new circle. The members are thoroughly pleased by the readings, and give a hearty support to the work, writing many interesting papers. At a recent meeting it was decided that the circle be hereafter known as the “Hurlbut” circle, in recognition of the assistance which has been received by them from the able C. L. S. C. worker, the Rev. J. L. Hurlbut.—At PHILADELPHIA, PA., so a note informs us, the “Arcadia” circle was organized in February, with a membership of five, and a promise of gathering in more. —On February 16th, through the efforts of Mrs. Dr. Seeley, a circle of the C. L. S. C. was organized at JEFFERSON, OHIO, which at present numbers thirteen members. It is the first circle organized in the place, although two of the members are graduates of ’82.—The C. L. S. C. of ELLSWORTH, OHIO, has never before been noticed in THE CHAUTAUQUAN. A circle of seven members was organized there more than a year ago. Each made an effort to enlarge the circle this year, but succeeded in obtaining but one new member. It is a country place, and the members are scattered, but meet occasionally; although they can not meet often they are busy workers.

This little note from IOWA explains itself: “We are glad to announce an organized local circle in IOWA CITY, of eleven members. At our first meeting we received an invitation from the Nineteenth Century Club, to attend a lecture given by the president of the State University. Subject: ‘Our

National Constitution.' We are all enthusiastic over the C. L. S. C., are working now for a large membership to begin fall work in time."——At HOPKINTON a circle was formed last October of fourteen members. A bright, interested circle it is, too, quite up to the times in the variety and quality of its fortnightly exercises. The special days afford much pleasure to the circle. College day was spent in a half day's visit to their flourishing local college, Lenox.

At the confluence of the Vermillion and Missouri rivers in the town of VERMILLION, DAKOTA TERRITORY, a new circle came to life in November last. The circle has an active membership of twelve, consisting of lawyers, teachers, printers, university students, milliners, business men and their wives. Among the special features of the circle may be mentioned that of thorough interest in the readings; special effort to acquire accuracy in pronunciation; the utmost freedom in conversing about, and discussing questions that incidentally arise during the evening's reading; essays on the important facts of the subjects considered; and the roll call, the responses thereto being made by reciting mottoes, wise sayings, proverbs, quotations from the poets, brief descriptions of foreign countries (the assignments for this exercise having been made at a previous meeting). At its next meeting the responses will consist of three minute biographical sketches of eminent American statesmen.——Still another new circle sends greetings from the heart of the Rockies. A class of three has been formed at GUNNISON, COLORADO, and neither the small number nor their far-away home dampens their ardor. They are "greatly interested, and feel a thousand times repaid."

Several of our senior friends from NEW YORK have come in with how-do-you-dos and cheerful news this month. The C. L. S. C. readers in the "Flower City" (ROCHESTER) have not been idle during the past months, and although, like most Chautauquans, they are busy men and women—teachers, professional and business men, housekeepers and students—still they find time to keep abreast with the prescribed reading, and do not fail to attend the meetings of the circle. The circle is known as a section of the Rochester Academy of Science. By affiliating with this body they secure an excellent hall for a place of meeting, as well as increased dignity and importance, and frequently members of other sections are attracted to their meetings as interested spectators. Their circle was reorganized early in the fall, starting off with nearly twice as many members as last year. They hold

meetings twice a month, and the interest and attendance are constantly increasing. The leaders in the Academy of Science, at first somewhat adverse to connection with them, are only too glad now to welcome the circle in their monthly meetings, and the vice president of the circle is now the corresponding secretary of the Academy. There are about fifty active members, and many others are quietly pursuing the course of reading. In character they are quite cosmopolitan, representing extremes in age and character, as well as every C. L. S. C. class, from that of '82 to that of '88. The president is an old Chautauquan, and although an active business man, never misses a meeting; other leading members are quite as punctual in their attendance.—At ANDOVER the circle of nine is steadily working away, and with good results, too, for their work has brought this experience: “We all agree that the C. L. S. C. has brought a blessing and inspiration into our lives, and we give to all its projects our undivided and unswerving loyalty.”—At SOUTH LANSING the C. L. S. C. has lost one of its most devoted workers. Miss Emma Morrison, a member of the class of '84, died at her home October 21, 1884.—Another bereaved circle is that of OLEAN. Nelson F. Butler, a warm admirer of the Chautauqua work, and a leader in the “Philomathic” circle, was taken from them February 20, 1885.—“Les Huguenots,” of NEW PALTZ, N. Y., was organized in 1883, since which time the circle has increased from fifteen to twenty-seven. The programs, prepared two weeks in advance, are very bright and interesting. The circle is faithful, and work promises well for the future.—Some excellent suggestions, and aptly called too, come from WEBSTER’S CORNERS, N. Y., where the “Iota” class of Orchard Park entered upon its second year’s work last October. It is at present composed of fourteen earnest members, the classes of '84, '86, '87 and '88 being represented. The aim of the class has been to make its meetings as informal as possible, and this year it has succeeded. Among their exercises are roll call, responded to by quotations or facts, talks on some given topic, select readings, pronouncing contests, and the question box. To vary the program a “basket of facts” is sometimes substituted for the usual question box. Sometimes they have conundrums on Greek History. One feature of a recent program which gave an excellent drill, besides affording much amusement, was a Greek memory test, consisting of twelve facts from Greek History. At first the leader gives but one fact, the class repeating it. As each additional fact is given, the ones previously given are repeated in reverse order. For instance, after the

twelfth is given, all are repeated in this order—12th, 11th, 10th, 9th, ... 2d, 1st. They have also had “An Historical Lingo,” commencing about 900 B. C., and giving prominent facts in Greek History down to the year 145 B. C., when the Romans controlled Greece.

Several PENNSYLVANIA items are at hand, too. The “Emanon” circle, of WEST PHILADELPHIA, now in its second year, is meeting with good success. The members of this circle have been delighted with the studies ever since the organization—no one regrets, they say, having joined the circle. While actively engaged in the literary, historical, and other studies, they pay more attention to the scientific studies, probably because they have more advantages in that direction. The circle has access to a very fine microscope, and is one of the circles to whom Mr. Hall, of Jamestown, N. Y., sends slides with instructions regarding them, and the preparation of the same. Again, they have a good outfit of chemicals and chemical appliances for experimenting in chemistry; also the use of stereopticon views, to illustrate some studies. And while thus well equipped in various instruments and appliances to help in their studies, it should be added that they are largely—indeed, altogether—indebted to their instructor, Mr. John S. Rodgers, for the explanation of these branches of study.—At LOCK HAVEN the circle has been enjoying a good winter, and prominent in their work has been chemistry, many experiments having been performed for them by an interested friend.—At the Y. M. C. A. parlors of HARRISBURG a meeting was held on March 20th, the program of which we have received. It has some very pleasant features.

An excellent method of work has been adopted at BAYONNE CITY, NEW JERSEY. Each member makes a specialty of some subject in the course, and is prepared to furnish an article on the subject at any meeting when called upon, and also to answer any questions on that subject from the question box. The “Pamrapo” circle has ten members and one officer, a president—they do, however, have an extra official, a journalist, who is appointed at each meeting.

As enjoyable a C. L. S. C. banquet as we have heard of this year was that held in AKRON, OHIO. The circle entertained its friends royally on this occasion, some one hundred of whom were seated at the supper table. A happy surprise of the affair was an unexpected visit from Chancellor

Vincent, who responded to the toast, "Chautauqua." A fine speech was made by President Lewis Miller. "This is an age of quick things," he said, giving an apt illustration of his remark by referring to his telephone talk with Dr. Vincent, at Cleveland, forty miles away, but for which conversation, voice to voice, Dr. Vincent would not have been the guest of the Akron C. L. S. C. Because one could talk with Pittsburgh by telegraph—by the quick medium which was the product of this latter day—did not obviate the necessity for the longer or slower trip by rail, requiring hours. In fact, the telegram might be only a preliminary to the trip by car. As this is an age of quick things, so it is an age of condensed things. The student sweeps over 1,000 years of history—of great events—in the story of an hour. This is the work of the C. L. S. C. It takes these broad, quick views of the great events in the world's life. And because the C. L. S. C. student makes this general survey, we are not to infer that he is content with that. It is the hour's study in the history of the Roman empire that precedes the trip to Rome.

"Though we are so late in reporting the existence of our little circle, known as the 'Philomaths,' of ACKLEY, IOWA, we are confident that none other has been carrying on the season's work with more enthusiasm than our own. September 19, 1884, we organized with an enrollment of ten; since that time the number has increased to fourteen—all 'Plymouth Rocks.' Each member leads a busy life, yet we meet each Friday evening, and are convinced that we can not spend our few spare minutes more profitably than in following the C. L. S. C. readings. Our programs, prepared by an efficient board of three members, are based on the proposed programs of THE CHAUTAUQUAN, and are published each week in the town papers. The quotations selected as responses are brought into the circle on uniform slips of paper and are preserved in a 'Mark Twain Scrap Book.' In course of time we shall possess a very choice collection of 'gems.' Bryant and Milton days were observed in their turn, as was also a Burns day. We are all delighted with the work, and our only regret is that the wave reached us no sooner."——"We have organized in our village—BLANCHARD—a local circle of the class of '87, of eleven members. This year we have nineteen members, one being a lady seventy-four years old, a graduate of the class of '82. We hold our meetings weekly, members answering by quotations. We pursued THE CHAUTAUQUAN plan of questions and answers last year very

successfully, and are proceeding in the same way this, although our programs vary according to the option of the leader. We usually have written questions on the readings in THE CHAUTAUQUAN. The work has proven pleasant and profitable.”——Another IOWA circle from which we are very glad to hear is that of GRUNDY CENTER. They had the misfortune to have their goodly membership of fifteen of last year dwindled down to five when they started last fall, but their enthusiasm was too much for discouragement. They have “caught up” again, and now are a democratic assembly composed of three ministers, Methodist, Baptist and Presbyterian, and their wives, one doctor, four lawyers, an editor, and two school teachers, a banker’s wife, two merchants’ wives, a county officer and two farmers.

It is a matter of congratulation that the CHATTANOOGA, TENNESSEE, branch has joined the “local circles.” The cheery letter giving their past history is a guarantee that we will get the lines from them giving accounts of their future progress which they promise to send from time to time: “We have a very active little circle in Chattanooga, and think we have accomplished a fair amount of work in a short time, and under somewhat adverse circumstances. In the latter part of March, 1883, seven of us met and organized a class. We hoped and expected to increase our numbers in a short time, but resolved to pursue the course of reading and cling together, whether successful or unsuccessful in our efforts to induce others to do likewise. A month passed before we got fairly to work, the Chautauqua term being then two thirds gone. We preferred, however, doing double work and studying during the summer months to waiting until the following October to commence. We completed the first year’s reading in December, and were ready to begin the third year the first of last October. Our meetings are intensely interesting, for we are all in love with the course, and intend to finish it. Our silence respecting our circle and its work is attributable to the fact of our work and the jealous economy of every moment of time. We hope, however, to forward an account of our progress from time to time.”

“We can not do without the Chautauqua movement here,” so writes the secretary of the circle at KAHOKA, MISSOURI. The class of workers there is large, including twenty-two regular members, beside many local ones. They are studious and regular, and as a result interested. Last year this circle held

an open session in June, which was very successful, and they are looking forward hopefully to the next one.

The *personnel* of the circle at COLUMBUS, NEBRASKA, is very striking, and, we think, decidedly an advantage. Here is what the secretary says: "One farmer, one teacher (our pioneer, all honor to her), one book-keeper, and two housewives. We are also decidedly cosmopolitan; one hailing from Switzerland, another from Alsace, one from Nova Scotia, and two from Ohio. One Nihilist, four woman suffragists (the ladies included), four prohibitionists, but all enthusiastic Chautauquans. What we lack in quantity we make up in quality, versatility and power. Our Longfellow anniversary was a right pleasant affair, and instructive withal. 'Ah, that's the way you literary people entertain yourselves!' exclaimed one aged visitor. Our town is not a 'literary' one, by any means. Saloons, skating rinks and ball rooms seem to crush all upward tendencies. It is evident that a mingling of people from the four quarters of the globe has a depressing effect on public morals here."

At SALT LAKE CITY, UTAH, much more interest is taken in the Chautauqua readings this year than ever before. The circle has regularly observed memorial days. It meets every Tuesday evening. The roll call, responded to by quotations from different authors, is always profitable and entertaining. The Rev. T. C. Iliff, pastor of the Methodist Church, is the president. He is an enthusiastic leader, and frequently entertains them with accounts of his travels in many of the places mentioned in the Greek studies. In its platform the circle is broadly Chautauquan, four churches being represented. A class in the "Spare-Minute Course" has lately been organized in Salt Lake City, composed mostly of pupils from the various schools in the city. Excellent work is being done.

We are heartily glad that our loyal Chautauqua worker, Mr. Burnell, brought out the SEATTLE friends who were consenting to hide their light under a bushel. Here is a second come forward to vindicate Seattle's C. L. S. C. honor. However, it must be said, in order to in turn vindicate Mr. Burnell, that his work on his western tour was evangelistic, that his efforts to aid the C. L. S. C. was an extra labor of love, done because his heart was so warm toward Chautauqua, so zealous for her welfare that he was glad to use any effort to extend her usefulness. He was in Seattle only a few hours

and was driven with work all the time. It is not strange that he did not find the workers which now come so valiantly to the front. As we said before, we are glad Mr. Burnell has “brought them out.” And here is the second vindication. It contains much excellent news about the work in that section: “The article from K. A. Burnell is entirely behind the times. Three active circles are in Seattle, with an average attendance of forty in all. The University of Washington has just arranged a series of twelve lectures, six on ‘Chemistry,’ and six on the ‘Greek College Course.’ Professor G. O. Curme, Professor of Greek, Latin and German, is an earnest worker and enthusiastic lecturer on Greek history and literature, and four of the professors are actively engaged in the course. An executive committee of five from each circle, and two from the university faculty, have organized to hold a Chautauqua Assembly on Puget Sound the coming summer, and the teachers of the public schools are in full sympathy with the movement. The first public movement in the Chautauqua course known to me was the formation of a literary society for the study of American authors, in the Seattle Baptist Church, September, 1883, at which THE CHAUTAUQUAN was regularly read for one year. This society resolved itself into a Chautauqua circle the first of September, 1884, and engaged Prof. G. O. Curme, Prof. C. B. Johnson and J. C. Sundberg, M. D., to lecture before its members. Other circles were formed, and a general society centered in the university, resulting in the present combination, as above stated. I think there will be twenty circles in Puget Sound next year.”

THE C. L. S. C. CLASSES.

CLASS OF 1885.—“THE INVINCIBLES.”

“Press on, reaching after those things which are before.”

OFFICERS.

President—J. B. Underwood, Meriden, Conn.

Vice President—C. M. Nichols, Springfield, Ohio.

Treasurer—Miss Carrie Hart, Aurora, Ind.

Secretary—Miss M. M. Canfield, Washington, D. C.

Executive Committee—Officers of the class.

Class badges may be procured of either President or Treasurer.

The thousands of men and women who belong to the Chautauqua Class of 1885 are beginning, we think, to feel the pressure of the responsibility of their position—which is something more than that which has rested upon the shoulders of their honored predecessors; that the expectations of the Chautauqua public are becoming higher, year by year; that what was equal to the great occasion in former years will not meet the hopes that will be entertained by the masses of people who will be gathered on the grounds this year. The readings must not only be faithfully and intelligently completed, so that the diplomas may be earned and received, but it should be the purpose of every member to answer, entire, the list of questions submitted, and answer them correctly. And an organized attempt should be made to secure the attendance of all members of the class who can possibly be present on Commencement day. We should have a larger class present in front of the Golden Gate than has ever appeared in that conspicuous place.

Members should commence their preparations at once, and so arrange matters as to enable them to do their part in swelling the ranks of a class which is to honor itself by the work of the year, and by the demonstration of strength and spirit which shall bring it to the front at the supreme hour!

For Chautauqua is growing! And its career is onward, and upward, and outward! It has planted itself in hundreds of cities and villages throughout the land, and in some other lands, and in thousands of social circles it has shown an influence and potency that is not only wholesome, but inspiring and wonderful. Well may the class of 1885 sing in behalf of the whole fraternity:

“No pent-up Utica contracts our powers;
The whole boundless continent is ours!”

And the whole world, too! For Chautauqua is not only reaching outward, in all countries, but the peoples of all climes and zones are beginning to reach toward Chautauqua. Her representatives have their hands upon the machinery which moves many of the country's most important enterprises; the new administration could not carry on its work without Chautauquans, and we can not now tell whereunto this great thing will grow. For the end is not yet. Chautauqua has only made a fair beginning. Let us hope that Dr. Vincent and Mr. Lewis Miller will live to experience a long series of annual surprises at the wonderful developments and achievements of Chautauqua!

The work of generating, cultivating, and executing the first of these surprises belongs to the class of '85. Classes of former years have done so well that it will require thought, and effort, and scheming, and coöperation, and energetic pushing on the part of our class to do better! There must be hard studying, and close figuring on expenses, and a vast deal of management, and a world of rallying and enthusiastic work, if we are to have the best and the largest of classes at the foot of the Chautauqua Acropolis this year! But let us have it! Let the 'Eighty-Fivers, of all classes, sexes and colors, flock toward Chautauqua early in August, from Oregon, and California, and Texas, and Florida, and Canada, and England, and China, and India, and the Soudan, with a common impulse, and inspiration, and a common purpose to honor their *alma mater* and the cause of popular,

intellectual and moral culture, and growth, and progress, which she represents.

NOTES FROM MEMBERS.—One lady, mother of two little children, writes: “I only wish every young mother in this land could see her way clear to try this course, not only for her own pleasure, but the influence it would have on her home.”

Another from Massachusetts, “the solitary ’85” in a circle of fourteen, writes: “I read alone for a year, then succeeded in starting a circle. I have had some advantages of education, but this C. L. S. C. has made my life very different from what it would have been without it.”

From Philadelphia, likewise, report comes of good work by the ’85s, who organized the “Ivy” circle of that city.

An “Invincible” asks, “Why can not the class of ’85 have a seal for reading the course of biographies, etc., which was provided for the ’84s last year?”

A classmate now residing in Kansas challenges all his fellow-students that, being born in the year in which the battle of Waterloo was fought, he is the oldest—*no, the youngest*, who will claim his diploma this year among the “Invincibles.” Does any one dispute the honor? His letter, together with many others, will be read at our class gatherings at Chautauqua, the coming season. Let us all be there.

Those of the ’85s who can not possibly be at Chautauqua this summer, but who can visit, if for only one day, Ocean Grove, N. J., will be pleased to learn that during the Sunday-school Assembly exercises at that popular resort, C. L. S. C. Recognition services will be held, July 29th, and then those who have won their parchments can not only obtain them, but also hear eloquent words of congratulation from Dr. Vincent, who has consented to be present. All who expect to be present, and desire their diplomas,

should send their names immediately to the Rev. E. H. Stokes, D.D., the president of Ocean Grove, or the Rev. B. B. Loomis, Superintendent of Instruction for Ocean Grove Assembly, at Troy, N. Y.

It is proposed that the “Invincibles,” after their graduating exercises are over, publish in a small volume the Baccalaureate Sermon by Chancellor Vincent, Oration by Bishop Warren, Class Poem by Mrs. Frank Beard, the Memorabilia of the Class Meetings, and whatever else may be deemed of interest. Such a book could be issued and bound in cloth, in class colors, for fifty cents each, or seventy-five cents in gilt edges, if (500) five hundred copies are desired. If enough names are received by Miss M. M. Canfield, our secretary, Washington, D. C., before July 1st, arrangements will be made to issue the book.

Many inquiries are still made, notwithstanding several explanations have been given in this column, as to the purpose of the class fund. It is that we may present to our *alma mater* a suitable remembrance of the “Invincibles.” Just what it shall be will be decided by the class at Chautauqua, at such time as the largest representation may be on the grounds. Every true '85 should send their contribution at once to Miss Carrie Hart, Treas., Aurora, Ind, as it is very desirable to have as little of this business to do at Chautauqua as possible; we want all the moments then for the ever-to-be-remembered “good time coming,” when we meet as a class at the Hall in the Grove.

The Rev. J. H. Vincent, D.D., Chancellor of the Chautauqua University, will conduct Recognition services for the benefit of '85s who can not be at Chautauqua, at Ocean Grove, N. J., July 29th, and at Framingham Assembly in July—date to be announced in next CHAUTAUQUAN.

CLASS OF 1887.—“THE PANSIES.”

“Neglect not the gift that is in thee.”

OFFICERS.

President—The Rev. Frank Russell, Mansfield, Ohio.

Western Secretary—K. A. Burnell, Esq., Chicago, Ill.

Eastern Secretary—J. A. Steven, M.D., Hartford, Conn.

Treasurer—Either Secretary, from either of whom badges may be obtained.

Executive Committee—The officers of the class.

The Allegheny, Pa., circle, Class of '87, is gaining a reputation for enthusiasm and thoroughness of work. They hold regular meetings, have printed programs, and sometimes are entertained at times of meetings or excursions, at a good hotel.

The Toledo, Ohio, circle, with the Rev. H. M. Bacon for its president, issues a beautiful program of the whole course of winter meetings, with a list of the memorial days, put in handsome shape.

The officers of the class are in receipt of many pleasant letters, speaking among other things of many letters which have been written to Pansy containing suggestions about the class memorial book which is expected from her pen, and which will receive most enthusiastic welcome from every member of the class.

As the dreary winter weather is likely at last to give way to spring, Pansy blossoms are appearing on paper in preparation for their appearance in the memorial Pansy bed, which it is hoped will be a treasured improvement in the Chautauqua grounds.

Williamsburg, Kansas, writes of a circle of forty members, all but two of whom are members of '88. These two delightful Pansy blossoms, a minister and his wife, represent the class of '87.

Here is a C. L. S. C. inspiration from a lonely but enthusiastic reader in New Virginia, Iowa: “I have never had the benefit of a circle, and could seldom attend if one were here, having four small children, the eldest of whom is not yet seven, and besides, I am trying to fill my mission as the wife of a pastor whose work takes him much away from home. After the little ones are tucked up in bed, and the good-nights are spoken, I find it delightful to rest with my book and reading for an hour or more. I am determined that my husband with his studies shall not leave me far behind. Two years is a long look ahead, but I am planning for that one trip to Chautauqua, when I shall hope to grasp the hand of many an ’87.”

The Lower Oswego Falls circle, New York, is doing a most excellent work by downright hard study, in their class meetings.

The circle in Hartford, Conn., with Secretary Steven for its president, issues a fine program for its meetings, which already for this winter run into the twenties.

A new circle has been organized in Chicago, called the Oakland circle. All of its members belong to the Pansy class, though that can not account for their all being ladies. They are doing a most excellent work.

Great gifts are not given to all,
Great tasks from all not required.
The Master is just—faithful use
Of talent is all desired.

“Neglect not the gift that is in thee,”
Oh! heed well that resonant call;
Teach, write, or speak, at its bidding—
To thy work, ere the shadows fall.

—Margaret Heath, Class ’87.

Miss Eliza Gummage, a member of the class of '87, recently died at her home in Lewiston, Me. She was a devoted and enthusiastic Chautauquan.

MASSACHUSETTS.—I am a member of the Pansy class, but am entirely alone in my reading. My attempts to form a circle have not yet been successful. Not because our people are not interested in all good work, but they have some “first loves” in the way of clubs, the proceeds from which are appropriated for good at home or abroad. Still I hope that very soon they will expand their hearts and take in the C. L. S. C., the benefits of which are so many. I read with much interest all items from our class. In fact, I think the Pansy column is the first thing I look for upon the arrival of THE CHAUTAUQUAN. I am not only alone in my reading, but have not the acquaintance of a single member of the great family of “Chautauquans.” However, I am far from discouraged, and look forward to meeting a goodly array of Pansies in '87.

Miss Flora Warren Potter, a member of “Pansy” class ('87), C. L. S. C., and of “Union” circle, of Washington, D. C., died in that city on the evening of March 20th, at the residence of her brother-in-law, Geo. H. Walker, Washington correspondent of the Cleveland *Leader*. Becoming a member of the C. L. S. C., she foresaw the advantages and possibilities which it opened out before her, and though an humble toiler in the work, none loved it more than she. At five o'clock on the Sabbath afternoon following her death, at the regular Chautauqua vesper hour, funeral services were held in the Union M. E. Church, the members of “Union” circle being present in a body and rendering the music, some new and choice selections, on the occasion.

In the past winter the C. L. S. C. lost a devoted member of the class of '87 in Miss Maggie R. Elwell, of Salem, New Jersey. Most appreciative resolutions of condolence and respect were sent the bereaved friends by the Salem circle.

CLASS OF 1888.—“THE PLYMOUTH ROCKS.”

“Let us be seen by our deeds.”

CLASS ORGANIZATION.

President—The Rev. A. E. Dunning, D.D., Boston, Mass.

Vice Presidents—Prof. W. N. Ellis, 108 Gates Avenue, Brooklyn, N. Y.; the Rev. Wm. G. Roberts, Bellevue, Ohio.

Secretary—Miss M. E. Taylor, Cleveland, Ohio.

Treasurer—Miss M. E. Taylor, Cleveland, Ohio.

All items for this column should be sent, in condensed form, to the Rev. C. C. McLean, St. Augustine, Florida.

We give name, locality, and number of members of new circles formed: “Kate F. Kimball,” Minneapolis, Kan., 12; “Delta,” Norwich, Conn., 16; Orlando, Fla., 20; “Evening Star,” Torreyville, N. Y., 7; “Oird,” Oird, Mich., 27; “Lawrence,” Chillicothe, O., 18; “Souri,” Blair, Neb., 22; “Progressives,” Danielsonville, Conn., 16; “Colton,” Colton, Cal., 16; “Gleaners,” Zumbrota, Minn.; “Euclid,” Vicksburg, Miss., 18; “Olympic,” Yarmouth, Me., 14; “Adelphic Union,” Holden, Mo., 21; South Manchester, Conn., 50; oldest member seventy-four, youngest sixteen; West Lebanon, N. Y., few members; Carpinteria, Cal., 8; Hopkinton, Ia., 12; Tiffin, O., organized with 6, now 13; “St. Johns,” Toledo, O., 26, was organized by a “lone member.” In Barrie, Ontario, Canada, the enthusiasm of 5 enrolled 24 more; smallest attendance has been 16. Two teachers of Brainard, Neb., have failed but twice during the severe winter, to meet every week. Their walk is several miles to the home of an invalid, who, with them, composes a circle, “The Triangle.” “Straight Line,” Matawan, N. J., 2. These object to name, saying “it speaks of poultry.” Can not the class suggest names and have them given in our column, and be voted upon, selecting the one receiving the highest number of votes? Portland, Ind., circle desires a change of name. One from Darlington, Ind., also objects to name. One member from Chicago, Ill., suggests change of motto to “Let us be doers of the Word, not hearers only:” Matt. v:16. “Evening Star,” Terryville, N. Y., “started late, but worked hard and caught up, and have not yet had a dull meeting.” “Oird,” Mich., writes: “After starting, no one wishes to turn back.” Quite a number have written regarding their “Longfellow” day. One

circle, “Delta,” of Norwich, Conn., sends a poem respecting their “Washington” day. Want of space crowds out this, and much more.

THE PLYMOUTH ROCKS.

Why should we take with such pretense
A name so great? and in what sense
Should we be likened to the stone
Which to the Pilgrims first was known?

This answering thought then came to me;
“The teachings of our class should be
A stepping-stone for coming youth,
From seas of doubt to shore of truth.”

So, when the ocean waves of wrong
Shall dash about us fierce and strong,
We may not fear, nor be dismayed—
Truth never in the grave was laid.

And when the surging waves subside
And calmed shall be the raging tide
We, like the “rock,” may firmly stand
To “welcome home” the storm-tossed band.

These principles our lives will lead,
If to our name we give good heed;
True helpfulness to others give,
And firmness for the right to live.

Our influence will then appear,
Proving more clearly year by year,
The motto which our class now leads,
“Let us be seen by our (good) deeds.”

—*Marietta S. Case.*

Many write of the “unbounded pleasure” they find in our reading course.

“Adelphic Union,” Holden, Mo., says their circle has neither “flaw nor break.” They have sent us a program of a symposium. Each member assumed a Greek name and wore an appropriate costume. The Greek idea was carried out even to the “Master of Revels.”

“Kate F. Kimball,” Minneapolis, Kansas, sends us their *menu*. The invitation to this tempting repast we could not accept, though the food was prepared in accordance with THE CHAUTAUQUAN’S directions. We must also decline the kindly invitation to the reception tendered our president, the Rev. A. E. Dunning, by the Congregational Sunday-school Superintendents’ Union, of Boston. The card is itself a treat.

Make your items for this column very brief, as we are not allowed much space. Write *no more words* than are necessary, and yet state everything of interest.

ILLINOIS.—“The North Side C. L. S. C., of Chicago, held a public examination and reception at the parlors of the Grace M. E. Church, recently; it was a great success. There were about 150 persons present, and they evinced great interest in the exercises. We belong to the class of ’88, and this is our first reception, and we all feel delighted over our success, and we will have another in a short time. We know that our meeting will be of interest to all Chautauquans, particularly those of Chicago and vicinity.”

From a packet on the Tensas River comes a bit of history telling how one new member has been added to the class of ’88: “Having complained to a passenger on board my steamer that time hung heavily on my hands during a portion of each trip, and asked her to suggest a remedy, she immediately named the Chautauqua school, and advised me to become a member of the class of ’88, and gave me a list of questions I would be obliged to answer. As my education has necessarily been limited, I thought favorably of the scheme of self-improvement, as a relief to the monotonous long watches.” The master of the packet is now a member of the class of ’88.

Way down in Texas, at Hempstead, is a faithful band of fifteen C. L. S. C. workers of the class of '88. They are college folks, the president of the institution—Soule College—to which they belong being the president of the circle. They are very enthusiastic over the course, and do a great deal of work.

EDITOR'S OUTLOOK.

GENERAL GRANT'S ILLNESS.

The pathetic interest surrounding the illness of General Grant recalls the intenser, but not more persistent, emotions of the days when President Garfield lay dying. The strong and sustained popular interest in the illness of General Grant is shown by the constant attention of the press to the theme. No day is allowed to pass without a telegraphic bulletin informing the nation how well or ill the distinguished patient slept the last night, and reporting any change for the better or the worse. We are all gathered about the invalid chair, in which the illustrious sufferer spends his wearisome days; and any word which drops from his lips flies on the wires all over the country. It is not a passion, or a folly, or a nullity. It is a piece of modern Providential education. What distinguished patient, ever before Garfield had so large and so *near* an audience of sympathy? And who does not see in the strong-flowing tide of sympathy for General Grant another lesson of the same kind? Eminence has been honored before; but these thought-laden wires take us into the very chamber of the patient and set us all upon muffled words of regret, and pity, and sorrow. We are learning, hardly knowing it, how eminence claims our regard and commands our attention, and how rapidly we can forget our criticisms and our antagonisms when Death knocks at the great man's door. There is nothing political, nothing sensational, in this illness. No public fortunes, or hopes, or fears hang upon the event. General Grant is dying; that is all; but the man who has filled so large a space in his country's history, and dwelt so long in the world's eye, can not die without quickening all pulses, and awakening every soul to pitiful attention. Great worlds of mysterious human powers of interest and sympathy seem to open before us, and invite us within the awe and solemnity of their spiritual skies. It is better—we see why, now—to go to the house of mourning than to the house of feasting. No day of triumph in General Grant's life had such an uplifting and educating power as is borne along in the arms of these days' sickness and sorrow. Dying so among us,

the illustrious patient does indeed die for us—his death lifting us into better life.

There are many smaller lessons. What an education in gunshot wounds we got at the bedside of Garfield! What a window into one of the awfully mysterious diseases we are looking through at the bedside of Grant! Before the end comes, cancer will have parted with much of its mystery. What doctors know the nation will know; and the education will save human lives; perhaps impel men to closer and more effectual search for the causes and remedies of this terrible disease. A human interest, such as we are feeling throughout the nation and the world, has a stimulating power which no man can measure. It may be that out of Grant's dying of cancer may come discoveries of permanent and universal value to mankind.



THE UNITED STATES SENATE.

Many circumstances have combined to fix attention at the present time upon the Senate of the United States. We have seen a Republican majority confirm the appointments of a Democratic President with the calmness and dignity of a Supreme Court passing upon questions of law. The event is not a new one, but it is still an impressive one, quite as much so as the change of the presidential office on the fourth of March. We then saw a defeated party resign the control of the government with decorum and civic reverence. We waited with bated breath for the first conflicts between the President and the Senate. No conflict came. Appointments were confirmed by the Senate without partisan bickerings or lamentations. The dignity of the Senate was seen to be a splendid and honorable reality. We catch other lights shining upon the Senate by going back a little. Last year, when the English people were "mad to the verge of insanity" because the House of Lords refused to pass a bill extending the franchise, the English magazines and newspapers overflowed with commendations of our Upper House. In the light of their own trouble, the English saw how happily our fathers had founded an Upper House upon enduring foundations, and how deftly they had combined popular representation with conservative privilege in our Senate. No other American institution ever had so much first-rate advertising as this Senate of ours had last year. The praise came from the

best foreign sources; from men deeply versed in history, rich in public experience, and renowned for candid and sober judgment. What the historian sees in our Senate is an Upper House which reposes upon the elective principle, and is in fact constituted indirectly by the same vote which fills the Lower House—and therefore as truly from the people—but which has at the same time a distinct and original and impressive office in the government. The result was achieved by the fathers with a few well-aimed strokes of political art. For example, *the Senate never dies*. There are many Congresses, but there is only one Senate. There is at this time no House of Representatives; there never was an hour of constitutional history when there was not a Senate. This was achieved by a simple provision that one third of the Senators should retire biennially. The Senate must “advise,” that is to say, confirm all the President’s appointments. It is his council for considering and completing treaties with foreign powers—and without the consent of the Senate the President can make no treaty. And if the people through the House of Representatives impeach the President, it is in that august presence that he must appear and plead in his defense. The Senate has exercised all these peculiar functions during the century past, and in all of them it has displayed in the main those special qualities which the framers of the constitution sought to enlist in the service of government. It has even tried a President in the intense heat of controversies begotten in civil war—and acquitted him in the face of the clamorous dominant party to which the majority belonged. Its record is full of striking triumphs over the bitterness of party spirit. It has been judicial in its temper many a time when the air was full of rancorous strifes and malignant personalities. It has sloughed off the partisan sores caused by factional poison; and though it was seriously endangered by such devices as “the courtesy of the Senate,” it has refused to be used for narrow and selfish ends. The wranglers have passed out of sight; the Senate never dies.

The judicial character of the body is apparent in its methods of discussion. There is no “previous question” gag upon debate. A conspicuous unfairness to a member is impossible. A neat way of showing self respect by respecting brother Senators is the custom of confirming, without reference to a committee, any nomination of a Senator to an office. All other names must go to the committees; the names of Senators are honored by immediate action. Members of the Lower House are, not without reason,

jealous of the power and “arrogance” of the Senate. But the people enjoy the breadth and decorum of their Senate. It can be trusted to *judge*, to put its candid opinion into all the peculiar functions which it exercises. In the field of politics, senatorial action may be very like any other human behaving in such environments; but in the special judicial tasks of the Senators there is the serenity and probity of a court room. And yet the Senate has its dangers, and it has in recent years been close to the perilous edge of the precipice. A tendency to venal methods of electing Senators came in after the civil war, and a number of rich nobodies have disgraced the high office. It is believed that there are now a few Senators whose purses are far longer than their heads, and in some of the wrangles over recent elections of Senators, the power of money has been very freely talked of and boasted of. But in this matter the worst is over. The election of Mr. Evarts in New York is a proof that legislatures can be lifted above money influence; for if any senatorships would invite special efforts to win them by venal arts, they are those of the Empire State. When that state chose last winter one of its most eminent citizens—a man known rather as a great man than as a politician—it set an example which will have no small influence over other states. The Senate has in both parties men of great ability. There is undoubtedly a growing desire and purpose that only great men be clothed with senatorial honors. We are well past the civil war and reconstruction periods, and as we advance into happier conditions we are likely to take an increasing popular pride in our unique, original and successful Upper House. There is nothing like it in the world. It is the most conspicuous work of American political genius. The more the people realize it, the more pains will they take in filling it with great men.

SUMMER AMUSEMENTS.

We are, as a people, growing in taste for amusements. Some of the manifestations of this taste are not of an entirely satisfactory character; but there are other aspects of a very agreeable nature. Our summer amusements are in the open air. We have not yet learned to play outdoors in the winter; but we are slowly learning of the Canadians, and it is not improbable that Southern people will by and by come north in the season of short days to play in the frost of our most Arctic states. We have commented in a former

number of THE CHAUTAUQUAN upon the advantages of the Canadian winter sports. Our summer sports are in a more advanced state of development. Base ball, Lacrosse, lawn tennis and croquet are established institutions, while we are only experimenting with tobogganing. It needs no argument here to satisfy people who think at all that amusements should be in the open air and require exercise enough to increase the strength and expand the lungs. Exercise is a farce unless the heart is put into it; and play is unwise if it is not healthful. Play is primarily a demand of the physique; its value to the mind begins with the refreshment of the body by wholesome use of the muscles. The summer amusements in which women take part are above all just criticism. Croquet and lawn tennis have no doubtful elements. The exercise they afford will not content an athlete, but they are adequate to the wants of sedentary people in warm weather. There is no doubt that many persons would be greatly benefited by such games. They are too sedentary; they live too much indoors; they are too closely tied to a routine of thought or feeling. The open air, the mild exercise, the social chat, would give them a change of feeling and an agreeable exercise. Nor is there any conceivable avenue of approach for moral dangers. It is not wise for any one to make a business of croquet or of any other amusement; but the danger of excess is not worth considering. It will occur so seldom and have such limited consequences that the moralist need not post sentinels upon croquet grounds.

The “manly sports” are less satisfactory. Cricket, base ball and Lacrosse have the disadvantages following: First, they are exhibitions and public rather than social; second, they require violent exertion in hot weather; third, they are accompanied by gambling and other unmanly vices. Cricket and Lacrosse are not open to the last objection to any considerable extent. Those who engage in them are for the most part gentlemen; they are, so to say, the aristocratic games, while base ball is the great popular athletic game. It is not very old. When men now fifty were boys they did not know the modern game, though they did know a much simpler and far less strenuous practice with a soft ball—a game also called base ball. There is some reason to fear that this form of athletics is being ruined by immoral attachments and environments. Of boating we make special mention because it *requires* no overstrain—since boating need not mean boat-racing—and, indeed, is open to none of the objections urged against the exhibitory

public games. On Chautauqua Lake, in the Assembly season, boating is one of the most healthful and enjoyable recreations. The exercise can be adjusted to strength, arrested at any time or prolonged at pleasure, and the boatload is enough for pleasant society—which may, of course, be selected to suit one's tastes. Of this amusement we have only one regrettable feature to mention—it can be enjoyed only where there is convenient water.

Of the vigorous sports, we are compelled to speak with some reserve. We doubt the wholesomeness of athletics; they involve excess of exertion, and the gambler is the curse of base ball games. The other games may escape the influence of the demoralization, and cricket, polo and Lacrosse become great American exhibitory games of strength and skill. But the quiet social amusements ought to flourish among us, and indefinitely increase. The pleasure row-boat, the croquet and lawn tennis grounds, deserve our special attention. Let us play a little more. We can spare the time, for we shall live longer; we can well afford the hours, because the other hours will be worth more to us. In these quiet games we get refreshment of body and of spirit.



THE MODERN TREATMENT OF THE SOLDIER.

Decoration Day has many happy outlooks upon the humanities. Philanthropy may use it as a finger-board indicating the direction of our modern progress. The soldier's grave is its special theme; it as clearly suggests the happier fate of the modern soldier as compared with his ancient brother. Great nations have always honored their dead soldiers; it is only in modern times that nations have given their whole hearts to the living soldiers. In the long wars between France and England from the twelfth century onward, the armies had no surgeons, and medical supplies were unknown. The medical equipment of a modern army is costly and ample; and that no man may die unnecessarily, woman hangs on the verge of battle to nurse the wounded, sheltered and safe under a red cross or a red crescent. In the old navies of England and France, the men were slaves who had been captured in their own lands and sent to suffer in crowded bulkheads of ships, or in the galleys, steaming with the most abominable odors. A French duchess in the sixteenth century wrote of that "living hell," the many-oared

galley war ship of the Mediterranean. One can not recall the horrors of any battle on sea or land with composure, but the whole life of soldier and sailor in public service was in the old days full of the horrors of battle fields.

It is often said that war will eventually be stopped by the increased and perfected effectiveness of engines of war. It may well be doubted on general grounds; but it is specially true that humanity has robbed war of many of its terrible aspects; it may well be that those who open again the gates of mercy are competing successfully against those who “shut the gates of mercy” on mankind.

The modern treatment of the soldier is conspicuous in providing for his comfort. Why should England buy canned meats for her soldiers? Some crusts would have sufficed the providing spirit of an ancient general. The British army in the field must be well fed or there will be a great noise about the ears of the government. Let it be written home that the biscuits were stale, or the army went without its supper, and the newspapers will roar out the indignation of the nation. It is an immense task; but it must be accomplished; the modern soldier must have his regular meals with certainty, and the food must be good. The Mahdi has no such cares or duties. His soldiers must forage and browse as best they can. The superior power of the civilized soldier lies as much in his regular feeding as in his discipline—the feeding is an element of his discipline. The soldiers must also be comfortably clothed and sheltered. Woe to the commander who exposes his men to needless hardships. The country will not allow its loyal and brave defenders to suffer a needless deprivation or hardship. If commissaries are careless or venal, the nation will pillory them in eternal infamy. The soldier must have, even in the far off desert, many of the comforts of home or the country will know the reason why. And when the battered veterans come home, how the air rings with huzzas, how tender the pity for the wounded, how liberal the pensions for the widows and orphans of those who did not come home! Neither Cyrus nor Alexander had any such pension rolls. Rome idolized her armies, but she let them starve abroad, and forgot their families at home. This whole line of treatment means more than we can express in words. It is a very real and royal worship of the nobility which we see in the soldier. Often he is a sorry human creature, but it is almost a profanation to say so. We idolize him and his office. He is our defender, our chivalric knight, our personation of the

flag over us, and of the civilization in us. But—but—what chance does this treatment of the soldier afford for the Day of Universal Peace? Will a sword ever become a pruning hook so long as it is glorified by such a symbolism and illumined by these soft lights of pity and reverence? Let us not take too gloomy a view of the effects of our philanthropy toward the soldier. The causes of war probably lie out of the range of these influences. Wars would still be, if they were still as diabolically merciless as they were in the mediæval days when a war galley was “a living hell.” Peace is a question of universal civilization; and the pity we yield to the soldier is one of the undying agencies of universal civilization.

THE LIGHTING OF TOWNS.

Street lighting is a modern invention. The history is imperfect, but Alexander Dumas gives credit to the tradition that Naples was first lighted in the seventeenth century by the cunning of a popular and sagacious priest, who induced the people to burn votive lamps before the numerous images of St. Joseph, the patron saint of the city. In the ancient towns people went about at night with lamps; and in mediæval times crimes of vengeance and greed found shelter and safety in the gloom of unlighted thoroughfares and bridges. When lighting began with oil lamps, the situation was not much improved; the feeble glimmer of the lamp-wicks only made certain corners less gloomy. When gas lights began to be used the millennium seemed to have come, and gas was expected to abolish midnight crimes. Until about a score of years ago, there was general satisfaction with gas light. Very satisfactory results were obtained in small towns by the use of petroleum, and the only formidable difficulties were those arising from the high cost of gas in towns of moderate wealth. It almost doubled the tax-levy, and when this bill did not materially decrease the cost of a police force the tax-payers murmured. Still, the work of lighting went on, and as soon as a town became ambitious, its citizens demanded street lights of some kind. The general result has been an immense increase of the aggregate outlays for this purpose. If we take into account the growth of towns and the extension of public lighting, it is safe to estimate that the public lighting bill of the world is twenty times as large as it was fifty years ago.

The invention of electric lights has, by the superior efficiency of this method, rendered oil and gas unsatisfactory; and the electric lamp furnishes three or four times as much light as gas at the same cost. But there are a dozen or more methods of using the electric lamp, and it may be doubted that we have yet reached the end of our inventive wits in this field. It is quite probable that the electric lamp of the next century will cost far less than any now in use. We are yet in the infancy of electrical invention, and it may be wise for communities to suffer a little longer the evils of darkness in order to obtain the best appliances for public lighting. The time is at hand when all towns will have street lamps; the inventors are busy and hopeful, and a little cautious patience in the public will probably stimulate rather than discourage invention. It is a good trait in our people that they want the newest device, at whatever cost; but on the other hand the ability of A to stock the market with an inferior article discourages the efforts of B to devise a superior one. The plant for lighting a town is expensive, and can not easily be thrown aside for a better one. Besides, we are in some danger of hatching a new brood of monopolies to plague us with unreasonable exactions.

We need street lights much more than our fathers did. In large towns—and in many small ones—the din of toil does not cease when darkness comes on. There is a steady increase of night occupations. Some of these occupations are of high convenience, such as the pharmacies, the printing offices, and the depots of travel. Others are means to profitable ends for individuals. In a great city a multitude of people use the streets at night. The market gardener must be in his stall before day dawn. The daily bread is baked or distributed to depots of sale in the night time; a thousand small trades are plied in the darkness to provide the tables of the families with the necessities and luxuries of life. The result is a growing demand for artificial sunshine, and this demand will be amply met in a near future. The bright lights will do what the feeble lights partially failed to do. The night will cease to be the hour of crime. If one will but think of it, a marvelous change has come over the world since petroleum was discovered in Western Pennsylvania—which was, as it were, but yesterday. Then we had tallow dips in all but the largest towns for all lighting purposes, except when extravagant people burned on rare occasions the costly illuminating oils. To

make noonday in a whole town at midnight would have seemed a foolish dream thirty years ago. The world moves—into the light.

EDITOR'S NOTE-BOOK.

The world was never kept busier studying its geography than just at present. All quarters of the world are demanding attention. In China the Tonquin trouble has assumed such proportions that the daily papers have come out with a map of the disputed country. The Afghan difficulty has set us to locating Herat and learning how to pronounce the barbarous names; the Soudan is pinned up on everybody's wall or tucked into their note-book; the revolution in Central America demands that we familiarize ourselves with a country we never did know much about, while the Oklahoma boomers of the West and rebellious Manitoba keep us interested enough in home affairs not to forget how our boundaries lie.

One of the most popular places in Washington during inaugural week was the National Museum. During the 2d, 3d, 4th, 5th and 6th days of March 23,000 visitors were registered. Going to the inauguration, like "goin' to the Fourth," is becoming an American custom. To express the peculiar habits of those who surrendered themselves to the festivities, Washingtonians say that they have gone "inaugurating," a noun which we may be obliged to put into the dictionary if we continue to make so much of our political moultings.

A useful improvement in letter delivery is to go into effect on July 1st. A ten-cent stamp is to be provided, which, attached to a letter, entitles it to immediate delivery in all cities having 4,000 inhabitants or over, within the carrier limit of any free delivery office, or within one mile of the postoffice. It is a perfectly practical scheme, and it is apparent that there is a demand for a quicker means of communication than an uncertain and delayed letter delivery, for a class of letters which are unsuited for telegraph or telephone.

The "Bird's-Eye View of Forestry" which appears in this issue of THE CHAUTAUQUAN from the pen of the Rev. S. W. Powell will be read, we trust,

with attention. Mr. Powell is an authority on the subject, being corresponding secretary of the New York State Forestry Association recently organized at Utica, New York.

There is something new in bills of fare. For several seasons *littérateurs* have rejoiced in *menus*, with quotations. The “allusive” *menu* takes its place now. Here is a sample from a mid-Lent luncheon party in London: “Beauty draws us with a single hair” turned out to be jugged hare; “My Lord, the early village cock,” curried spring chicken; “Sing me songs of Araby,” coffee.

The year 1884 did nothing brilliant for astronomy. Nine new asteroids were added to the list, giving us a family of 244. Six new comets were noted, but none of them created much of an excitement save in astronomical circles, and even there they were rather disappointing.

Russia threatens to completely outstrip America in the production of petroleum. Their richest petroleum region has but 400 wells, while in America there are over 23,000; but one well of the 400 is declared to produce, in a day, more than all our daily production. Spouting or flowing wells throw out such mammoth quantities that the oil is allowed to run into the sea or is burned.

Everybody knows that we get our trained artisans from Europe; that our trades unions have discouraged apprentices until it is very difficult for boys to get any instruction in trades. What shall take the place of the former system of apprenticeship? It seems to us that the New York Trade Schools are framing a practical answer. The schools offer courses in bricklaying, plastering, plumbing, carpentry, wood carving, pattern making, stone cutting, and fresco painting; they furnish the best instructors procurable, and for such a cost that almost any young man can afford the instruction. These schools, if encouraged, will make it as possible for young Americans to become skilled mechanics as it is for young Germans or Englishmen.

The re-appointment of Postmaster Pearson, of New York City, will do more for civil service reform than many speeches in favor of the measure. The wholesomeness and reasonableness of appointing men because they are competent to do the work, and not because they belong to a particular party, will be more forcibly demonstrated to the country by a few such illustrations than by any other means.

In a recent crusade movement at Cornwall, N. Y., several ladies of the "Society of Friends" besieged a saloon, where they remained several days, praying and singing. Pepper was burned on the stove, the room was smoked full by the loafers, but with more valor than discretion the ladies staid in spite of every insult. It is difficult to see what has been gained. The saloon had law on its side, and the good women were arrested for trespass and fined, while the leading men of the Society published a card declaring that they did not "approve nor consent to all their unwise practices." "Wise as serpents, harmless as doves," is the only motto for those who contend effectually with the liquor traffic.

Some one has set afloat, in the midst of all the attention which has been given to the Washington monument, a touching story of a monument erected seventy-five years ago to the "Father of his Country," by the inhabitants of Boonesboro, Md. It was purely a labor of love. Near the town stands South Mountain, and on the most conspicuous point a site was chosen, where the farmers of the vicinity hauled and laid with their own hands the rocks which they themselves had quarried. Labor and time were given willingly until the work was complete. The humble, eloquent tribute still stands, a witness to the honest devotion of a faithful people.

The Afghan frontier difficulty between England and Russia came in like a lion, and bids fair, at this writing, to go out like a lamb. War was announced to hang by a thread. The Russians were declared to be advancing into the territory of the Ameer; England to be ready with an ultimatum, which might be accepted or not, as Russia pleased. Announcements were made that no such vast stores of ammunition, unbounded supplies of provisions, and altogether gigantic preparations for a ferocious war had ever before been made. British consols and Russian securities went down, and

American wheat went up. Undoubtedly the war cry has been fostered in England—a shrewd maneuver of the ministry—to take the attention from the Soudan trouble, and there is but little doubt now that the negotiations in progress will be successful. England has her hands full already, while Russia is not so hot-headed as to rush into a war without counting its cost.

A pleasant surprise has stirred literary circles this past month. A favorite magazine contributor for several years has been Charles Egbert Craddock, whose striking, original stories, full of freshness and keen observation have been constantly becoming more popular. “Mr.” Craddock kept himself quietly in St. Louis until his literary position was well established, and then went to Boston to make the acquaintance of his publishers. What was the astonishment of the latter to find that this contributor was a lady, Miss M. N. Murfree by name. The revelation was almost “too good to be true,” for no one had suspected the vigorous writing to come from other than a masculine mind. The surprise has greatly increased her popularity, of course.

The striking public spectacle of 100,000 visitors gathered to witness and to swell the pageant of the 4th of March is not yet at an end. The month goes, but it still leaves in Washington hundreds of office seekers, who have before themselves the belittling, wearing, unmanly business of *etching* their way into public service. The way in which most of the appointments have thus far been made signifies very plainly that this work is at a discount, and that we may reasonably hope to soon see the office seeker starved out.

Niagara is to be preserved. The bill which passed the New York State Assembly recently, providing for the preservation of the banks of the rivers from the works of the vandals, proves conclusively how quick we Americans are to do the right thing when we are fully persuaded what is right. Most of our wrongs against good taste and our depredations against rivers and forests are rather to be attributed to a lack of thought than, is usually the case, to be laid at the doors of avarice.

We are pained to record the death of our able contributor, Mr. Richard Grant White. For several months Mr. White has been seriously ill, though making a brave effort to continue his labors. The excellent series of articles on Good English, which have appeared in the Required Readings in this volume of THE CHAUTAUQUAN, from his pen, was interrupted by his illness, and now his death leaves a vacancy which President Wheeler kindly comes in to fill for our readers. Mr. White's work for THE CHAUTAUQUAN has very deeply interested him. As late as March 18th, he wrote us: "I may be obliged to abandon the series entirely, but this I should greatly regret." His expressions of interest in our work have been encouraging and hearty. Mr. White was only sixty-three years old at his death. He first gained public distinction as a musical critic in the *Courier and Enquirer*. He was subsequently attached to the staff of the *World*, the *Albion*, and the *Times*; and of late years he has been widely known as a writer on English topics, and especially as a critic of social and philological subjects. The news of his decease will be received with sincere sorrow by the wide circle of his friends and professional associates.

The Rev. Joseph Leslie, an honored minister, who for about fifty years served churches within a few hundred miles of Chautauqua Lake, died at his home in Cattaraugus, New York, March 13th. He was a pioneer preacher in the grove at Chautauqua many years before the Assembly was held there. As a faithful preacher, a man of fine character and sunny disposition, he has made a strong impression for Christianity on the people among whom he lived. He was a trustee of the Chautauqua Assembly, and in his death the Board lose one of their most honored members.

Mr. T. S. Arthur, author and publisher, died at his home in Philadelphia, on March 6th. Mr. Arthur is well and widely known as the author of one of the most effective temperance stories ever written—"Ten Nights in a Bar Room." Temperance people, particularly, owe him a kind remembrance for his vigorous works against strong drink.

Another writer of a widely popular book died the past month, Miss Susan Warner, the author of "The Wide, Wide World." A healthy, vigorous

story it is, and its continued popularity for a third of a century is an almost unknown phenomenon in American novels.

Think of it! On the first day of April there were eighteen inches of snow covering the Assembly grounds at Chautauqua, and the robins were singing in the trees.

A most remarkable work has been accomplished in the last twelve years by Mr. Anthony Comstock and his associates. When he put his hand to the work in 1872 there was a systemized business for spreading vile literature and vile pictures over the land. Out of 201 books published in New York, the plates of 199 have been seized, 230,955 pictures, 1,402,444 circulars and leaflets have been destroyed, and 982,010 names which the managers of this infamous business had collected from the catalogues of schools and seminaries seized. In 1877 there were six hundred open gambling dens and nine lotteries in New York City; to-day there is not an open saloon or lottery in the city where gambling can be done or lottery tickets purchased. The character of criminal papers has been so restricted by law that two out of the four worst papers scattered through the country have died of enforced respectability. The circulation of a third has been reduced from 125,000 to 67,000, and that of another has fallen fifty per cent.

There is probably no doubt in the mind of the magic-working electrician that the horse car of the future, and, indeed, the railway car, will be run by electricity, and the public has seen so much of the wonderful that it is quite ready to believe in anything promised. Already a very successful experiment with a tram car has been made in Millwall, England, where it has been proven that the electric machine necessary has in weight an advantage of five to one compared with a steam or air locomotion, that the speed can be increased much more easily than with horse power, and if necessary, a propelling power equal to sixteen horses can be gained. The changing of batteries requires less time than the change of horses; the arrangement of bells and lamps is much superior, and the cost per mile just one half. Another consideration suggests itself to the member of the Society for the Prevention of Cruelty to Animals—it will prevent much wicked abuse of horses.

There is a society in London which has undertaken to furnish two-cent dinners to poor children. The first thought would be that such a scheme must fail entirely of paying its own way, and sooner or later collapse if not endowed. The organization sends out a report, however, which declares that it is a financial success—such a success, in fact, that a second society has undertaken to furnish one-cent dinners to the very poor children. This latter enterprise, it is believed, will nearly pay its way. Here is one of the worthiest schemes for the philanthropists of our cities and larger towns, and one which can be conducted without taxing anybody's pocket-book.

Mr. Richard Proctor says: "One of the most remarkable inventions of the age is the ammoniaphone." The inventor, Dr. Carter Moffat, has for years believed the beauty of Italian vocal tones was due to something in the air of Italy. Visiting Southern Italy he made over seventy-five analyses of the air and dew, and finally became convinced that its peculiar characteristic was its saturation with ammonia and hydrogen peroxide. He has spent nine years in perfecting an instrument for inhaling vapors. Mr. Proctor says after testing the instrument: "One draught of air was inhaled, when, to our great astonishment, the intensity of the voice was about doubled, while its cleanness was as greatly increased." The inventor claims that the "employment of the ammoniaphone according to direction Italianizes the voice, and makes a weak voice or a drawing room voice strong, rich, clear, and ringing."

C. L. S. C. NOTES ON REQUIRED READINGS FOR MAY.

ENGLISH AS A UNIVERSAL LANGUAGE.

[1.](#) “Max Müller.” (1823- .) An English philologist, and one of the most eminent living orientalists. He prepared an edition of the “Rig-Veda” (a collection of ancient Hindoo literature), with a commentary. This stupendous work is composed of six volumes, and each volume contains more than 1,200 pages. He has published many philological works, and has done more than any other one scholar to awaken a taste for the science of language.

[2.](#) “The Aryan tribe.” See in THE CHAUTAUQUAN for October, 1884, the first article, “Why we Speak English.”

[3.](#) “Sanskrit.” The literary language of the Hindoos, the Aryan inhabitants of India.

[4.](#) “Dark Ages.” The name given to the time of intellectual depression in Europe, extending from the fifth century to the fifteenth. During this time some periods were darker than others, the darkest occurring about the seventh century.

SUNDAY READINGS.

[1.](#) “Gil-fil'lan,” George. (1813-1878.) A Scottish minister and author; has published many works.

[2.](#) “Herbert,” George. (1593-1632.) All English divine and poet. He was generally known as “Holy George Herbert,” his life being most exemplary and zealous.

[3.](#) “Prof. Upham,” Thomas Cogswell. (1799-1872.) An American author. For two years he was pastor of the Congregational church in Rochester, and in 1825 he accepted a professorship in Bowdoin College.

CHEMISTRY.

[1.](#) “Adams,” John Couch. (1819- .) An English astronomer, who shares with Leverrier the honor of having calculated the place of the planet Neptune before it had been seen.

[2.](#) “Leverrier,” Urbain John Joseph, lŭh-vair-yā. (1811-1877.) A French astronomer. His “Tables of Mercury” and some memoirs led to a friendship with Arago, and opened to him the door of the French Academy. He then began the study of the disturbances of the planets, which led to the discovery of Neptune. By a remarkable coincidence the existence of this planet was discovered at the same time, and independently, by Adams.

[3.](#) “Foucault,” Leon, foo-kō. (1819-1868.) A French natural philosopher, who for years turned his attention exclusively to optics, and was very successful in mechanics. He invented an electric lamp which was adopted by natural philosophers for physical experiments, and was used for lighting large factories.

[4.](#) “Brob'dig-nag,” also written Brobdingnag. The imaginary country described by Dean Swift in “Gulliver’s Travels,” which was inhabited by a race of giants.

[5.](#) “Perihelion,” per'i-hēl,yun; “Aphelion,” af-hēl'yun.

[6.](#) “Kepler,” Johann. (1571-1630.) A distinguished German astronomer and mathematician, to whom the world is indebted for the discovery of the laws that regulate the movements of the heavenly bodies. He was appointed to the chair of astronomy in Gratz University, in Styria, but in 1598 was dismissed because he professed the reformed religion; he was afterward recalled. He was an earnest disciple of Copernicus, and published in seven volumes the “Epitome of the Copernican Astronomy,” which was placed in the list of prohibited books by the inquisition. He was the author of many other works.

7. “Dr. Maskelyne,” Neville. (1732-1811.) An English astronomer.

8. “Dr. Cavendish.” See Notes in THE CHAUTAUQUAN for October, 1884.

9. “Archimedes,” ark-i-mē'des. (B. C. 287-212.) The most celebrated mathematician of antiquity. Cicero and Livy both refer to him in their writings, the former to his aptness in solving problems, the latter to his ingenuity in the invention of warlike engines. The law referred to was, that a body plunged in water loses as much weight as is equal to the weight of an equal body of water. King Hiero suspected that a gold crown had been alloyed with silver, and asked Archimedes to test it. He was trying to find some means by which he could decide the matter, when going one day to the bath tub he found it full, and immediately saw that as much water must run over the tub as was equal to the bulk of his body. He saw at once a method for determining the matter of the crown, and crying “*Eureka, eureka*; I have found it, I have found it,” he ran home.

THE EYES BUSY ON THINGS ABOUT US.

1. “Turner,” Joseph Mallord William. (1775-1851.) An English painter. His greatest fame was acquired through Ruskin, who in “Modern Painters” gives a full account of his works. As an artist he is distinguished by the strong lights and high colors of his landscapes.

2. “Ruskin.” In addition to what was said of him in the Notes in THE CHAUTAUQUAN for April, the following is given: His books upon art are the most eloquent and original ever written. His “Modern Painters” was revolutionary in its spirit, and roused the hostility of conservative art lovers. Its design was to prove the superiority of modern landscape painters, and particularly Turner, over the old artists. Its high merits gave it a fixed place in literature.

3. For Galileo, Humboldt and Hugh Miller, see former Notes in the present volume of THE CHAUTAUQUAN. “Newton,” Sir Isaac. (1642-1727.) An English philosopher. He was educated at Trinity College, Cambridge, and in 1669 accepted the mathematical professorship in that institution. While there he made his three great discoveries, of fluxions, the nature of light and colors, and the laws of gravitation. He was the author of many

works, chief among which is the “Principia,” containing his theory of the universe, which brought him fame and riches. His generosity was remarkable; he frequently entertained in a royal manner his many friends, and his kindness and courtesy toward foreigners was very marked.

“Agassiz,” Louis. (1807-1873.) One of the most eminent naturalists of the century. He was born in Switzerland, and educated in the universities of Zurich, Heidelberg and Munich. He made great researches in the field of science in different lands, and published extensively on subjects of natural history. From 1846 his biography belongs to the scientific history of the United States. For some years he was professor in the scientific school founded at Cambridge by Mr. Abbot Lawrence. For two years he held the professorship of comparative anatomy in the medical college of Charleston, S. C., and in 1868 was appointed professor in Cornell University, in Ithaca, N. Y. His eminence as a scientific man was early recognized in Europe. He was made a member of the Academy of Science in Paris, and of the Royal Society of London.

4. “Vergers,” ver'jers. 1. An attendant upon a dignitary. 2. A beadle in a cathedral church; a pew opener or an attendant.

5. “Cathedral at Pisa.” One of the most remarkable structures in the world. It was commenced in 1068 and finished in 1118. It is built of white marble, and its noble dome is supported by seventy-four pillars. It contains some celebrated works of art, and these, together with its variegated marbles and stained windows, add much to its attractiveness.

6. “Count Maurice of Nassau.” (1567-1625.) The second son of William of Orange. He was in his seventeenth year at the time of his father’s assassination, and was soon after proclaimed governor by the states of Holland and Zealand. His talents, as a general, in the troublous times that followed, surpassed all expectations.

7. “Sir Samuel Browne.” (1776-1852.) An English engineer who brought into use both chain cables and iron suspension bridges.

8. “James Watt.” (1736-1819.) A Scottish inventor. He began life as a mathematical instrument maker. In this capacity he was employed in the university of Glasgow from 1757 to 1763. He devoted his evenings to the

study of modern languages and music. In 1758 he began his experiments with steam as a propelling power, which led to the invention of the steam engine. Among his other inventions are a micrometer, a copying machine, and a machine for making drawings in perspective. For some years he devoted himself to land surveying and superintending the works on canals, to the improvement of harbors and the building of bridges. He possessed an extraordinary memory, and a more than superficial acquaintance with many sciences and arts.

[9.](#) “Sir Isambert Bru-nel'.” (1769-1849.) The well known executor of the Thames tunnel. He was a Frenchman, and was intended for the church, but manifested so strong a liking for the physical sciences, and so great a genius for mathematics, that he was allowed to follow his natural bent and adopt the profession of a civil engineer. During the French Revolution he emigrated to the United States, where he engaged in many great works. After a stay of some years he went to England and there invented a number of useful machines, and was steadily employed upon important architectural and engineering works. His greatest achievement was the construction of the tunnel under the Thames River, which after many difficulties and disasters was completed in 1843, eighteen years after it was commenced. He was a member of the Royal Society, and also of the French Academy.

[10.](#) “Cuvier,” kŭ'vyā. George Leopold Christian Frederic Dagobert, Baron. (1769-1832.) An eminent French naturalist. Such was his talent and such the perseverance with which he followed up his examination and inquiries, that he was soon looked upon as one of the best zoölogists in Europe. His fame reached the ears of Napoleon, and he bestowed upon him the most important offices in the department of public instruction. France is indebted to him for the finest osteological collection in the world; and the world is indebted to him for the great addition he has made to the science of zoölogy.

[11.](#) “Buffon,” George Louis Leclerc. (1707-1788.) A French naturalist. He studied law, but never practiced it, being strongly inclined to scientific and mathematical studies. Euclid was his constant pocket companion. After having traveled extensively, he settled down in France and devoted himself to study and to his literary works. His fame rests upon his “Natural History,” in thirty-six volumes, which has been translated into almost every

European language. It is a wonderful work in the extent of its information and its eloquence, though it is often inaccurate, and is full of wild theories.

[12.](#) “Cuttle-fish.” A molluscous shell-fish. The shell is internal, and is of a friable, calcareous substance, much used for making pounce, tooth powder, and for polishing purposes. The fishes are provided with eight arms and two long tentacles, all of which are ranged round the head and provided with suckers, which take such fast hold of objects that the limbs will sometimes tear away before they will let go. The animal is provided with an ink bag as a means of defense; when attacked it instantly darkens the water with this black fluid, and so conceals itself, and often makes its escape. They have very large eyes, which are designed for use in the night, as the animal seems to shun the light of day. It is not easily caught, and is one of the pests of Scottish fishermen, as it frequently devours the fish which have been caught in their nets.

[13.](#) “Crom'ar-ty Firth.” A landlocked inlet of the North Sea, on the northeast coast of Scotland. It is a fine harbor, eighteen miles long, running southwest, and from three to five miles broad. The largest fleet could be safely sheltered within it.

[14.](#) “Am'mon-ites.” A genus of fossil shells, somewhat like the *Nautilus*. They are found in all fossiliferous rocks from the transition strata to the chalk. They vary in size from those that are exceedingly small to those three and four feet in diameter. Some of them resemble in form the coil of a ram's horn, and others a snake coiled up. For a long time they were taken by the common people to be petrified snakes. They are so abundant in Burgundy that in some places the roads are paved with them.

ANIMAL BIOLOGY.

[1.](#) “Bar'na-cles.” A kind of shell fish, now recognized as belonging to the *Articulata*. They are provided with a long, flexible footstalk, by means of which they adhere either to fixed or floating objects. At the summit of this stalk are shelly valves, five in number, which enclose the principal organs of the animal. This shell opens and closes to admit of its spreading out and retracting a net-like organ, by means of which the animal catches its

food, which consists of small crustacea. On emerging from the eggs, young barnacles are free, and are furnished with organs of locomotion, and with large eyes, but in a short time a change occurs in them. They assume the form of their parents, and attach themselves to some place of residence. In warm climates they are exceedingly abundant, and often fasten themselves in such numbers to the bottom of a vessel as to retard its progress.

2. “Fourteen-footed Crustacea.” The beach-fleas which are found so commonly among weeds, belong to this order, and the mantis shrimp.

3. “Ten-footed Crustacea.” This order is represented by the shrimp.

4. “Cephalo-thorax,” seph'a-lo tho'rax. Head-chested. The first segment of the animal contains the head and the chest.

5. “Carapace,” car'a-pace.

6. “Epidermis,” ěp-i-derm'is. The thin, semi-transparent covering over the true skin. It is readily seen in the occurrence of blisters, as the fluid is always contained between it and the dermis, or true skin. It extends over the whole body, even the front of the eye.

7. “Acarina,” ă-ka-ree'na.

8. “Pedipalpi,” ped'i-pal'pĩ. The word is derived from two Latin words, meaning a foot, and to touch softly.

9. “Araneina,” ar-a-nĩ'na.

10. “Centipedes,” sen'ti-pēdes. The word means hundred-footed.

11. “Chitine,” kī'tēn.

12. “Chilognatha,” kī-log'na-thä.

13. “Chilopoda,” kī-lop'a-dä.

14. “Neuroptera,” new-rop'te-rä.

15. “Orthoptera,” or-thop'te rä.

16. “Hemiptera,” hě-mip'te-rä.

17. “Coccus Cacti,” coc'cus cac-ti.

18. “Nopal,” nō'pal. A plant of the genus cacti; the Indian fig.

19. “Cortes,” or Cortez, Fernando. (1485-1554.) The Spanish conqueror of Mexico.

20. “Coleoptera,” cō-lē-op'te-ra.

21. “Imago,” ī-mā'go.

22. “Elytra,” el'ī-tra.

23. Ornaments made from the sheaths of beetles. In the National Museum at Washington are many articles made by Indians and trimmed with beetle wings. There are leather capes and straps decorated with them, and head-dresses on which rows of the wings are sewed together edge to edge. Besides these, many little fancy ornaments are made of them. They may also be seen in large millinery stores, as they bid fair now to come in vogue as decorations for ladies' bonnets.

24. “Lepidoptera,” lěp-i-dōp'te-ra.

25. “The silk worm.” This insect is a native of the north of China, and a large part of the raw silk for Europe and America comes from that country. The silk worm was brought into the south of Europe in the sixth century, whence the insects, being found profitable, gradually spread into Italy and France, in both of which countries the production of silk has long been an important industry. The worms, when properly cared for, do remarkably well in this country, where, in an early day, considerable attention was given to silk culture. In colonial times the government encouraged the industry, and the production was considerable. In all the middle and southern parts both soil and climate were found favorable, and there was fair prospect of success; but for some reason the production of raw silk has fallen far behind other American industries, and certainly is not now in a flourishing condition. As late as 1844 the production was 396,700 pounds, worth \$1,400,000. It has been much less since. But with our superior natural advantages, and the very fine quality of silk that can be produced, if ever the price of labor in other countries is raised to near the same it is here, it will be profitable, and capitalists ready to invest largely in the business.

[26.](#) Such statements as these call to mind the following doggerel couplet:

“Big fleas have little fleas upon their backs to bite ’em
And these again have other fleas; and so on, *ad infinitum*.”

[27.](#) The habits of ants form a most interesting study. The males and females are provided with delicate glistening wings, the infertile females, or neuters, are wingless. The latter are divided into two classes—the workers or nurses, and soldiers. There is on the part of the fertilized females a disposition to desert the colony, but the workers, who are always on the lookout for any such manifestations, prevent it if possible. The nurses take all the care of the eggs, which are so small as scarcely to be visible to the naked eye, and which the mother never notices unless she is left alone; they also care for the young ants. When the proper time comes they cut them from the cocoons in which the pupæ envelop themselves, but from which they are unable to extricate themselves without help. Winged ants are seen most frequently in the autumn, and the greater part die before cold weather. Ants feed mostly on the sugar found in vegetable substances, and on the secretions of the *aphides*, or *plant lice*, called honey dew, which is found smeared over the leaves of plants. Some kinds of ants catch these *aphides* and carry them to their cells where they carefully provide for them in order that they may have the honey dew for food. Thus in their way they keep cows. The workers also have the care of building the habitations of the colony, forming the streets and chambers, repairing them, and fortifying them against the weather. The soldiers are larger, and are provided with stronger jaws. They do the heavier parts of the work, and the fighting for the colony. Some species of ants are slaveholders. They attack other colonies, and if not repelled, carry off the eggs and cocoons, which they care for, and when they are hatched and grown they are compelled to life service for their captors.

[28.](#) “Ascidian,” as-sid'i-an.

[29.](#) “Lamprey,” lam'pry.

[30.](#) “Marsipobranchii,” mar-sip'o-brank'ī.

[31.](#) “Lamprey pie.” Lampreys were formerly held in high esteem for the table, and it was an old custom for the city of Gloucester annually to present a lamprey pie to the sovereign. Worcester, also, is famous, for its pies and potted lampreys.

[32.](#) The American lamprey likes best shallow places in rapidly flowing streams where there are pebbly bottoms. Out of the pebbles it builds its circular nest of stones, which vary in size from a hen’s egg to a cannon ball. It carries the stones in its mouth. The eggs are laid in these nests, and the young remain here until able to care for themselves. For full account of nest building fishes see the Christmas number of *Harper’s Monthly* for 1883.

[33.](#) “Nictating membrane.” “A thin membrane at the inner angle of the eye, capable of being drawn across the ball beneath the lid, as in birds and some ruminant animals; the third eyelid.”—*Webster’s Dictionary*.

[34.](#) “Operculum,” ō-per’cu-lum.

[35.](#) “Vascular,” vas’cu-lar. Consisting of vessels. The vascular system in animals contains the arteries, veins, and like parts.

[36.](#) “Elasmobranchii,” e-las’mo-brank’ĭ.

[37.](#) “Teleosti,” tē-lē-os’ti.

[38.](#) “The electric eel.” These animals are found in several of the rivers of South America. They are three or four feet in length, though a few have been found measuring six feet. The viscera lie close to the head, and all the rest of the body is taken up by the electrical apparatus, which consists of four batteries, two on each side. These batteries consist of horizontal membranous plates, intersected by delicate vertical plates; the spaces contain a glutinous matter. The batteries are supplied with two hundred and twenty-four pairs of nerves on each side. Humboldt gave much study to these eels, and wrote a graphic description of how the Indians captured them by driving horses into the water occupied by them. The powers of the fishes were exhausted in shocking the horses (some of which died from the effects), and the eels were caught. It is said, too, that the Indians sometimes caught wild horses by driving them into the water and capturing them while they were under the influence of the shock. Faraday calculated that the eel emitted a force as great as the highest charge of a Leyden battery of fifteen

jars, having a coated surface of 3,500 square inches. The most powerful shocks are felt by touching the head of the eel with one hand, and the tail with the other.

[39.](#) “Blind fishes.” These are fishes found in caves, especially in the Mammoth Cave of Kentucky. They possess organs of touch so delicate that they are able to pursue and overtake fishes with eyes, that stray into their domain. It is very difficult to capture them, their sense of hearing being as acute as that of touch. They are nearly colorless and present a ghostly appearance in the water. They vary in length from two to five inches; they are viviparous.

[40.](#) “Sargossa Sea.” The name given to that part of the Atlantic Ocean lying between 25° and 36° north latitude, and west of the Azore islands, which is covered by a kind of seaweed distributed in great masses by the Gulf Stream. Humboldt speaks of it as “that great bank of weeds which so vividly occupied the imagination of Christopher Columbus, and which Oviedo calls the seaweed meadows.” On his first voyage Columbus passed through this sea, which caused great alarm to his companions, who thought there must be rocks or shoals near. The quantity of the weed is such as often to impede the progress of vessels.

TALK ABOUT BOOKS.

The histories of James Anthony Froude are best appreciated by and adapted to those who already have a good knowledge of history. They are like studies in higher mathematics, which always demand a careful preparation in the branches preceding them. All who read Thomas Becket^[B] will readily assent to this. Without some knowledge of the “Constitutions of Clarendon,” one could hardly gather from this book what the beginning of the trouble was about, and would lose much of the enjoyment to be had in the fine analysis of the event. The first chapter contains a few incidents illustrative of the spirit of the times; then comes a very brief sketch of the famous archbishop, up to the time of the rupture of friendship between him and the king from that time until his murder in the cathedral of Canterbury, his life and characteristics are very fully drawn. The book lacks entirely that which no good book should ever be without—a full index.

Abridged dictionaries have been among the most unsatisfactory works which we have ever owned. They never cover the ground. A fresh attempt to make a complete, compact work has resulted in a book that no one need hesitate to recommend. It has been revised from Webster’s unabridged dictionary, and the editing has been subject to President Noah Porter. Several plans have been adopted for saving spaces, which neither cheapen the work nor injure the quality. The abridgment has been accomplished, we believe, after carefully comparing the abridged and unabridged works, without sacrificing either pronunciation, definitions or derivations. The aid of examples and synonyms is lost in the smaller work. The invaluable appendix of the larger work is very adequately represented by a pronouncing vocabulary of Biblical, classical, mythological, historical and geographical proper names. The cost of the book places it in the reach of almost every one. We have felt for a long time that there was no really desirable dictionary of low price which we could recommend willingly to our C. L. S. C. readers. This work will fully meet their needs, and we take pleasure in calling attention to it.^[C]

A study of frontier life and government is to be found in “Mining Camps.”^[D] As one reads the book the old saying “One half of the world does not know how the other half lives,” recurs again and again to the mind. That such great organizations should have been in existence, governed by local laws devised by themselves to suit the necessities of their condition, and carried up to a high state of development, while other parts of the country were almost in ignorance concerning these commonwealths, seems hard to understand. The book is not designed as a technical history of mining. Ancient and mediæval mining systems are examined, and the development of their institutions carefully traced. The greater part of the book is devoted to the study of camps in the remote West.

The “Historical Reference Book”^[E] at once takes its place among those works which cause one to wonder how he ever did without them. It comprises a chronological table of universal history, a chronological dictionary of universal history, a biographical dictionary, and geographical notes. Great care has been taken in the biographical dictionary to select all the names of those men who have a strong claim to distinction, and from the list, which is necessarily limited, those have been omitted whose renown is fleeting. For those who can not provide themselves with cyclopædias, large dictionaries, and books of reference, we know of no work better calculated to meet their needs, and those who have these other helps at hand will find this the most convenient for brief notices. It is especially adapted to the use of students.

In “Workday Christianity”^[F] the names of tradesmen, as “The Carpenter,” “The Potter,” etc., have been used as the subjects of chapters, and the history of each calling is briefly given, from the earliest Bible times down to the present. They are then used as figures, and around them are draped moral lessons from which may be gathered many useful suggestions. There is, however, a gloomy outlook spread before the Christian; his life is made to seem only as “a life of work, of trial, of tears and fears, of conflicts fierce and long.” In the author’s denunciations of hypocrisy, style, cant and caste in the churches, he inconsistently pays them the high honor of allowing them to overshadow all else. He never sees the true, the beautiful and the good existing there also. Such sentences as the following have a wrong tendency and do harm: “It was not considered a disgrace in those days to ply a trade.” “How many rich young ladies would scorn to associate

with the sons and daughters of our workmen.” For some strange cause there is a large class of laboring people who are always debasing themselves by supposing other people feel above them. They are constantly snubbing themselves, in the fear that somebody is going to do so. This feeling should never be fed by a religious book. The author stands on the wrong side of many questions he attempts to handle.

The most pleasing observations of nature at present being contributed to our literature are those by John Burroughs. Most writers in their descriptions of the outside world are one-sided. They see the landscape but forget the sounds. Burroughs never does this. He catches everything: the dew, the color, the sound, the accent of the country-folk, the lay of the land, the build of the plow. In his “Fresh Fields”[\[G\]](#) the effect is exactly what the walk through the fields would have been. A vivid, fresh, constantly changing panorama is spread before you. The style suits the shifting scenes. It is not “fine writing,” but it is clear, plain and appropriate; like the corduroy trousers, short coat, and top boots which form the outfit for tramps like those of Mr. Burroughs, it is not elegant, but exactly “the thing.” While the observations of flower and bird and sky are so exact and pleasing, there is much “humanization of nature.” He is not so enamored with the fields that he can not take a genuine interest in men. The most delicious story we have read for a long time is his “Hunt for the Nightingale.” No knight in fiction ever followed his lady-love more eagerly than does this ardent wooer his Philomel, and it has been a long time since we have been more eager to have a story turn out well.

In “Letters to Guy”[\[H\]](#) boys will find an interesting book. These letters are written from Australia, by a mother to her son left at home in England. They tell of the voyages from one place to another, of the places visited, of the people, and of the natural history of the country. They are written in a bright, racy style, and are so homelike that any boy could easily forget they were penned by a titled lady, and imagine they might be his own mother’s letters to himself.

In “How to Get On in the World”[\[I\]](#) will be found a full account of the life and literary works of William Cobbett. In his preface the author says: “It is thought that an account of the life and writings of one of England’s most powerful writers and most remarkable characters, with some of the

best productions of his pen, can not fail to be useful.” And a very useful and entertaining book he has succeeded in giving to the public. The making it serve the double purpose of biography and autobiography affords, as is always the case, a pleasing variety. His early history, his experience in the British army, in the United States, and as an editor, his trial and imprisonment for the libels he placed on government and individuals, and all of the leading events in the stirring life of this great political writer are clearly set forth. There is also a full account of his works, which are very numerous. Better than any theoretical treatise on this subject is the history of this self-made man, conquering difficulties and winning successes along the lines in which he sought it.

A story of the times of Wyckliffe is given under the title “Dearer than Life.”^[J] One of the best means of doing good now in use is that of teaching the young people useful lessons in the form of these attractive historical novels. In this one, the fortunes of a family who were for a long time divided in their opinions concerning the doctrines of the great reformer are narrated, and are so closely interwoven with the real history of the times that there can be no skipping of the facts for the sake of the fancy.

Of the recent text-books published for use in schools, on physiology and hygiene, none deserves higher commendation than “Our Bodies, and How we Live.”^[K] The lessons are all so arranged and expressed as to awaken and hold the attention of the scholar, and can not fail, especially in the hands of a skillful teacher, to make this important study an exceedingly interesting one. The effects of strong drink on different parts of the system are carefully shown. The numerous illustrations are very clear, and so well labeled that they perfectly supplement the lessons and leave no chance for misunderstanding or mistake. The book contains a glossary and an index.

Two little books by Charles Kingsley,^[L] put into the hands of children who have been taught to love good reading—and indeed the books of themselves would teach any child to do this—would prove a treasure-house to them. The prefaces alone, with their cordial, sympathetic greeting, their natural, straightforward statements, and their spirit of love and reverence, are worth the price of the books.

Any one who has had experience in arranging tableaux knows how true it is that there is a false and a true way of producing effects. Not knowing how to drape, to select colors, to arrange a group, to copy this or that, spoils many tableaux and discourages managers. We are glad to find a suggestive book on this difficult art.[\[M\]](#) Without any theorizing the authors teach us how to do by plunging *in medias res* and producing the tableaux before our eyes. The book describes twenty-four tableaux, but the variety of subjects is such that study of them furnishes a very complete drill for producing any desired effect.

A good game will occasionally fill a niche in an evening, in a way entirely its own. We believe we have found two such in Miss Alice M. Guernsey's Shakspeare Game,[\[N\]](#) and Elements and Compounds.[\[O\]](#) The games are pleasing variations of the well known game of "Authors." The latter is particularly novel in its arrangement, and local circles who want to fix in mind the troublesome "compounds" will find it very useful.

It is a very convenient thing for a reader of history to have at hand a chart which gives in brief the synchronological events of nations. So many charts of this kind, however, are cumbersome, that the trouble of using is almost as time-taking as that of consulting books. A chart without this drawback is the "Concentric Chart of History"[\[P\]](#) which Dr. Ludlow has recently published. It can be held in the hand when in use, and folds up into small compass. It contains all the facts which readers ought to go to a chart for, and some interesting items on the useful arts, on sculptors, artists, and literary characters. Altogether it makes a very convenient reference table for a reader of history.

The "Common School Compendium"[\[Q\]](#) is a little volume, intended, the author says, "to serve several purposes—to provide graduates of high schools and colleges a quick means of reviewing the work of early school days; to give to teachers a reliable hand-book of knowledge they are expected always to have at command, and above all to provide that large class of young people who are striving in the privacy of home to master the difficulties of a systematic course of study, a work that should do away with the necessity for large numbers of text-books." It outlines, and gives brief lessons in, geography, arithmetic, grammar, natural history and history. It will be found by all to be a valuable reference book.

GORDON IN THE SOUDAN.—“I have certainly got into a slough with the Soudan; but looking at my banker, my commander-in-chief, and my administrator, it will be wonderful if I do not get out of it. If I had not got this Almighty Power to back me in His infinite wisdom, I do not know how I could ever think of what is to be done. With terrific exertions I may in two or three years’ time, with God’s administration, make a good province, with a good army and a fair revenue and peace and an increased trade, and also have suppressed slave raids, and then I will come home and go to bed and never get up again till noon every day, and never walk more than a mile.”[\[R\]](#)

BOOKS RECEIVED.

Evolution and Christianity, or an Answer to the Development of Infidelity of Modern Times. By Benjamin F. Tefft, D.D., LL.D. Boston: Lee and Shepard. New York: Charles T. Dillingham. 1885.

How to Do It. By Edward Everett Hale. Boston: Roberts Brothers. 1884.

Weird Tales. By E. T. W. Hoffmann. A new translation from the German. By J. P. Bealby, B.A. In two volumes. New York: Charles Scribner’s Sons. 1885.

Words and Ways; or, What They Said, and What Came of It. By Sarah J. Jones. New York: Phillips & Hunt. Cincinnati: Cranston & Stowe. 1885. Price, \$1.00.

Edward Arnold as Poetizer and Paganizer, Containing an Examination of the Light of Asia for its Literature and for its Buddhism. By William Cleaver Wilkinson. Funk & Wagnalls. New York: 1884.

The Clerk’s Manual of Rules, Forms, and Laws for the Regulation of Business in the Senate and Assembly of the State of New York. Including Croswel’s Manual. Albany: Weed, Parsons & Co. 1885.

Consolation. A Special Collection of Standard Hymns, Tunes, and Chants for Funeral and Memorial Services, together with suitable “Gospel Songs,” New and Old, designed to Comfort those Who Mourn. Edited by James R. Murray. Cincinnati: Published by John Church & Co.

Serapis. A Romance. By George Ebers. New York: William S. Gottsberger, Publisher. 11 Murray Street. 1885. Price, paper cover, 50 cents.

A Railroad Waif. By Mrs. C. B. Sargent. Cincinnati: Cranston & Stowe. New York: Phillips & Hunt. 1885. Price, 75 cents.

Mind-Reading and Beyond. By William A. Hovey. Boston: Lee and Shepard. 1885.

The Three Pronunciations of Latin. By M. M. Fisher, D.D. LL.D. New York: D. Appleton & Co. 1885.

The Story of the Resurrection By William H. Furness, D.D. Philadelphia: J. B. Lippincott & Co. 1885.

Composition and Rhetoric. By G. P. Quackenbos, LL.D. New York: D. Appleton & Co. 1885.

Organic Chemistry. By Ira Remsen. Boston: Ginn, Heath & Co.

Continuity of Christian Thought. By A. V. G. Allen. Boston: Houghton, Mifflin & Co. 1884. Price, \$2.00.

The Hallam Succession. A Tale of Methodist Life in Two Countries. By Amelia E. Barr. New York: Phillips & Hunt. Cincinnati: Cranston & Stowe. 1885. Price, \$1.00.

The Lenâpè and their Legends. Library of Aboriginal American Literature. By D. G. Brinton, M.D. Philadelphia. 1885.

The Open Door. The Portrait. Two Stories. By the author of A Little Pilgrim, and Old Lady Mary. Boston: Roberts Brothers. 1885. Price, 75 cents.

Tales from Shakspeare. By Charles and Mary Lamb. Edited for the use of Schools. Boston: Ginn, Heath & Co. 1885.

[B] Thomas Becket. By James Anthony Froude. New York: Charles Scribner's Sons. 1885. Price, 50 cents.

[C] Condensed Dictionary of the English Language. Edited under the supervision of Noah Porter, DD., LL.D. By Dorsey Gardner. New York and Chicago: Ivison, Blakeman, Taylor & Company. Price, \$1.80.

[D] Mining Camps. By Charles Howard Shinn. New York: Charles Scribner's Sons. 1885. Price, \$2.00.

[E] The Historical Reference Book. By Louis Heilprin. New York: D. Appleton and Company. 1885. Price, \$3.00.

[F] Workday Christianity. By Alexander Clarke. Cincinnati: Cranston & Stowe. New York: Phillips & Hunt. Price, \$1.00.

[G] Fresh Fields. By John Burroughs. Boston: Houghton, Mifflin & Co. 1885. Price, \$1.50.

[H] Letters to Guy. By Lady Barker. New York: Macmillan & Co. 1885. Price, \$1.50.

[I] How to Get On in the World, as Displayed in the Life and Writings of William Cobbett. By Robert Waters. New York: R. Worthington, 770 Broadway. 1885. Price, \$1.50.

[J] Dearer than Life. By Emma Leslie. New York: Phillips & Hunt. Cincinnati: Cranston & Stowe. 1885. Price, \$1.00.

[K] Our Bodies, and How we Live. By Albert F. Blaisdell, M.D. Boston: Lee and Shepard. New York: Charles T. Dillingham. 1885. Price, 60 cents.

[L] The Heroes. Price, 30 cents. Madam How and Lady Why. Price, 50 cents. New York: Macmillan & Co. 1885.

[M] Artistic Tableaux, with Picturesque Diagrams and Descriptions of Costumes. Text by Josephine Pollard. Arrangement of Diagrams by Walter Satterlee. New York: White, Stokes & Allen. 1884.

[N] Quotations. A Shakspearean Game. By Alice M. Guernsey. Chicago: S. R. Winchell & Co. Price, 25 cents.

[O] Elements and Compounds. A Chemical Game. By Alice M. Guernsey. Chicago: S. R. Winchell & Co. Price, 25 cents.

[P] Ludlow's Concentric Chart of History. By James M. Ludlow, D.D. New York City: Funk & Wagnalls. Price, \$2.00.

[Q] The Common School Compendium. For Home Students and Teachers. By Mr. L. J. Lanphere. Chicago: Fairbanks & Palmer. 1885.

[R] Chinese Gordon, the Uncrowned King. His character as it is portrayed in his Private Letters. Compiled by Laura C. Holloway. New York: Funk & Wagnalls. 1885. Ribbon-tied. Price, 25 cents.

CHAUTAUQUA, 1885.

Chautauqua is the original recreative and educational summer resort on Chautauqua Lake;

Chautauqua is the center of an elegant and literary social life;

Chautauqua is the first of many similar movements in all parts of the land, and the one from which they have received their idea and inspiration;

Chautauqua is the seat of the world-wide "C. L. S. C." (the Chautauqua Literary and Scientific Circle), which enrolls more than fifty thousand readers, and provides more than thirty distinct courses of reading and study for persons of all ages and degrees of culture;

Chautauqua is a place of rest and recreation; with grounds, high, dry, perfectly drained, clean, delightful; with three lovely natural plateaus rising from the lakeside to an elevation among the very highest on the lake. The sanitary regulations are scientific and effective. The healthfulness of the place is not excelled in America.

Chautauqua has a charming hotel, the Hotel Athenæum, one of the most elegant and substantial summer hotels on the continent. Its lovely outlook on the lake, its ample piazzas, spacious halls, parlor and dining room render it equal to any hotel outside of New York City.

Chautauqua provides cottage-boarding at all rates, and persons preferring cheap board to that of the more expensive and elegant Hotel Athenæum can easily find it.

Chautauqua is CHAUTAUQUA. The name of the ground is CHAUTAUQUA. The landing is CHAUTAUQUA. The postoffice is CHAUTAUQUA. The express office is CHAUTAUQUA. It is not "Point Chautauqua" or "Chautauqua Point," or "Chautauqua Lake," but simply CHAUTAUQUA, N. Y.

Chautauqua is the children's paradise. Games, romps, bathing, boating, calisthenics, roller skating under judicious control, bonfires, concerts, stereopticon exhibitions, a splendid museum of oriental curiosities and pictures, a useful "hour-a-day" during the Assembly season (if children wish it) of lessons, story-telling, and songs—all these make Chautauqua a most charming resort for children.

SUNDAY SCHOOL NORMAL WORK.

At the summer session of the Assembly the normal work is conducted in five departments, viz.:—

1. The Children's Class, for young people, taught by the Rev. B. T. Vincent.

2. The Intermediate Normal Class, for advanced scholars and teachers, also taught by the Rev. B. T. Vincent.

3. The Sunday School Normal Class, in the two sections of the Bible and the Sunday School, taught by the Rev. J. L. Hurlbut, D. D., and Prof. R. S. Holmes.

4. The Advanced Normal, conducted by the Rev. A. E. Dunning and the Rev. Frank Russell.

5. The Primary Teachers' Normal Class, for the instruction of teachers of little people. By Mrs. B. T. Vincent.

6. Among other exercises valuable and interesting to Sunday School workers are the following: Daily Bible Reading, under the direction of Dr. John Williamson, of Chicago, Ill.; Daily Devotional Services, led by the Rev. Dr. B. M. Adams, of New York; Occasional Question-drawers and Normal Councils, under the direction of Dr. J. H. Vincent; Sunday School Teachers' Meetings on Saturday evenings, and the great NORMAL ALUMNI REUNION on Thursday, August 13th.

Information concerning the Normal Course may be obtained by addressing either the Rev. J. L. Hurlbut, D. D., 805 Broadway, New York; or the Rev. A. E. Dunning, Congregational House, Boston, Mass.



**THE CHAUTAUQUA COLLEGE OF MODERN
LANGUAGES.**

PROF. J. H. WORMAN, PH. D., Director.

PROF. A. LALANDE, Associate, School of French.

The College of Modern Languages, under the direction of the distinguished teacher and author, Dr. J. H. Worman, will open July 11th, and continue in session for six weeks.

Concerning Prof. Worman, it is not necessary that anything be said in this announcement. As a teacher he is unexcelled in this country. As an author of school books in language he is widely known.

Prof. Lalande, a Parisian, a thorough Frenchman, a born teacher, captivates his pupils while he leads them on to a mastery of his native tongue through his aptness to teach, distinct enunciation, and personal enthusiasm.[\[S\]](#)

For full information concerning the COLLEGE OF MODERN LANGUAGES for the coming season, address as follows: German, Italian, and Spanish, Dr. J. H. Worman, Vanderbilt University, Nashville, Tenn.; for French, Prof. A. Lalande, Bridgeport, Conn.

[\[S\]](#) Indiana Cottage, 253 North avenue, Chautauqua, N. Y., will furnish for \$55.00 for the six weeks (July 11th—August 24th) of the Chautauqua Schools of Language, room and board, including all the comforts of a quiet home, with a private table to be presided over by Prof. A. Lalande, where nothing but French is to be spoken.

THE ACADEMIA OF LATIN AND GREEK.

(Summer Term of Six Weeks.)

Professor Shumway writes to the Chancellor of Chautauqua University:

My Dear Doctor Vincent:

It gives me great pleasure to be able to offer this summer, at Chautauqua, a course in Latin and Greek of unusual merit. Of the assistant teachers, Mr. Otto is already favorably known to our pupils of last summer, and to many correspondence students as an energetic and thorough teacher. Dr. Bevier

will be a great acquisition for Chautauqua. He was graduated from Rutgers with first honors, having also during his course won honors in Latin and Greek at the inter-collegiate contest. After graduation he studied at Johns Hopkins University (which conferred on him the degree Ph.D.), and then continued his studies in Europe. He was a student at the American School at Athens, Greece, and is now an enthusiastic and successful teacher. He is the author of a paper on the *Olympieion* (in the report of the School at Athens, published by Professor Goodwin, of Harvard).

Although our session in Latin last year began a week late, and we suffered from other disadvantages, I believe our numbers in Latin reached a total unparalleled in the history of Chautauqua.

What was, however, especially gratifying, was the improved quality of scholarship manifested by students.

For this summer we offer the following course:

1. ROMAN LAW (using the Institutes of *Justinian*) with information. Not only every lawyer, but every teacher of Latin to-day should familiarize “*thon*”self with Roman law, lying, as it does, *at the base of Roman civilization*.

2. THE LATIN OF THE EARLY CHURCH FATHERS.—Recent publication and discussion have brought into such prominence the influence of the early Latin Fathers on church doctrine that *every clergyman*, present or prospective, will do well to examine this question for himself.

3. COMPARATIVE PHILOLOGY (using Halsey’s Etymology; Ginn, Heath & Co.)—(Every student preparing to enter either of these three classes should *at once* communicate with the principal, that there may be no delay at the opening of the session, in securing apparatus.)

4. PLATO.—Apology and Crito, Tyler’s Ed. (Appletons.)

5. CICERO.—*De Natura Deorum*, Stickney’s Ed. (Ginn, Heath & Co.)

6. HOMER.—Odyssey.

7. VERGIL.—Æneid.

8. HORACE.—Chase's Ed. (Eldridge & Bro.)
9. CICERO.—Orations.
10. XENOPHON.—Anabasis.
11. CÆSAR.—*De Bello Gallico* (two hours per day).
12. BEGINNERS IN GREEK. Harkness's Text-Book, last ed. (Appletons.)
13. BEGINNERS IN LATIN (THREE HOURS PER DAY BY THE INDUCTIVE METHOD, WITH CONSTANT USE OF LATIN QUESTION AND ANSWER).

Latin students must have the "Hand-Book of Latin Synonymes." (Ginn, Heath & Co.)

Special rates will be made for correspondence pupils, and all are urged to attend.

I hope you will give us at Chautauqua zealous students, who will concentrate their work on Latin and Greek, and especially two classes: TEACHERS of Latin and Greek, and those who are absolutely BEGINNERS. A clear-headed student who doesn't know a word of Latin, can, by devoting six weeks to it, FIVE HOURS per day (*Beginners* and *Cæsar*) or ONE HUNDRED AND FIFTY HOURS in six weeks—quite as much time as the average school gives in one year—make decided progress.

It is thought that teachers of Latin and Greek will find of value not only the method, but also the inspiration which indubitably does arise when teachers gather.

It is perhaps hardly necessary to add that *the use of Latin not only in elucidating text, but also in discussing syntax, derivations, synonymes, history, geography, archæology, etc., is an essential feature of our work in Latin.* Worthy of the attention of teachers is the fact that our colloquial work is not the mere parrot-like repetition of phrases of the text.

Your ob't servant,

EDGAR S. SHUMWAY,
Principal of Chautauqua *Academia*.

RUTGERS COLLEGE, February 23, 1885.

THE CHAUTAUQUA SCHOOL OF HEBREW.

DR. W. R. HARPER, of the Institute of Hebrew, Director.

The Chautauqua School of Hebrew will open August 4, at 2 p. m., and continue until August 31, at 12 m.

The tuition fee is \$10.00. This sum includes admission to the grounds.

Elementary, Intermediate, Progressive, and Advanced Classes will be organized. For further information, correspond with Dr. W. R. Harper, Morgan Park, near Chicago, Ill.

THE CHAUTAUQUA COLLEGE OF ENGLISH AND ANGLO-SAXON.

The Chautauqua College of English and Anglo-Saxon is under the direction of Prof. W. D. McClintock, who has by steadiness, fidelity, tact, and rich scholarship commanded the respect and enthusiastic devotion of his pupils during several summers at Chautauqua. For particulars concerning the Summer School of English, address Prof. W. D. McClintock, Millersburgh, Ky.

THE CHAUTAUQUA TEACHERS' RETREAT.

The Teachers' Retreat is a three weeks' meeting of secular school teachers, opening July 11, 1885, for lectures, illustrative exercises, biographical studies, and scientific experiments, combined with the recreative delights of a summer vacation and the quickening influence of the summer school. The teachers in the "Retreat" for this season are: Dr. J. W. Dickinson, of Boston, Dr. J. T. Edwards, of Randolph, N. Y., W. C. J.

Hall, Esq., of Jamestown, N. Y., Prof. R. L. Cumnock, of Evanston, Ill., Prof. C. R. Wells, of Syracuse, N. Y., Prof. W. D. Bridge, of New Haven, Conn., Prof. Henry Lummis, of Boston, Prof. E. A. Spring, of Perth Amboy, N. J., Mrs. A. L. Blanchard, of New York City, Miss Mary A. Bemis, of Fredonia, N. Y., Prof. Walton N. Ellis, etc.

Lessons in experimental science, microscopy, kindergarten, elocution, the science and art of pedagogy, penmanship and book-keeping, mineralogy and geology, calisthenics, phonography, stenograph reporting, botany and forestry, drawing, painting, needle-work, clay modeling, voice culture, harmony, organ instruction, etc.

Tickets of admission to the Chautauqua Teachers' Retreat for the three weeks in July, \$5.00. This ticket admits to all general Amphitheater exercises, lectures, concerts, etc., and to the following: The special and general exercises of the "Chautauqua Foreign Tourists'" ideal excursion through Italy, brilliantly illustrated with the stereopticon; fourteen lessons in pedagogy; fourteen lessons in the practical application of pedagogical science; four tourists' conferences; four expositions of method in chemistry; one exposition of method in penmanship; one exposition of method in elocution; two admissions to each of the several classes in the Schools of Language; two lectures on school method, by Prof. Edward E. Smith, superintendent of schools in Syracuse, N. Y.; one exposition of method in standard phonography; one exposition of method in reporting by the stenograph; ten half-hour drills in school calisthenics, etc.

For circulars, address W. A. Duncan, Esq., Syracuse, N. Y.

THE AMERICAN CHURCH-SCHOOL OF CHURCH WORK.

REV. GEORGE P. HAYS, D. D., of Denver, Director.

Dr. Hays will open at Chautauqua this season a "School of Church Work," for the benefit of the laity in all denominations and in all branches of Christian activity. [See announcements in May issue of *Assembly Herald*.]

CHAUTAUQUA IDEAL FOREIGN TOUR.

Among the most instructive and entertaining features of the Chautauqua season is the annual “foreign tour” to some of the “lands beyond the sea.” The “Ideal Foreign Tour” this year will be made through ITALY. Foreign tourists’ conferences, parlor soirees, stereopticon illustrated lectures, and a large library of well-selected works on foreign travel, with a large variety of engravings, photographs, etc., will furnish abundant enjoyment and profit to all the members of the Teachers’ Retreat and the members of any department of the Schools of Language.

SCHOOL OF ORATORY.

PROF. R. L. CUMNOCK, Director.

The school will open on the 13th of July, and continue in session six weeks. The instruction in elocution will be thorough, practical and progressive. Four classes will be organized: Juvenile, General, Advanced, Ministerial.

- TERMS: I. Juvenile Class: \$10.00 for the session; \$7.00 for three weeks; — \$5.00 for two weeks; \$3.00 for one week.
- II. General or Advanced Class: \$12.00 for the session; \$8.00 for three weeks; \$6.00 for two weeks; \$4.00 for one week.
- III. General and Advanced Classes: \$20.00 for the session; \$14.00 for three weeks; \$9.00 for two weeks; \$6.00 for one week.
- IV. Ministerial Class: \$8.00 for three weeks; \$6.00 for two weeks; \$4.00 for one week.
- V. Private Hours, \$3.00. In sections of four, \$1.00 each per hour.

For further particulars, address Prof. R. L. Cumnock, Evanston, Ill.

Prof. Cumnock will give two public readings at Chautauqua during the season.

LESSONS IN ART.

Mrs. A. L. Blanchard, of the American Art School, New York, will conduct this department at Chautauqua the coming season. She will give thorough instructions in free-hand drawing, all branches of painting, crayon portraiture, and art needle-work.

TERMS:—	10		Lessons in Drawing	\$5.00
	12	”	” Painting, Mineral, Oil and Water Colors	10.00
	10	”	” Needle-Work	5.00
	12	”	” Out-door Sketching	10.00
	10	”	” Object Drawing and Perspective	5.00

THE ORGAN.

Isaac V. Flagler, of Syracuse University, will preside at the organ. A full program for ten organ recitals will be given in an early program. Many of the organ selections are not to be found in this country elsewhere, and will be played for the first time at Chautauqua. They were sent to Prof. Flagler by the composers, whose acquaintance he made in Europe.

The principal purposes of organ-playing will be demonstrated theoretically and practically by Mr. Flagler.

1. Playing for divine services.
2. Playing for concerts and exhibitions.
3. Playing accompaniments.

Especial attention will be given to manual and pedal technique, and the art of registration, or the employment of such stops as will display not only the different degrees of power, but also the utmost variety of tone color. Terms for a course of lessons on the organ at Chautauqua, \$10.00.

KINDERGARTEN.

This department will be in charge of Miss Mary A. Bemis, a pupil of Mrs. Kraus-Boelte, of New York, and for years teacher in this department at Chautauqua. An opportunity will be given, as last year, to observe the children under training by Miss Bemis. After each recitation, parents and teachers may receive practical instruction in the use of kindergarten material and the application of kindergarten principles to home and school life.

PHONOGRAPHY AND THE STENOGRAPH.

This department will be in charge of Prof. W. D. Bridge, of New Haven, Conn., for thirty years a shorthand writer, and for many years a practical shorthand reporter. For the last four years he has been engaged constantly with Dr. Vincent as shorthand secretary and assistant. Classes for beginners and advanced pupils in "Standard Phonography" will be organized. There will also be organized in this department of shorthand one or more classes in instruction on the "Stenograph," the newly invented machine for practical reporting. A competent teacher will be in attendance from July 11th to August 24th.

TERMS: \$10.00 for either department, "Standard Phonography," or the "Stenograph," in classes; seventy-five cents per hour in private.

For full information concerning this department, address Prof. W. D. Bridge, New Haven, Conn., up to July 1st; after that, at Chautauqua, N. Y.

THE CHAUTAUQUA MISSIONARY INSTITUTE

Will open Saturday, August 1st, and continue for several days, with conferences on important missionary topics, conducted by earnest men and women; with lectures, sermons and platform meetings.

MUSIC AT CHAUTAUQUA.

The arrangements for musical entertainment this year will exceed those of any former year in the history of Chautauqua.

The season of 1885 may justly be called the “summer of song” at Chautauqua. Among the attractions are the following:

A grand chorus under the direction of Prof. W. F. Sherwin, of the New England Conservatory of Music, in Boston; Prof. C. C. Case, of Akron, Ohio; and Prof. A. T. Schauffler, of New York City;

The famous Schubert Quartette (male), of Chicago, will be present from August 8th to 22d;

Miss Dora Henniges, of Cleveland, Ohio, *mezzo soprano*, will be at Chautauqua from August 4th to 18th.

Prof. Isaac V. Flagler, of Syracuse, N. Y., will preside at the great organ during the season;

Miss Adele M. Dodge, of Williamsport, Pa., will preside at the piano.

Mr. H. N. Hutchins, of Chicago, Ill., one of the greatest cornetists in America, will be at Chautauqua from August 4th to 18th.



THE FISK JUBILEE SINGERS.

The original company who so charmed Chautauqua three years ago with their matchless music, will be at Chautauqua July 11-18, and July 28-August 5. In this company are four of the original members of the earliest Fisk Jubilee Company, our old friends: Miss Jennie Jackson, soprano; Mrs. Maggie L. Porter-Cole, soprano; Miss Minnie W. Tate, contralto, and Miss Georgia M. Gordon, soprano.



THE PLATFORM.

The entire program has not been completed, but the following lecturers are engaged:

Dr. Geo. C. Lorimer, of Chicago, Ill.;

Dr. B. G. Northrop, of Connecticut;

Mrs. Mary A. Livermore, of Boston, Mass., who will lecture on Wendell Phillips, and a Dream of To-morrow;

Miss Kate Field, of Boston;

Dr. Geo. Dana Boardman, of Philadelphia, Pa.;

Rev. Robert Nourse, now of Washington, D. C.;

Philip Phillips;

Rev. J. W. A. Stewart, of Ontario;

H. K. Carroll, Esq., editor of *The Independent*, N. Y.;

Col. Homer B. Sprague, Boston, Mass., who will deliver two lectures on Shakspeare, and two on Milton;

Miss L. M. Von Finkelstein, of Jerusalem;

Bishop R. S. Foster, of Boston, Mass.;

Dr. J. M. Buckley, of New York;

W. M. R. French, the brilliant crayonist;

Dr. C. F. Deems, of New York;

Edward Everett Hale, of Boston;

Bishop Cyrus D. Foss, of Minnesota;

It is hoped that John B. Gough and Frank Beard may be present. Other names will be announced in due time.

The distinguished English orator and scientist, George Sexton, M.A., LL.D., will deliver several lectures on scientific subjects.

The full program of the CHAUTAUQUA ASSEMBLY and the SCHOOLS will be ready in a short time. Questions addressed to Dr. J. H. Vincent, Plainfield,

N. J., or W. A. Duncan, Esq., Syracuse, N. Y., will receive prompt attention.

THE FLORIDA CHAUTAUQUA.

The Florida Chautauqua is a success. Four months ago we had a dubious feeling that such an undertaking would fail of any real support in a clime which has always been so averse to adopting progressive ideas. Our healthy Chautauqua tree, we feared, would be enervated by tropical sunshine; but it has taken root with surprising readiness. And its growth is assured by the hearty northern support it is receiving. This support is a striking feature of Lake de Funiak. You see it in the pretty cottages that are being built about the grounds. They are generally owned by northerners. Wallace Bruce has a cottage there; Pansy is building one; Mrs. Harper, of Terre Haute, Ind., another; Dr. Hatfield, of Chicago, one, and Mrs. Emily Huntingdon Miller another. One delightful spot has been turned into an "Artist's Corner" by Joaquin Miller, Mr. Durkin, Harper Brothers' well known artist, and Mr. Gross, of Covington.

The attraction which Lake de Funiak has for literary and artistic people is easily explained. The country is enveloped in a mist of most fascinating story. Ponce de Leon and his warriors once searched its forest, and, perhaps, who knows, bathed in the lake's clear waters.

It has an ideal climate. The lake lies on a ridge eighty by thirty miles in extent, and three hundred feet above sea level. "Too cold to raise oranges here," the natives say, and sure enough it is, though east, at a lower altitude in the same latitude, orange groves are abundant. The beautiful LaConte pear, peaches, apples, and quinces, are the favorite fruits of this ridge. The result is that here in this overheated, indolent land, is formed an oasis with an even temperature, unknown to the mosquito, and unvisited by the cyclone.

No better place could be found for gathering the "material" in which the artist and the writer revel. These mammoth forests of pine, magnolia, cypress, palmetto, and oak, are broken by the settlements of a peculiar people. Northerners find here a fresh field of study for pen and pencil.

And it is a fresh field for the Chautauqua Idea. During the progress of the Assembly the people of the surrounding country were in a constant wonderment over the peculiar performances, but when they understood what was meant, their coöperation was the heartiest, and their interest was untiring. The earnest workers who have undertaken to introduce the Chautauqua plans, if they are still in the first stage, are yet sure of abundant results.

In arranging the Florida Assembly the effort has been to have everything truly Chautauquan. Naturally we think of the Auditorium first, and at Lake de Funiak the situation is superb. The lake, which is about a mile in circumference, some sixty-four feet in depth, and its water of extraordinary clearness and purity, has a setting of grassy banks which slope upward from the lake some fifty feet to the edge of the forest. Into this bank, looking out over the lake, is built a square auditorium, large enough to seat 4,000 people, enclosed and furnished with an iron roof. All of the various Chautauqua developments have found their way there. The platform, presided over by the Rev. A. H. Gillet, the C. L. S. C., the normal work, a school of Greek, a kindergarten, school of cookery, and an art school. Prof. Sherwin was there, presiding over the chorus. Messrs. Fairbanks & Palmer opened a bookstore. There were Chautauqua singers, songs, speeches, and ideas, and they all took root. The beautiful situation, the desirable company that is building the new town, the vigor of the management, and its sound financial backing, evidence the future of Lake de Funiak. What more beautiful southern home could those of us who migrate southward from this land of snow and ice wish, than under the pines of Ponce de Leon's fountain, surrounded by a band of the most earnest workers in the world, and in daily reach of the best thought which money and skill can bring together? Or if we can find time and money for but a month's study of Florida and her people, what more delightful headquarters?

SPECIAL NOTES.

A SCHOOL OF PEDAGOGY has been arranged in the CHAUTAUQUA UNIVERSITY. Its purpose is to assist the directors of popular education, and especially teachers, in the study—1st, of the subjects taught in the schools; 2nd, of the principles and art of teaching; and, 3d, of the history of education. Two courses of study and of reading will be arranged. One course may be completed in a year, the other in two years. Books to be studied and read will be suggested. Examinations for promotion and certificates will be made at the close of each year. The design of the course of studies and reading is to prepare school directors and teachers for the work of organizing and teaching the schools in accordance with the best methods. Any person may become a member of the School of Pedagogy by paying the Matriculation Fee of \$5, unless it has already been paid in connection with some other department, and the Tuition Fee of \$10. The Tuition Fee is a yearly fee. All fees are payable in advance, to R. S. Holmes, Registrar of the Chautauqua University, Plainfield, N. J., from whom all particulars in reference to the school may be obtained.

Miss Susan Hayes Ward, in her article in the March impression of THE CHAUTAUQUAN, spoke disparagingly of electro-silicon as a cleaner for silver. Some good housekeepers represent it to be both a very useful and safe cleaning material. In all such cases, however, the person using the article must be the judge. But in this case we favor the opinion of the good housekeepers.

The Easter cards of the season just past were very bright and beautiful, and many of them exceedingly rich. Prang's cards, as usual, took the lead in artistic design and fine finish. They issued a very large number of new designs, and some very taking novelties in satin and plush.

At Siloam Springs, Benton Co., Arkansas, another Chautauqua has been established. A joint stock company, with ample capital, has been organized, and a state charter secured. Prof. E. Dolgoruki was elected director-general. Siloam Springs is a place of 2,000 inhabitants, near the southwest corner of the state, one mile from Indian Territory. An amphitheater large enough to seat 2,500 persons is in process of erection, and efforts are being made to secure a good and diversified program for the coming session, which opens June 11, 1885, and continues three weeks.

In the March number of THE CHAUTAUQUAN, among the errata, appears the following: "Arann, the Rev. J. M., not *Araun*." It seems it is not right yet; it should be Avann.

Transcriber's Notes:

Obvious punctuation errors repaired.

Page 437, "transfered" changed to "transferred" (and transferred it to their own home)

Page 451, "crochety" changed to "crotchety" (crotchety but kind)

Page 455, "insistance" changed to "insistence" (ill-timed insistence)

Page 458, "tation" changed to "station" (the station agent was kind enough to say)

Page 459, "corrider" changed to "corridor" (a corridor nearly two hundred feet long)

Page 468, "fnnctions" changed to "functions" (the authority and functions of)

Page 474, "Broomfield" changed to "Bromfield" (36 Bromfield Street, Boston)

Page 479, “Eebruary” changed to “February”
(February 20, 1885)

Page 480, “wvies” changed to “wives” (two
merchants’ wives)

Page 487, “fnlly” changed to “fully” (when we are
fully persuaded)

*** END OF THE PROJECT GUTENBERG EBOOK THE
CHAUTAUQUAN, VOL. 05, MAY 1885, NO. 8 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.